STEPPERONLINE®

User Manual

DSY-RS Series

Low-voltage Servo Drive

User manual (V1.0)



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Chapter 1 Selection of Servo System

1.1 Drive specifications

Basic Specifications	**Model**			**DSY-200RS**	**DSY-400RS**	**DSY-750RS**	**DSY-1200RS**
	Continuous Output Current（Arms）			7.5	15	25	40
	Maximum Output Current（Arms）			24	48	80	120
	Power Supply Input			24VDC~70VDC			
	Environment	**Temperature**		Operation Temperature：0℃～55℃ (If Environment temperature over 55 ℃, plz lower it) Storage Temperature: -40℃～80℃			
		Humidity		0～90% RH (No dew environment)			
		Elevation		Elevation ≤1000m			
		Vibration/Strike		5.88m/s2 / 19.6m/s2			
		IP Grade / Contamination		IP20/ Contamination Rating 2			
	Control Method			MOSFET SVPWM Space vector pulse width modulation algorithm			
	Encoder			17-bit magnetic encoder；23-bit optical encoder			
	Input/Output Ports	**IO Signal**	**Input**	3 pcs (General)			
			Output	1 General DO Output, 1 Brake Output			
	Network	**RS485**		Modbus Communication			
		RS-232		Connect with PC to debug servo drive			

	Protection Function			Overcurrent, Overload, Overvoltage, Low Voltage, Over speed, Overheat, Encoder error, Communication error, Position deviation
	Control Mode			Position Control Mode; Speed Control Mode; Torque Control Mode; Mixed-mode Switching
	Position Control Mode	**Location Command Format**		PULSE/DIR CW/CCW A, B Phase Cross Pulse Function code parameters setting
		Smoothing Filter		Smoothing process for position command to ensure motor run more stably
		Electronic Gear		Two sets of electronic gear ratios available
		Feedforward Compensation		0 ～ 100% default resolution ratio: 1%)
	Speed/Torque Control Mode	**Command Form**		Internal speed/ torque command Function code parameters setting
		Speed Change Rate	**Current control Precision**	±5% (Rated torque)
			Overload Capacity	300% motor rated torque
			Temperature Fluctuation	25±25℃ (Rated speed)
	Common Spec	**Self-tuning Function**		Identifying inertia
		Over-distance (OT) Prevent Function		Slowing down to stop when P-OT、N-OT
		Error Feedback and Record		Red and green double-hole indicator, 9 set of records

1.2 Servo motors and drives model description

1.2.1 Drive Model Description

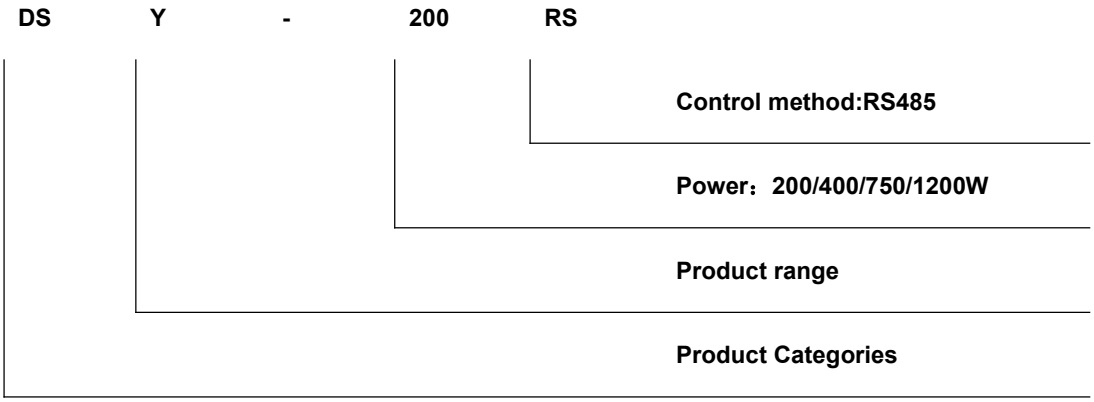


Figure1.1 Drive naming rule

1.2.2 Servo motor model description

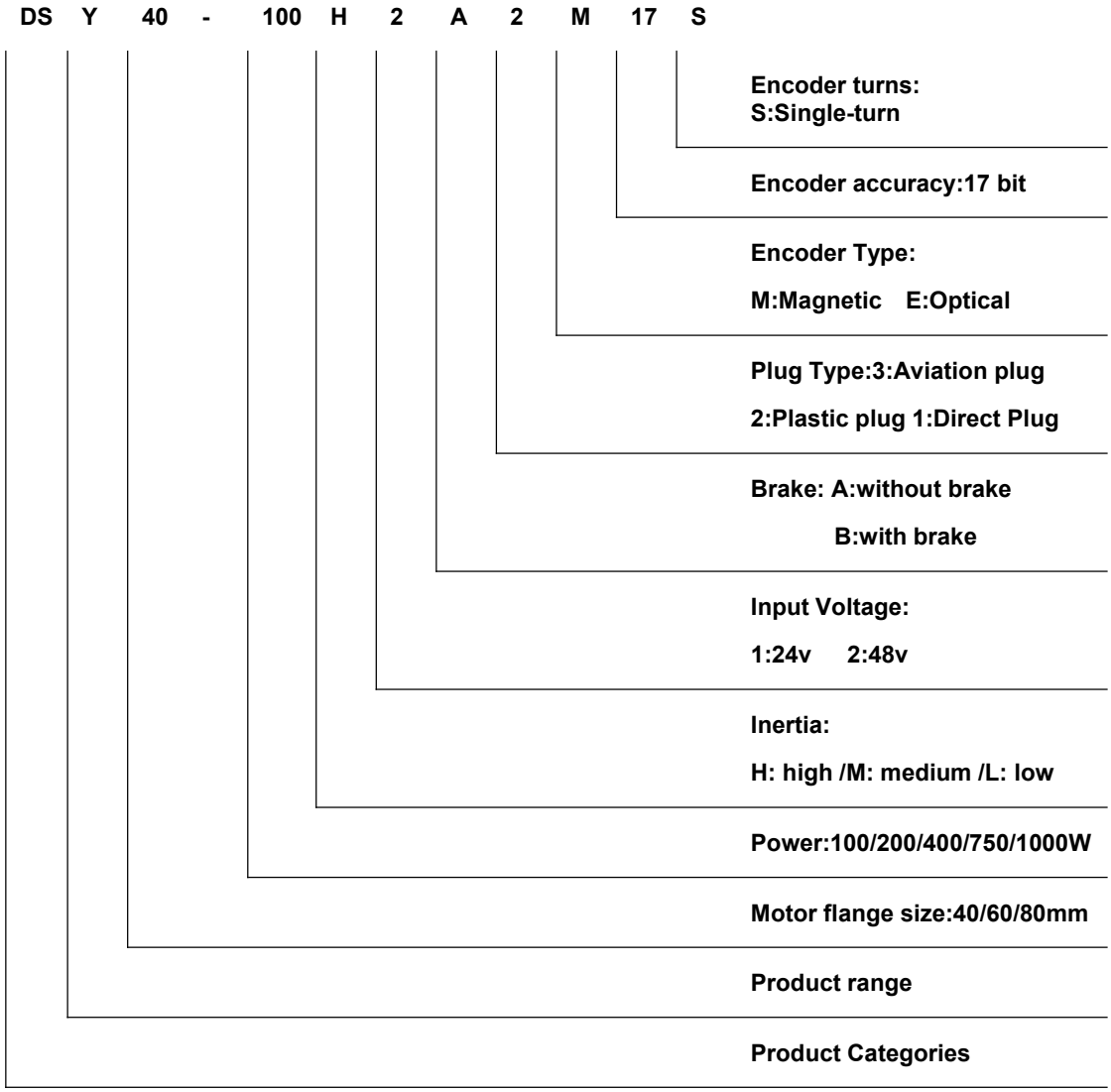


Figure 1.2 Motor naming rules

1.3 Servo drive and servo motor matching list

Table 1.2 Servo drive and motor matching table

Servo Motor Models		Flange	Power	Drive Model	Matching cable
				DSY-xxRS	
Small inertia	DSY40-100L2A2M17S	40	100W	200	P1-DS-xx E2-DS-xx
	DSY40-100L2B2M17S	40	100W	200	
	DSY60-200L2A2M17S	60	200W	200	
	DSY60-200L2B2M17S	60	200W	200	

Servo Motor Models		Flange	Power	Drive Model	Matching cable
				DSY-xxRS	
	DSY60-400L2A2M17S	60	400W	400	
	DSY60-400L2B2M17S	60	400W	400	
	DSY80-750L2A2M17S	80	750W	800	P2-DS-xx E2-DS-xx
	DSY80-750L2B2M17S	80	750W	800	
	DSY80-1000L2A2M17S	80	1000W	1200	
	DSY80-1000L2B2M17S	80	1000W	1200	

1.4 Matching motor parameters

Table 1.3 Motor parameter table

Motor spec.	Rated output KW	Rated torque Nm	Max.Torque Nm	Rated current Arms	Rated Speed min^{-1}	Rotor Inertia $Kgm^{2}\cdot 10^{-4}$
DSY40-100L2A2M17S	0.1	0.318	0.954	3.1	3000	0.046
DSY40-100L2B2M17S	0.1	0.318	0.954	3.1	3000	0.048
DSY60-200L2A2M17S	0.2	0.64	1.91	5.2	3000	0.27
DSY60-200L2B2M17S	0.2	0.64	1.91	5.2	3000	0.29
DSY60-400L2A2M17S	0.4	1.27	3.81	10.2	3000	0.49
DSY60-400L2B2M17S	0.4	1.27	3.81	10.2	3000	0.51
DSY80-750L2A2M17S	0.75	2.39	7.17	18.5	3000	1.42
DSY80-750L2B2M17S	0.75	2.39	7.17	18.5	3000	1.62
DSY80-1000L2A2M17S	1.0	3.2	9.6	24.5	3000	1.92
DSY80-1000L2B2M17S	1.0	3.2	9.6	24.5	3000	2.12

1.5 Matching cable

The naming rules for motor power lines are as follows：

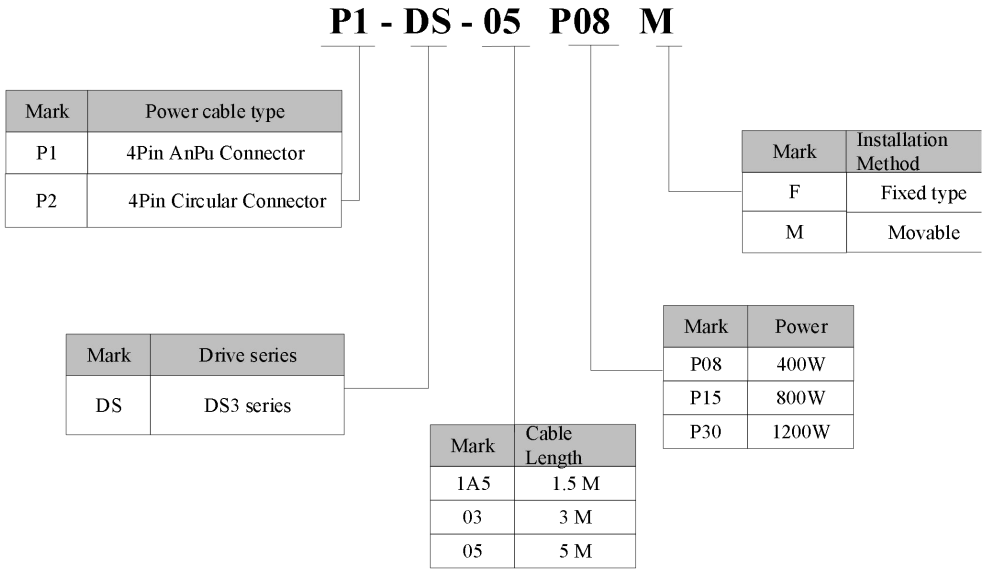
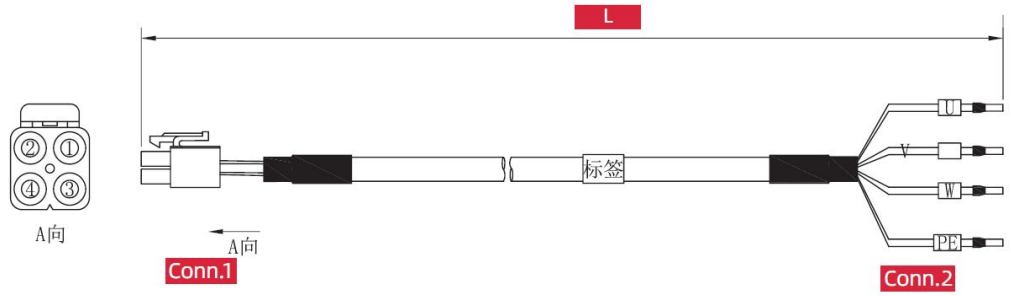
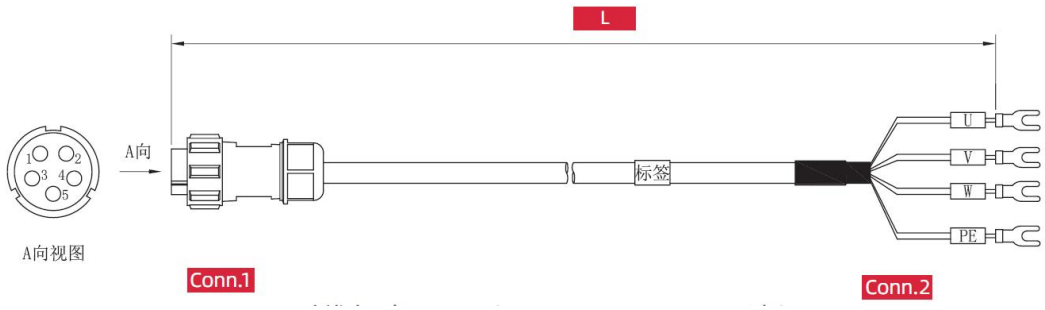


Figure 1.3 Power line naming rules

The structure diagram of the motor power cable is shown in the table below：



1.P1 AnPu Connector Cable（Lead Length L =1.5m、3m、5m）



2.P2 Circular Connector Cable（Lead Length L =1.5m、3m、5m）

Figure 1.4 Power cable structure diagram

The motor power line pins are defined as follows

Conn.1	Pin definition	Conn.2	Pin color
1	U	U	Red
2	V	V	Yellow
3	W	W	Blue
4	PE	PE	Yellow green

The naming rules for motor encoder lines are as follows:

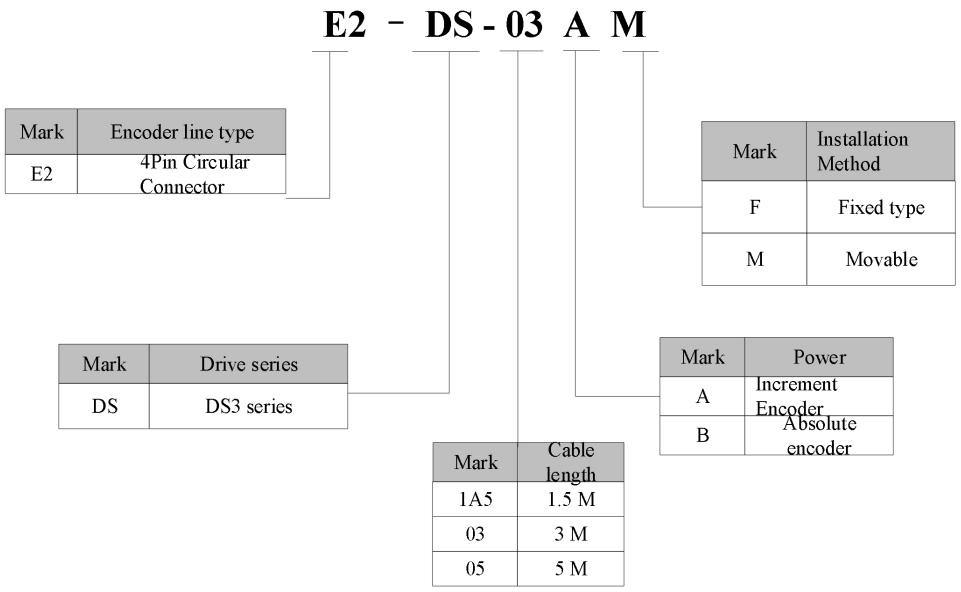


Figure 1.5 Encoder line naming rules

The structure diagram of the encoder cable is shown in the table below:

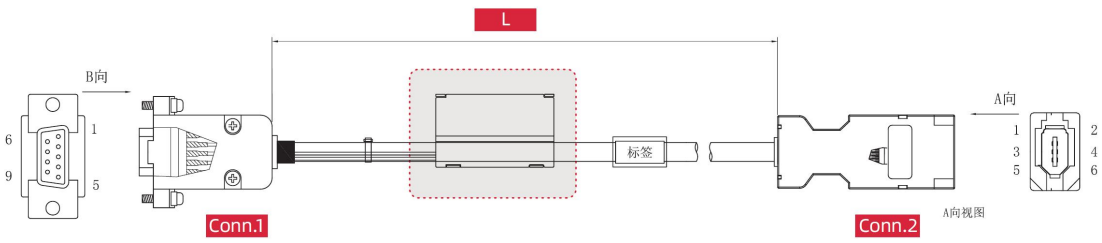


Figure 1.6 Encoder cable structure diagram (L=1.5m、3m、5m)

The motor encoder line pins are defined as follows:

Conn.1	Pin definition	Conn.2	Pin color
7	5V	1	Blue
8	0V	2	Orange
3	SD+	5	Brown
4	SD-	6	Green
Shell	PE	Shell	/
Battery	/	3	BAT+
		4	BAT-

Chapter 2 Dimensions and Product Installation

2.1 Servo drive dimensions

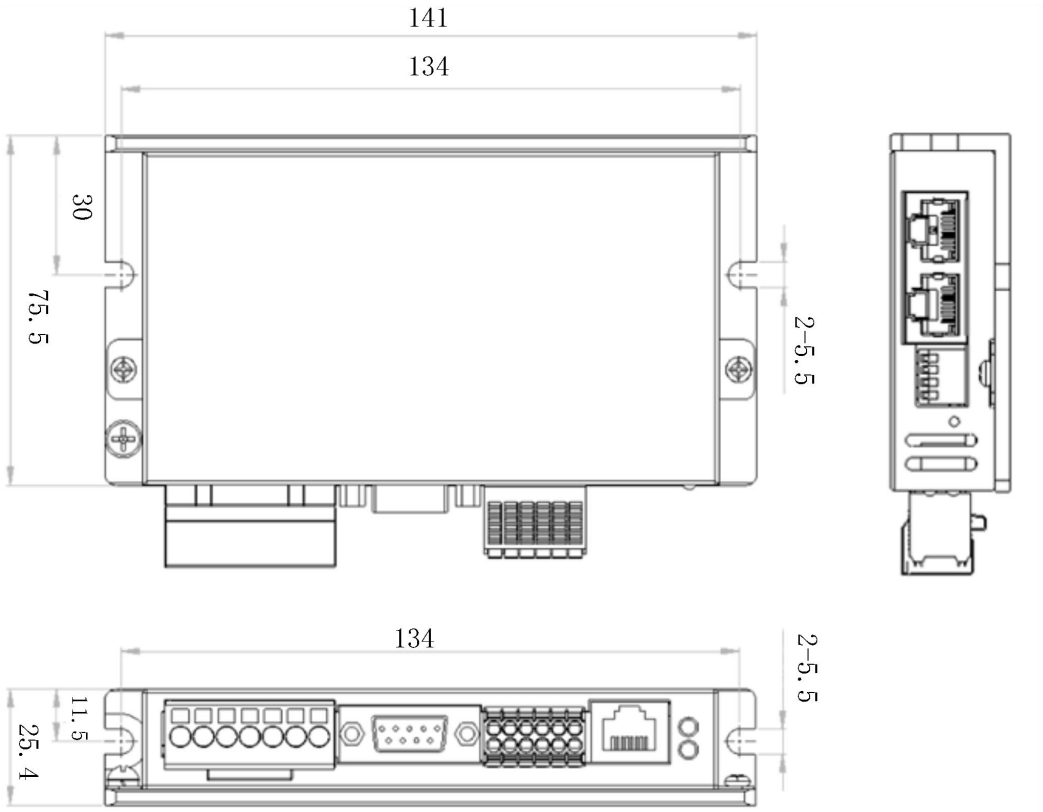


Figure 2.1 The picture of DSY-200RS/DSY-400RS drive structure

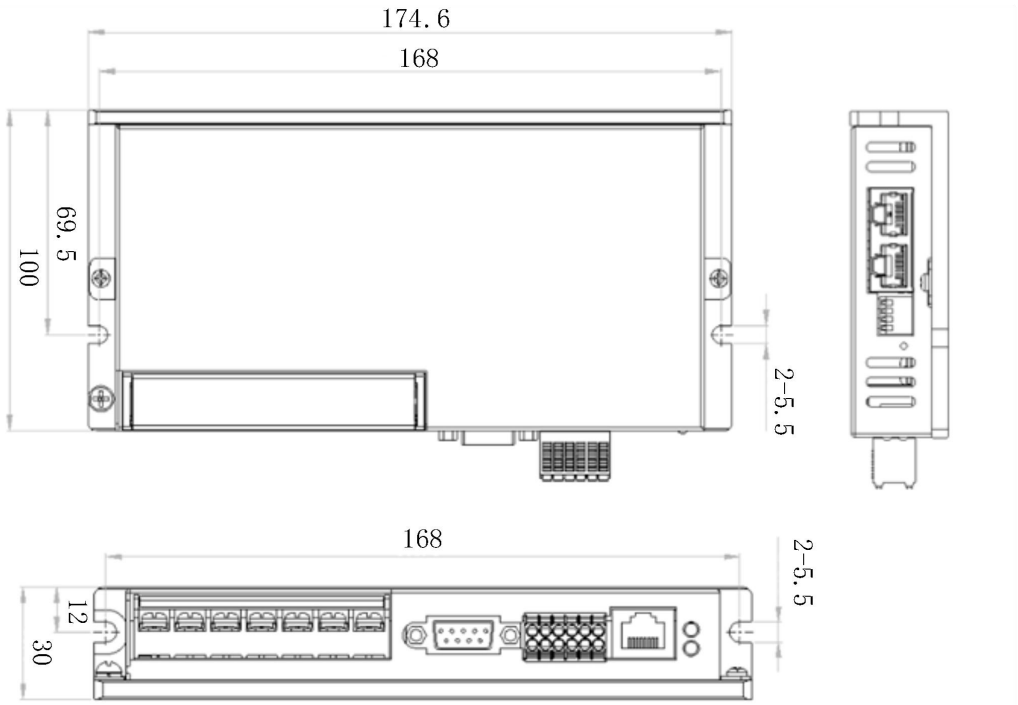


Figure 2.2 The picture of DSY-750RS drive structure

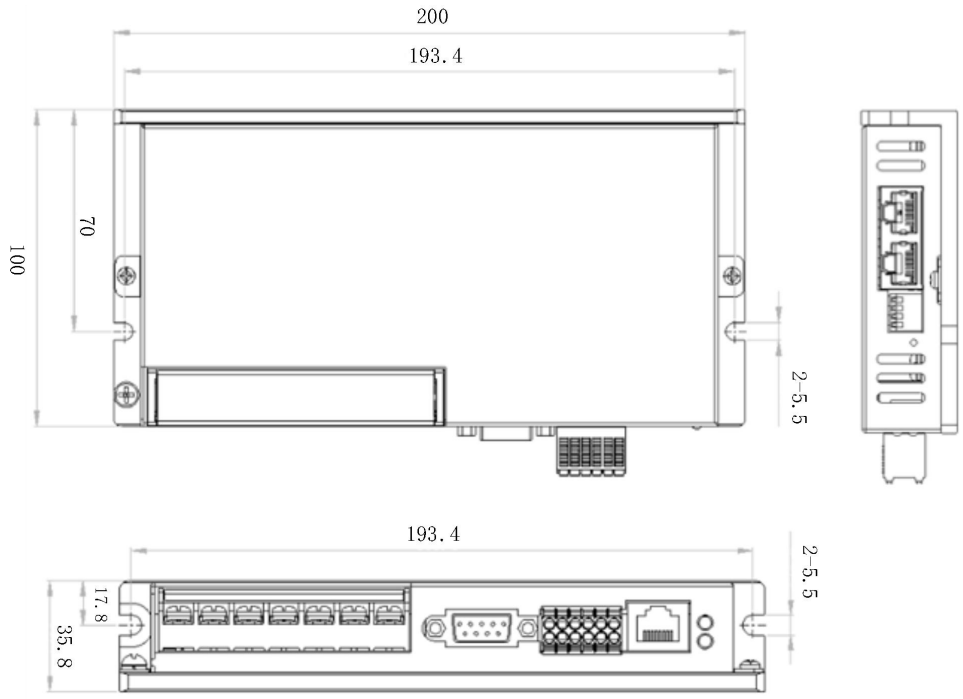


Figure 2.3 The picture of DSY-1200RS drive structure

2.2 Servo drive installation

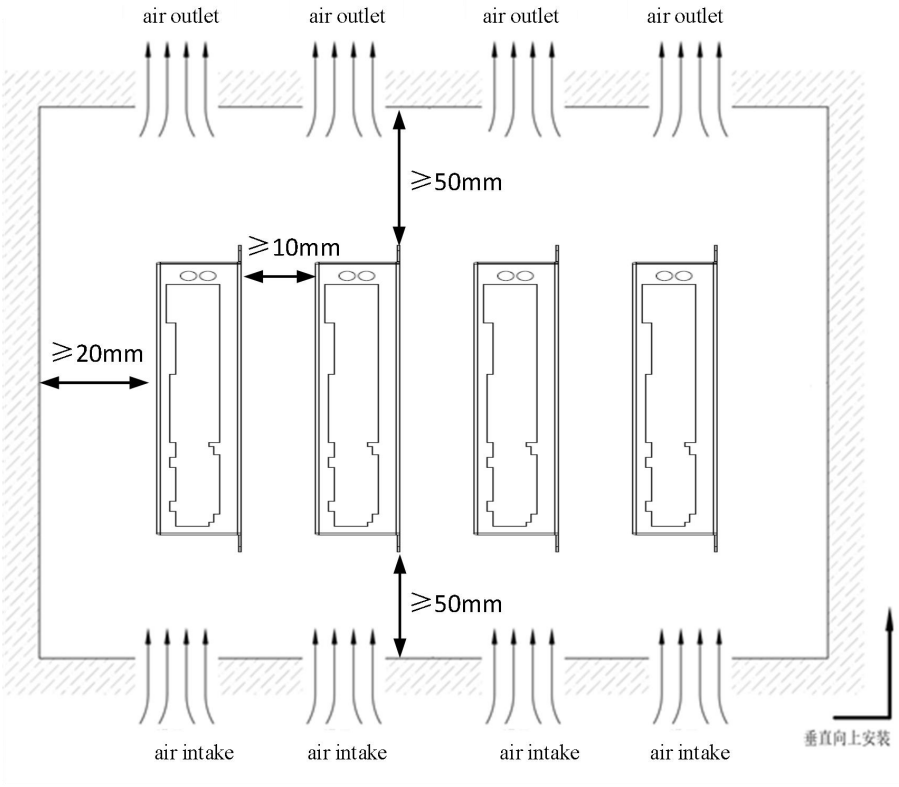
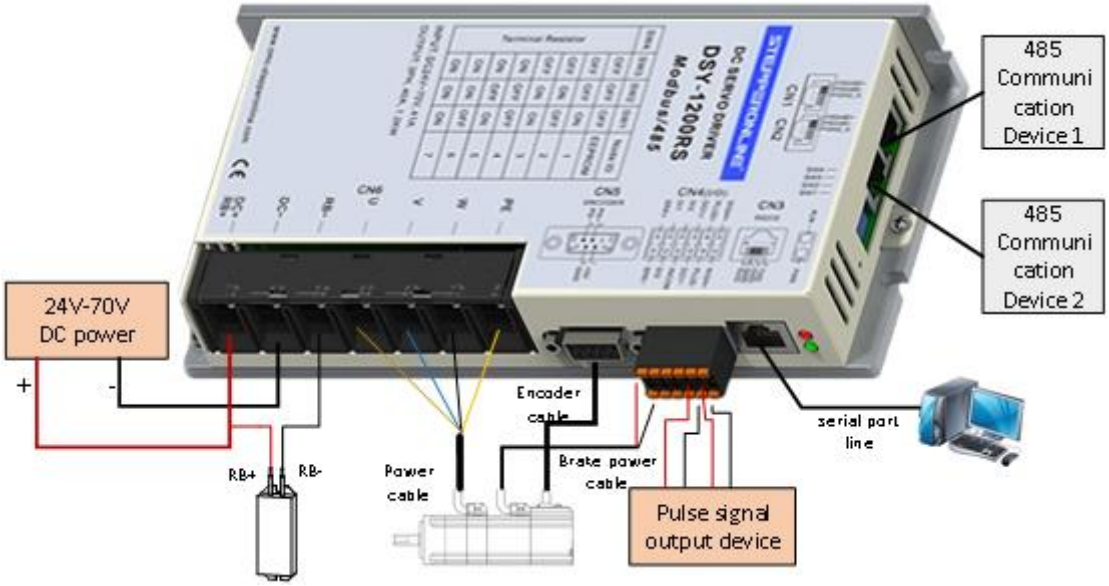


Figure 2.4 Servo drive installation

Chapter 3 Servo Drive and Motor Connection

3.1 System structure diagram



The names, functions and specifications of each part of the drive are as follows.

Table 3.1 Servo drive main circuit terminal name and function

No.	Name	Description
1	DC+、DC-	Input control circuit power according to specifications
2	RB+,RB- Regenerative Resistor Terminal	External Brake Resistor Connection Port
3	Motor power cable	Connect servo motor three-phase and ground wire
5	Communication port	Connected to PC via RS232 to monitor the drive, test and change parameter, etc
6	Communication control	Connected to RS485 communication device
7	Control port	Connect to host controller for IO signal control
8	Encoder feedback port	Connected to the motor encoder cable

3.2 Main circuit

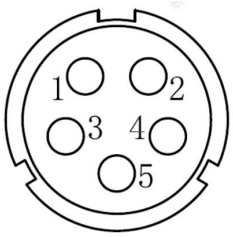
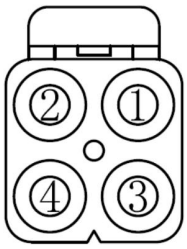
The functions, names and specifications of each part of the main circuit are as follows.

Table 3.2 Servo drive main circuit terminal function specifications

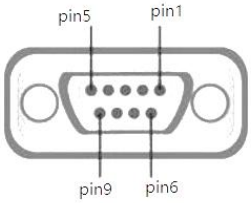
Name	Terminal mark	Function, specification
Power input terminal	DC+、DC-	Power circuit of the power supply according to the input specifications.
Regenerative resistor connection terminal	RB+,RB-	The external regenerative resistor between the RB + with RB -. For the external brake resistor, please buy separately.
Servo motor Power cable connection terminal	U、V、W	Connected to the U, V, W of the motor.
Ground Terminal	PE	Two grounding terminals are connected to the power grounding terminal and the motor grounding terminal. Be sure to ground the entire system.

3.3 Power cable connector

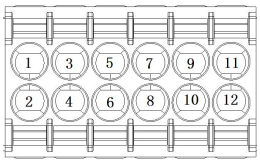
Table 3.3 Power cable connector

Terminal pin		Connector picture
5Pin circular connector		
Pin NO.	Signal name	
1	U	
2	V	
3	W	
4	PE	
4Pin AnPu connector		
Pin NO.	Signal name	
1	U	
2	V	
3	W	
4	PE	

3.4 Encoder connector

Pin definition					Picture
1	2	3	4	5	
		SD+	SD-		
6	7	8	9	PE	
	+5V	GND		PE	

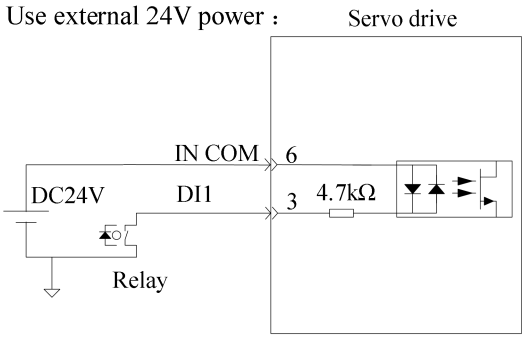
3.5 DIDO Port

Definition		PIN	Function
General input and output signal 	BRK+	1	Brake output
	BRK-	2	
	DI1	3	Digital input 1
	DI2	4	Digital input 2
	DI3	5	Digital input 3
	ICOM	6	DI power supply input port， 24V
	DO+	7	Digital output 1
	DO-	8	
	PULSE+	9	Pulse signal input（5V~24V, capatible）
	PULSE-	10	
	SIGH+	11	Direction signal input（5V~24V, capatible）
	SIGH-	12	

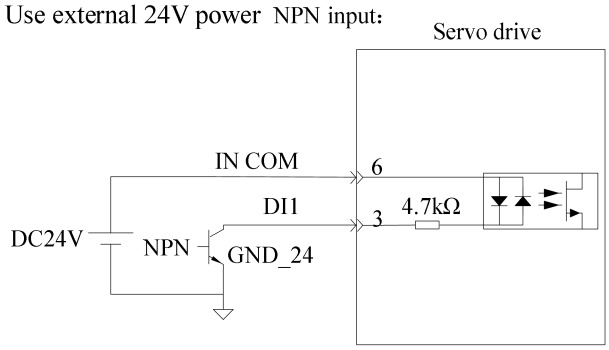
3.5.1 Digital input circuit

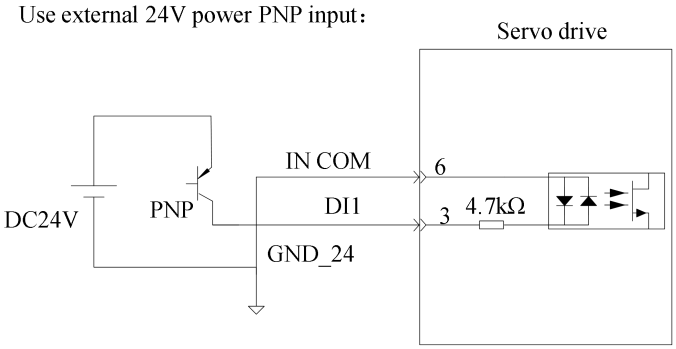
Take DI1 as an example, DI1~DI3 is the same.

a) **When the host computer is relay output:**



b) When the host computer is open collector output:



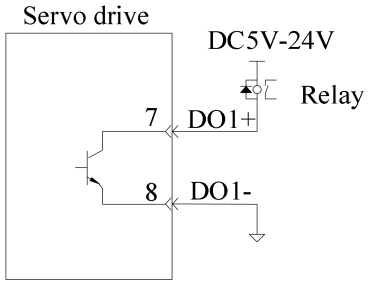


Note：

The mixing of NPN and PNP inputs is not supported.

3.5.2 Digital output circuit

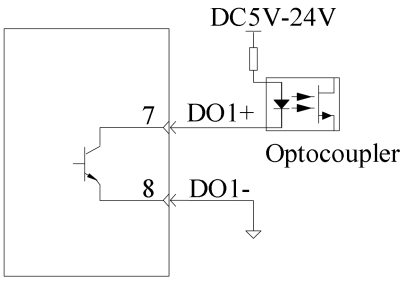
a) When the host computer is a relay input



Note：

When the host computer is relay input, be sure to connect the freewheeling diode in the correct direction, otherwise the DO port may be damaged.

b) When the host computer is optocoupler input



Note：

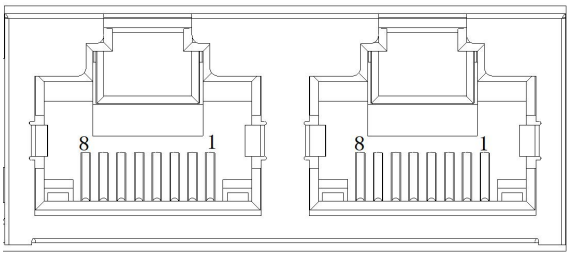
1. Be sure to connect a current limiting resistor.
2. The maximum allowable voltage and current of the internal optocoupler circuit of the servo driver are as follows:
   Voltage: DC30V (maximum)
   Current: DC50mA (maximum)

3.6 Communication signal port

3.6.1 CN1/CN2 Bus port CN1/CN2

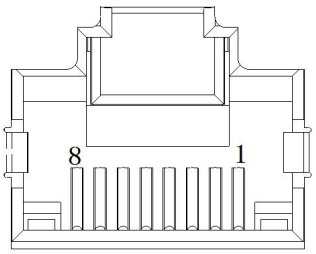
CN1 and CN2 is the 485 communication connector.

Table 3.4 communication connector pin definition

Pin	Signal name	Direction	Terminal
1	RS485+	Output	
2	RS485-	Output	
3	GND	Output	
4	--	unused	
5	--	unused	
6	--	unused	
7	--	unused	
8	--	unused	

3.6.2 Post computer communication port CN3

Table 3.5 communication connector pin definition

Pin	Definition	Function	Terminal
1	--		
2	--		
3	RS232-TXD	RS232 communication port	
4	GND		
5	GND		
6	RS232-RXD		
7	--		
8	--		
shell	PE	Shielding	

Chapter 4 Control

4.1 Basic control

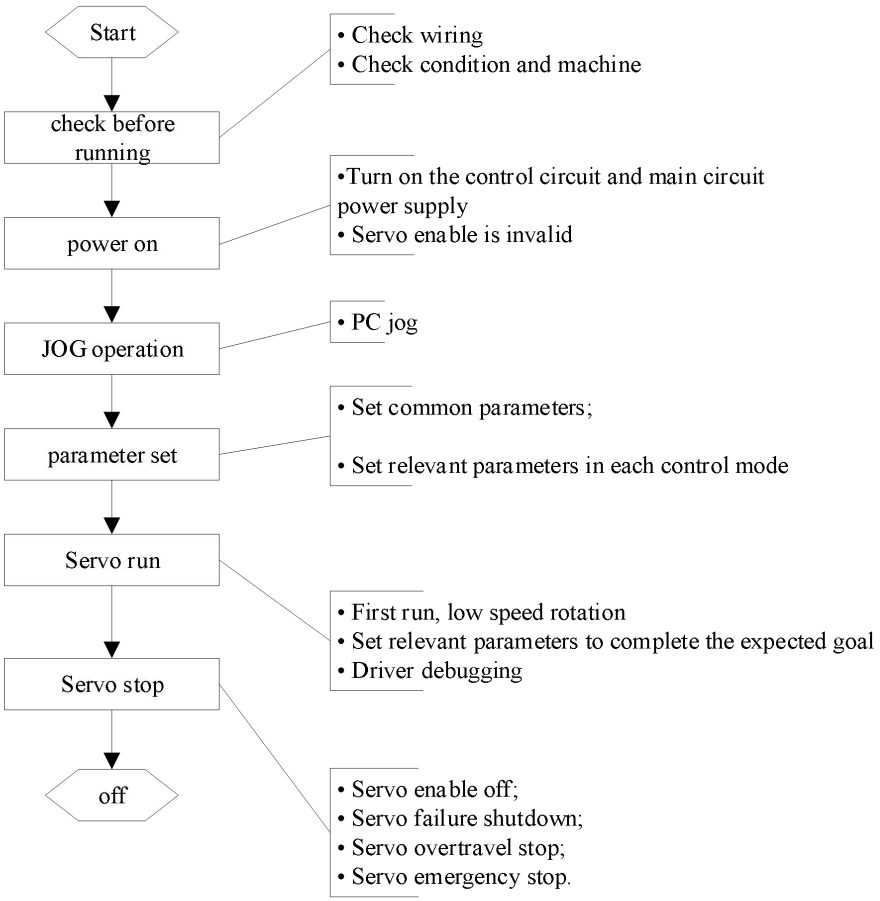


Figure 4.1 Servo setting process

4.1.1 Check before running

The following checks are required before the servo driver and servo motor run：

Table 5.1 Check steps before operation

Recor	No.	Description
Wiring		
□	1	The control circuit power input terminals (DC+, DC-) of the servo drive must be connected correctly.
□	2	The output terminals (U, V, W) of the main circuit of the servo driver and the main circuit cables (U, V, W) of the servo motor must have the same phase and be correctly connected.
□	3	The wiring of each control signal cable of the servo driver is correct: the external signal cables such as brake and overtravel protection have been reliably connected.
□	4	Servo drives and servo motors must be reliably grounded.
□	5	The stress of all cables is within the specified range.
□	6	The wiring terminals have been insulated.
Condition and machine		
□	1	There are no wire stubs, metal filings and other foreign objects inside and outside the servo drive that may cause short circuits in the signal wires and power wires.
□	2	Servo drives and external braking resistors are not placed on combustible objects.
□	3	The installation of the servo motor, the connection of the shaft and the machine must be reliable.
□	4	The servo motor and connected machinery must be in operable condition.

4.1.2 Power on

1）Turn on the control loop (DC+, DC-)

●After the control circuit power supply and the main circuit power supply are connected, the bus voltage indicator shows no abnormality and wait for the host computer to give the servo enable signal.

●If the driver's red alarm light is on, please check the wiring and eliminate the cause of the fault.

4.1.3 JOG operation

Please use the JOG mode in the host computer software to confirm whether the servo motor can rotate normally, and there is no abnormal vibration and abnormal sound when rotating. The JOG mode control panel is shown in Figure 4.2.

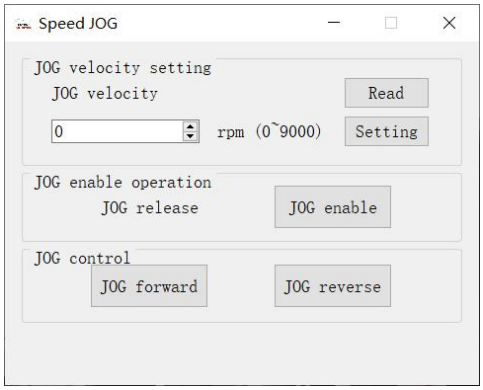


Figure 4.2 JOG mode operation interface

（the left one is the English interface, and the right one is the Chinese interface)

The JOG setting process is shown in the figure below.

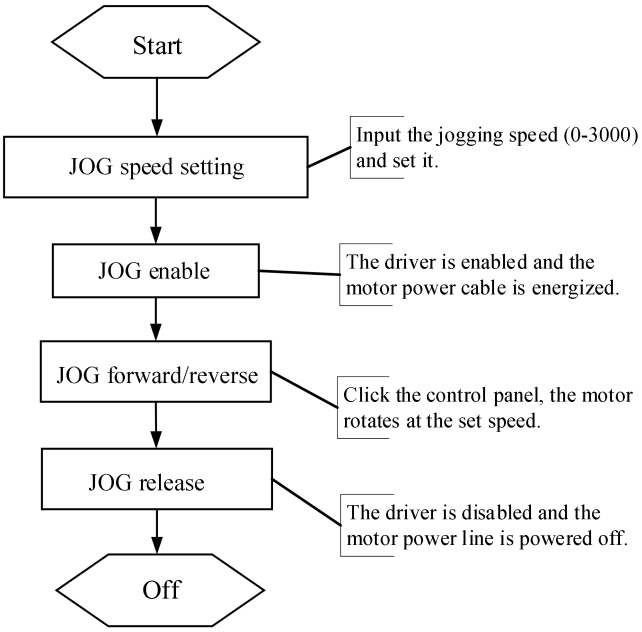


Figure 4.3 JOG control process

4.1.4 Rotation direction selection

By setting "rotation direction selection (P00.01)", the rotation direction of the motor can be changed without changing the polarity of the input command

☆Associated function code:

Function code	Name	SettingRange	Unit	Function	Type	Effective time	Default setting
P00.01	Direction of rotatio	**0**--CCW direction is the forward direction(counterclockwise) **1**--CW direction is the forward direction (clockwise)	-	Set the motor forward/reverse	Setting after motor stop	Re-power	0

Note:

When the rotation direction selection (P00.01) is changed, the output pulse form of the servo driver and the positive or negative of the monitoring parameters will not change.

4.1.5 Brake Setting

The brake is a mechanism that prevents the servo motor shaft from moving when the servo driver is in a non-running state, to keep the motor locked in position, so that the moving parts of the machine will not move due to its own weight or external force.

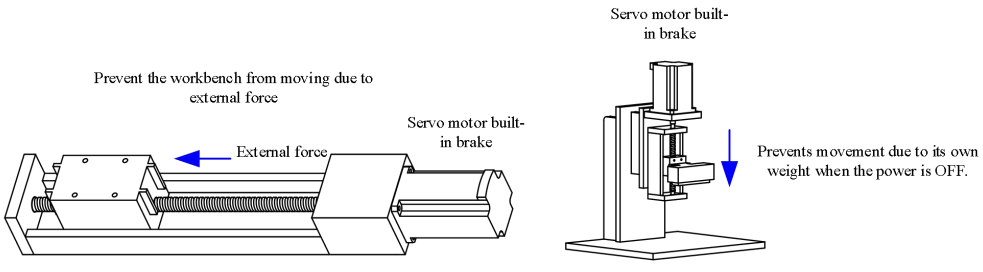


Figure 4.4 Brake application schematic

Note:

• The brake mechanism built into the servo motor is a non-energized fixed special mechanism, which cannot be used for braking purposes, and is only used to keep the servo motor in a stopped state.

• The brake coil has no polarity.

• After the servo motor stops, the servo enable (S-ON) should be turned off.

• When the motor with the built-in brake running, the brake may make a clicking sound, but it has no effect on the function.

• When the brake coil is energized (the brake is released), magnetic flux leakage may occur at the shaft end and other parts. Be careful when using instruments such as magnetic sensors near the motor.

a）Brake wiring

The connection of the brake input signal has no polarity, and the user needs to prepare a 24V power supply. The standard wiring example of the brake signal BK and the brake power supply is as follows:

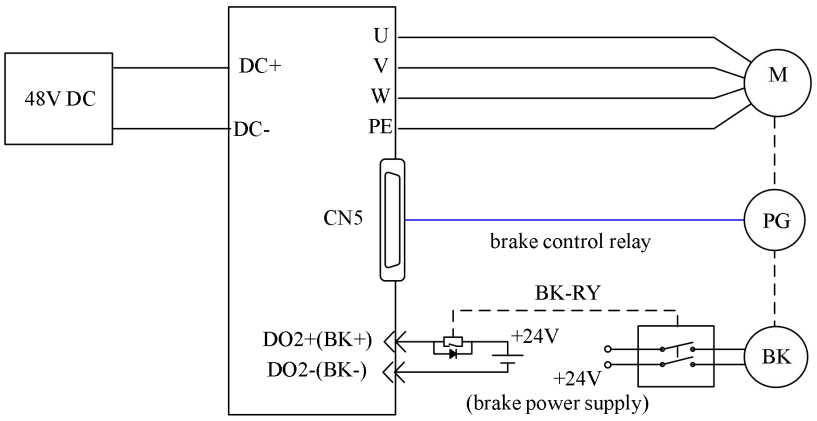


Figure 4.5 Brake Wiring Diagram

Note:

●The length of the motor brake cable needs to fully consider the voltage drop caused by the cable resistance, and the brake needs to ensure that the input voltage is at least 21.6V.

●It is best not to share the power supply with other electrical appliances, so as to prevent the voltage or current from reducing due to the work of other electrical appliances and eventually cause the brake to malfunction.

●When connecting the relay, please be sure to connect the freewheeling diode (recommended ultra-fast recovery diode, 1A, 1000V model: UF1010G), and the direction is correct, otherwise the DO port may be damaged.

●It is recommended to use cables above 0.5mm^2.

●Brake is the Homeal power consumption component. Brakes with different powers are installed according to the rated torque of the motor. The output current of 24V power supply inside the servo driver can directly drive the motor lock and drive current 1A.

b）Brake software setting

For a servo motor with a brake, the second DO terminal of the servo drive must be configured as function 11 (FunOUT.11: BK, brake output), and the effective logic of the DO terminal must be determined.

☆Associated function code:

Code	Name	Function Name	Function
FunOUT.11	BK	Brake output	Invalid, the brake power is disconnected, the brake acts, and the motor is in a position-locked state; Valid, the power of the brake is turned on, the brake is released, and the motor can rotate.

4.1.6 Brake setting

When the torque of the motor is in the opposite direction to the rotational speed, the energy is transmitted back to the driver from the motor end, which causes the bus voltage to increase. When it reaches the braking point, the energy can only be consumed through the braking resistor. At this time, the braking energy must be consumed according to the braking requirements, otherwise the servo drive will be damaged. DS3 series drivers need an external braking resistor, and the connection of the braking resistor is shown in Figure 4.6.

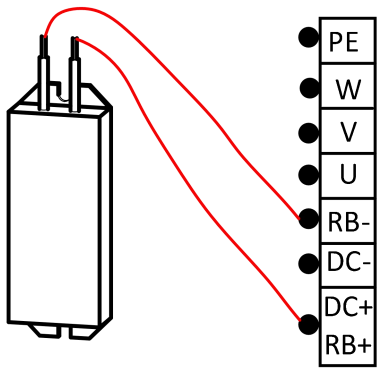


Figure 4.6 Wiring diagram of braking resistor

☆Associated function code:

Function code	Name	SettingRange	Unit	Function	Type	Effective time	Model

P0018	Energy consumption resistance setting	0--Use built-in power consumption resistor. 1--Use external power-consuming resistors and cool naturally. 2--Use external power-consuming resistors and force air-cooled. 3--No energy consumption resistor, relying on capacitance absorption.	1	0	Effective immediately	Setting after motor stop	PST
P0019	External resistor power capacity	1～65535	1W	Model parameters	Effective immediately	Setting after motor stop	PST
P0020	External resistance value	1～1000	1Ω	Model parameters	Effective immediately	Setting after motor stop	PST
P0021	External resistance heating time constant	1000~65535	1ms	Model parameters	Effective immediately	Setting after motor stop	PST
P0022	Energy consumption braking start voltage	0~410	1V	Model parameters	Effective immediately	Set when running	PST

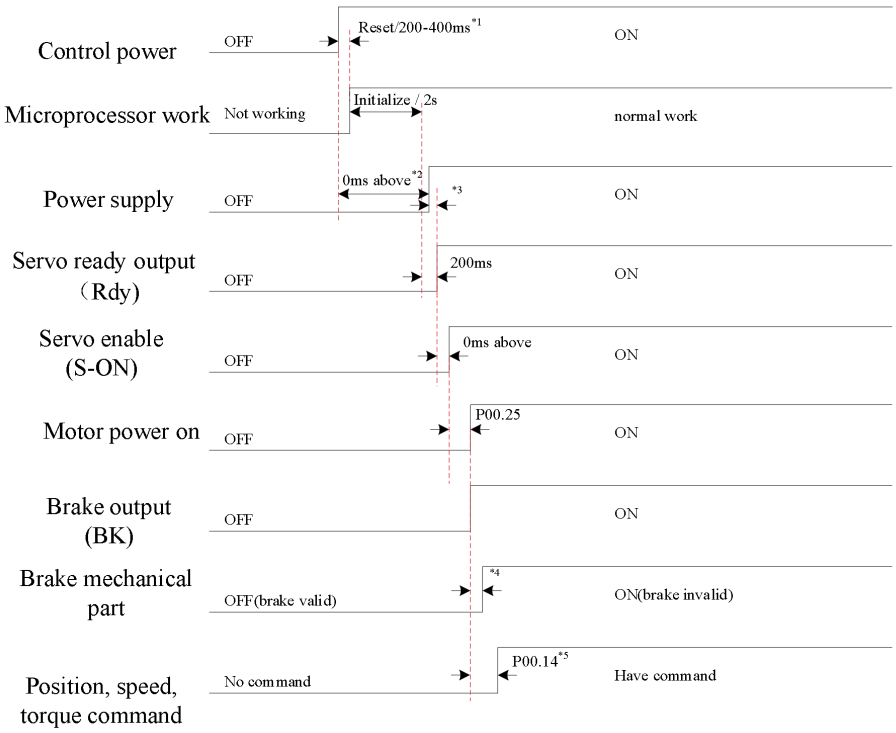
4.1.7 Servo operation

When the servo driver is in the enable state, the servo driver is in the running state at this time; but because there is no command input at this time, the servo motor does not rotate and is in the locked state. After the command is input, the servo motor rotates.

Table 4.2 Servo operation instructions

No.	Description
1	When running for the first time, set appropriate commands to make the motor rotate at a low speed, and confirm whether the motor rotation is correct.
2	Observe whether the direction of rotation of the motor is correct. If it is found that the direction of the motor is opposite to the expected one, please check the input command signal and command direction setting signal.
3	If the rotation direction of the motor is correct, you can use the host computer software to observe the actual speed P18.01 of the motor, the average load rate P18.02 and other parameters.
4	After checking the operation status of the motor above, you can adjust the relevant parameters to make the motor work in the expected working condition.
5	Refer to Chapter 5 "Operation Performance Adjustment" to debug the servo driver.

1）Power On Timing Chart



Note:

*1: The reset time is determined by the establishment time of the microprocessor +5V power supply;

*2: Above 0ms, it means that the time is determined by the actual moment when the main power supply is switched on;

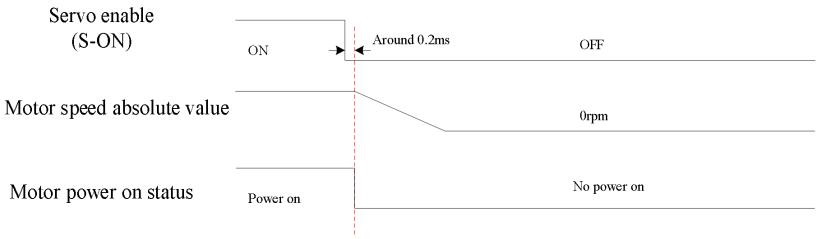
*3: When the control power supply and the power supply are powered on at the same time, this time is the same as the time from the completion of microprocessing initialization to the effective time of Rdy;

*4: For the delay time of the action of the brake mechanical part, please refer to the relevant specifications of the motor;

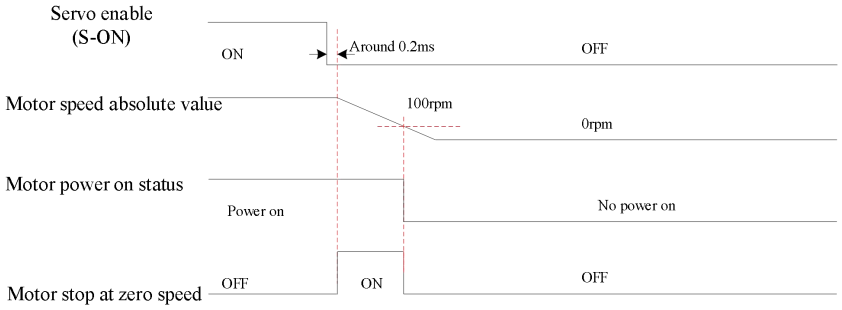
*5: When DO function 11 (FunOUT.11: BK) is not allocated, P00.14 has no effect.

2）Shutdown when servo enable OFF, warning or fault occurs timing diagram

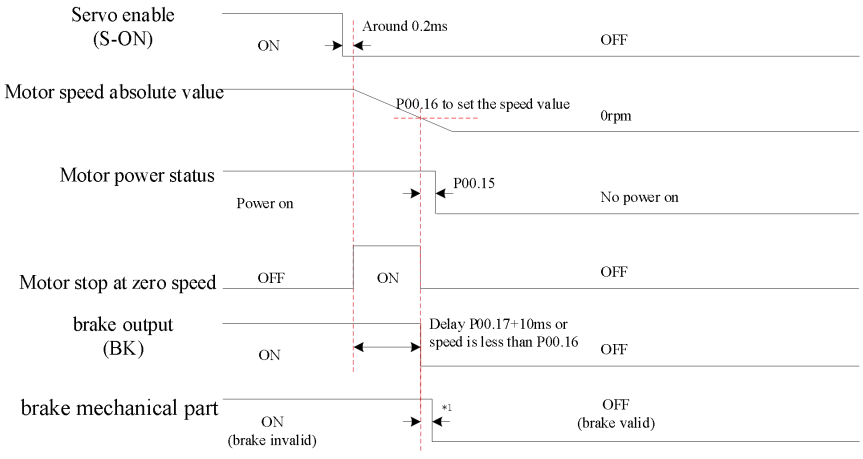
a） Servo enable OFF Non-brake: coast to stop, keep free running state (P00.10=0)



b） Servo enable OFF Non-brake: Stop at zero speed and keep in free running state (P00.10=1)



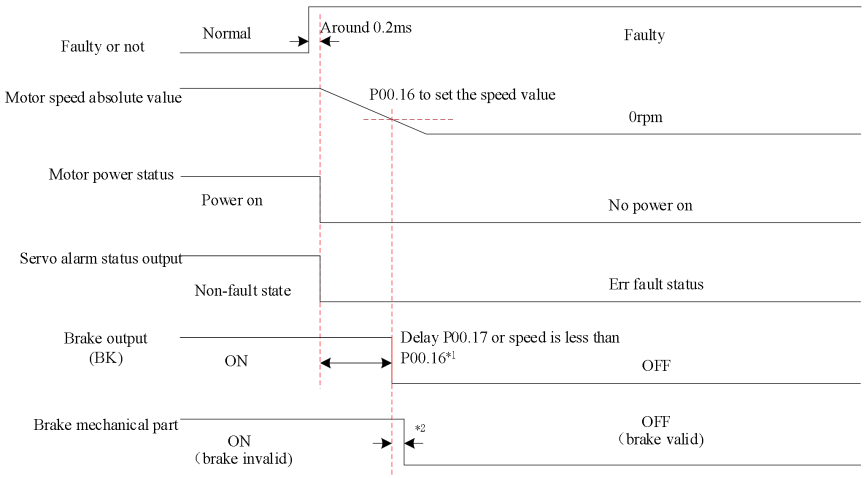
c） Servo enable OFF with brake: forced to stop at zero speed, keep free state



Note:

*1: For the delay time of the action of the brake mechanical part, please refer to the relevant specifications of the motor.

d）Type I faulty : free stop, keep free running state (P00.11=0)

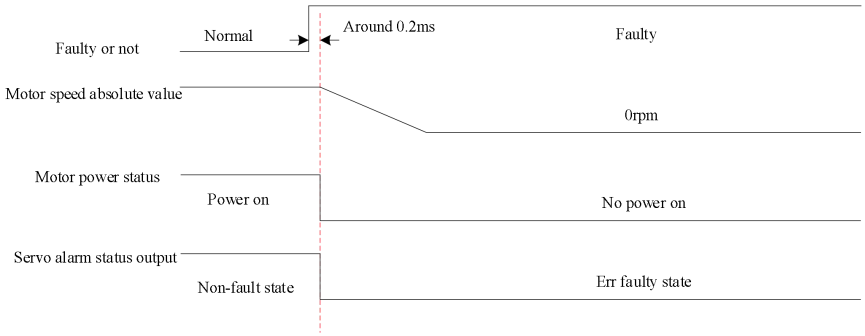


Note:

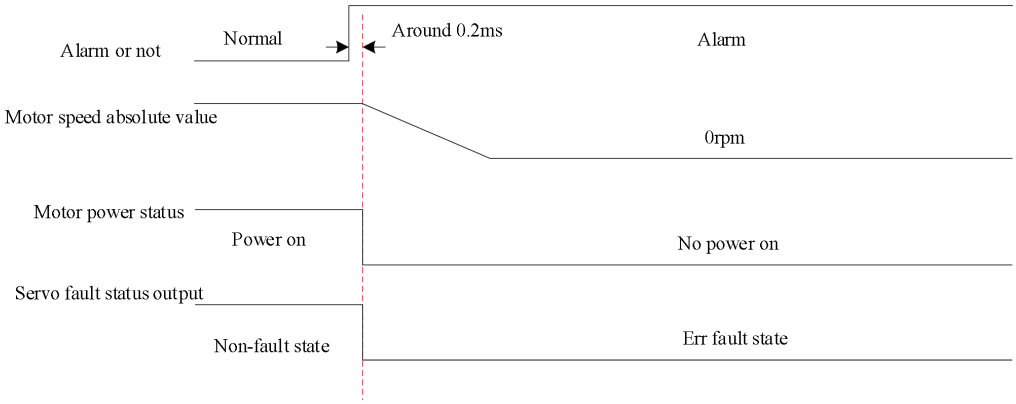
*1：When DO function 11 (FunOUT.11: BK) is not assigned, P00.16 and P00.17 have no effect.

*2：For the delay time of the action of the brake contact part, please refer to the relevant specifications of the motor.

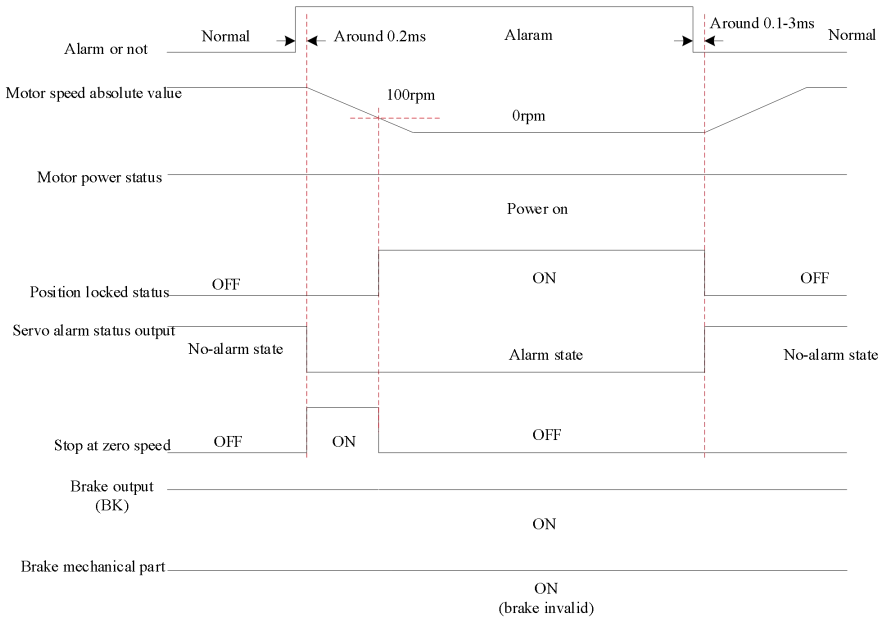
e）Type II faulty



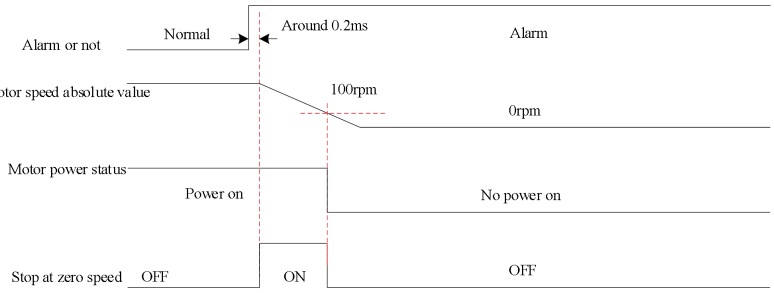
f）Overtravel stop warning non-holding brake: free stop, keep free running state (P00.13=0)



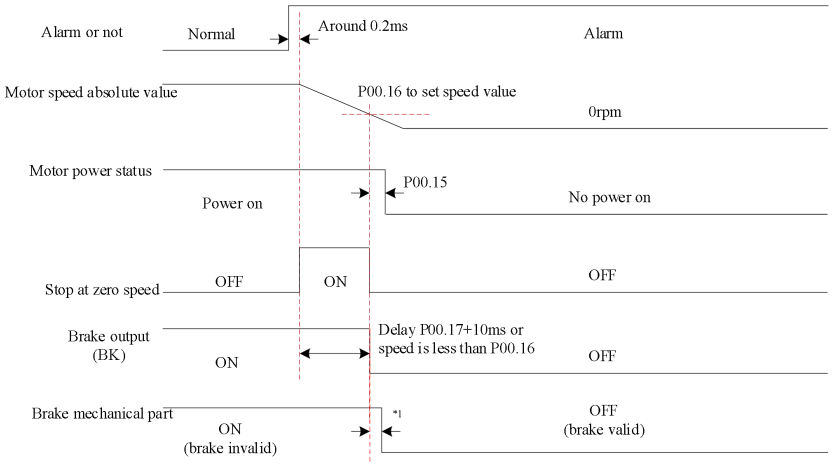
g）Overtravel, brake stop warning: stop at zero speed, keep position locked (P00.13=1)



h）Overtravel stop warning non-brake: stop at zero speed, keep free running state (P00.13=2)



i）Overtravel stop warning with brake: stop at zero speed, keep free running state (P00.13=0 or P00.13=2)

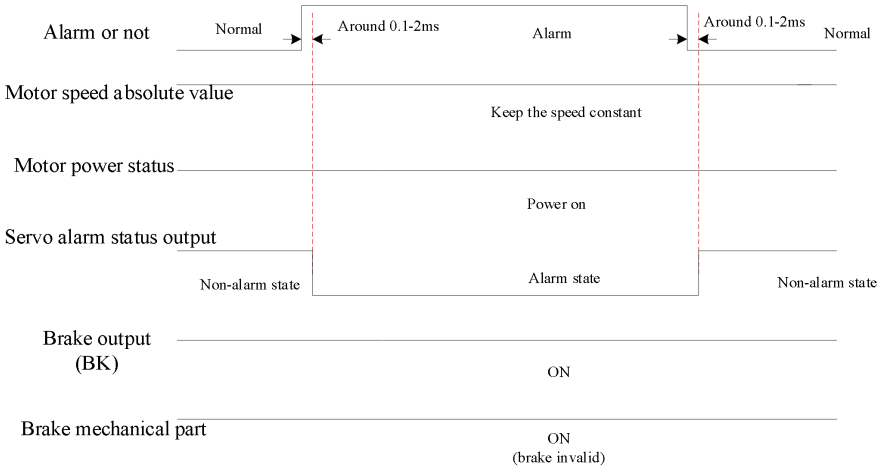


Note:

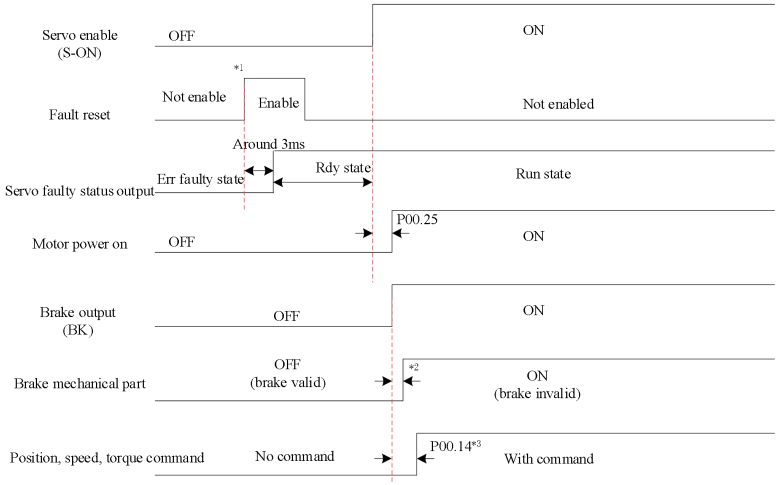
*1：For the delay time of the action of the brake mechanical part, please refer to the relevant specifications of the motor.

j）Non-shutdown warning：

Except for emergency brake, forward overtravel warning, and reverse overtravel warning, other warnings have no effect on the current status of the servo, as shown below.



k）Fault reset:



Note：

*1：DI fault reset signal (FunIN.2: ALM-RST) is valid when edge changes;

*2：For the delay time of the action of the brake contact part, please refer to the relevant specifications of the motor;

*3：When DO function 11 (FunOUT.11: BK) is not assigned, P00.14 has no effect.

4.2 Position control mode

4.2.1 Functional Diagram of Position Control

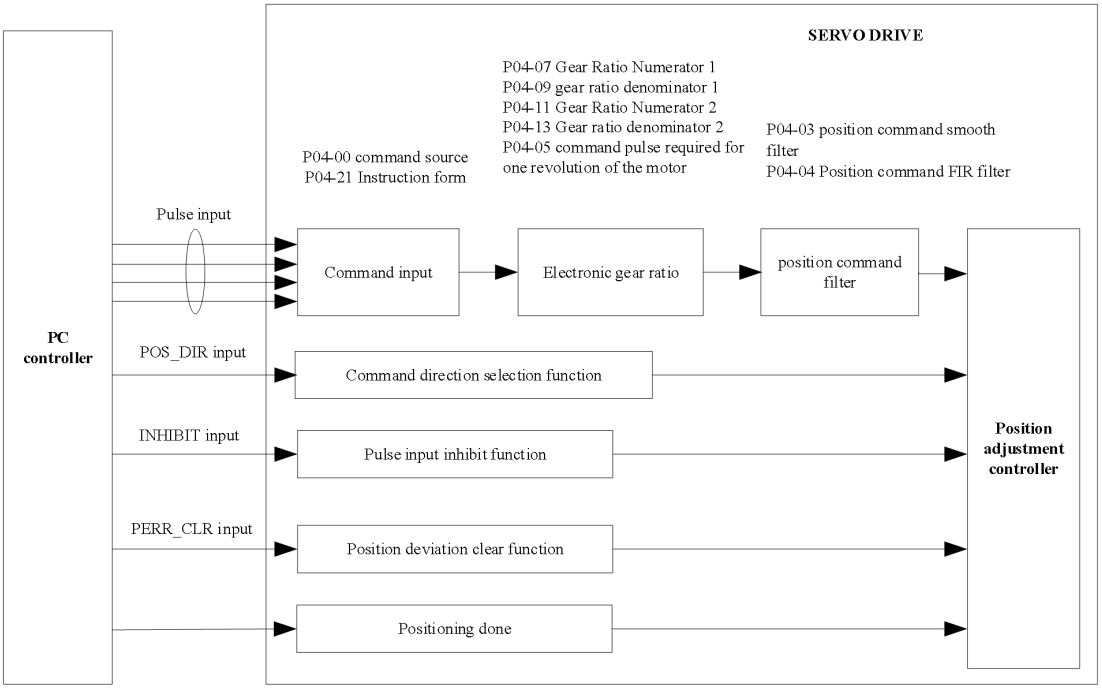


Figure 4.7 Position control diagram

The position mode is a common working mode of the servo drive, and its main steps are as follows:

1. Correctly connect the power supply of the servo main circuit and control circuit, as well as the motor power cable and encoder cable.

2. Disconnect the motor from the load, and conduct a servo JOG test run through the host computer to confirm whether the motor can run normally.

3. Configure pulse direction input, pulse command input and necessary DI/DO signals, such as servo enable, positioning completion, etc.

4. Make relevant settings of the position mode. Set the DI/DO used according to the actual situation, and refer to Group P02 for the function code. In addition, functions such as return to Home are sometimes set as needed.

5. Enable the servo, and control the rotation of the servo motor through the position command issued by the host computer. First make the motor rotate at a low speed, and confirm whether the direction of rotation and the electronic gear ratio are normal, and then adjust the gain.

4.2.2 Position control mode related function code setting

Parameter setting in position control mode, including mode selection, command pulse form, electronic gear ratio, DI/DO, etc.

4.2.2.1 Position command input setting

a) Source of position command

Set the function code P04-00=0, the position command comes from the low-speed pulse command, and it can also be set to other values according to the actual situation.

Function code		Name	SettingRange	Unit	Default setting	Type	Effective time	Mode
P04	00	Main position command source	0--low speed pulse command 1--reserved 2--step amount given 4--multi-segment position command given 5--communication given 6--reserved	1	0	Effective immediately	Setting after motor stop	P

b) Position command direction switching

By setting the DI function FunIN.24, you can use DI to control the direction switching of the position command to meet the situation where the direction needs to be switched.

Code	Name	Function name	Setting range	Mark
FunIN.24	POS_DIR	Position command direction	Invalid: no commutation Valid: reversing	The recommended setting for the logic selection of relevant terminals is: edge effective.

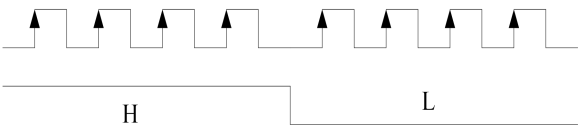
c) Pulse command form selection

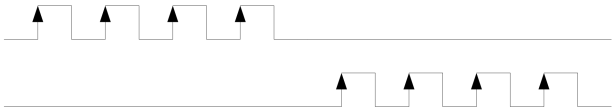
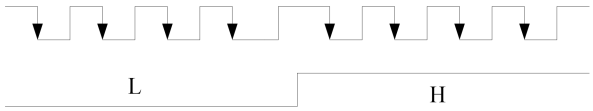
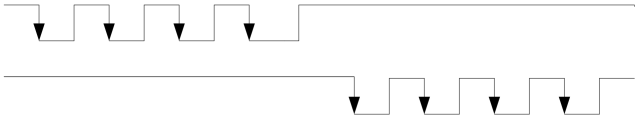
Set the function code P04-21 to select the form of the external pulse command, including three forms of "direction + pulse (positive and negative logic)", "orthogonal pulse" and "CW+CCW"

Function code		Name	SettingRange	Unit	Default setting	Type	Effective time	Mode
P04	21	Pulse shape	0-- pulse + direction, positive logic. (Defaults) 1-- direction + pulse, negative logic 2-- A phase + B phase orthogonal pulse, positive logic 3-- A phase + B phase orthogonal pulse, negative logic 4-- CCW+CW, positive logic 5-- CCW+CW, negative logic	1	0	Re-power	Setting after motor stop	P

The principles of the three pulse command forms are as follows.

Table 4.3 Principle of pulse command form

P04-21	Polarity	Pulse command form	Forward	Reverse
0	Positive logic	Direction+pulse		
2	Positive logic	AB phase quadrature pulse		

P04-21	Polarity	Pulse command form	Forward	Reverse
4	Positive logic	CCW+CW		
1	Negative logic	Direction+pulse		
3	Negative logic	AB phase quadrature pulse		
5	Negative logic	CCW+CW		

d) Pulse inhibit input

By setting the DI function FunIN.12, the pulse command input is prohibited.

Code	Name	Function name	Setting range	Mark
FunIN.12	INHIBIT	Pulse inhibit	Invalid: allow command pulse input Valid: inhibit command pulse input	Position commands are prohibited, including internal and external position commands. The logic selection of the corresponding terminal must be set as: level is valid.

4.2.2.2 Electronic gear ratio setting

Set the electronic gear ratio according to the actual situation of the machine and the host computer.

Function code		Name	Setting Range	Min. Unit	Default setting	Type	Effective time	Mode
P04	05	The number of command pulses required for one revolution of the motor	0 ~1048576	1P/Rev	0	Re-power	Setting after motor stop	P
P04	07	Electronic gear 1 molecule	1~1073741824	1	4	Motor resolution	Set when running	P
P04	09	Electronic gear 1 denominator	1~1073741824	1	1	10000	Set when running	P
P04	11	electronic gear 2 molecule	1~1073741824	1	4	Motor resolution	Set when running	P
P04	13	Electronic gear 2 denominator	1~1073741824	1	1	10000	Set when running	P

The schematic diagram of the electronic gear ratio is as follows:

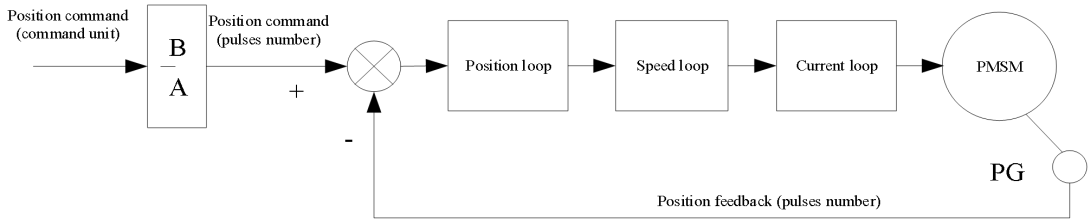


Figure 4.8 Principle of electronic gear ratio

a） When P04-05=0, the motor and the load are connected through the reduction gear, assuming that the reduction ratio between the motor shaft and the mechanical side of the load is n/m (the motor shaft rotates m circles, and the load shaft rotates n circles), the calculation of the electronic gear ratio is as follows.

$$\text{electronic gear ratio}\frac{B}{A}=\frac{P04-07}{P04-09}=\frac{\text{encoder resolution}}{\text{displacement of the load shaft for one rotation (command unit)}}\times\frac{m}{n}$$

This type of driver supports up to 2 sets of electronic gear ratios, and the electronic gear ratio switching function FunIN.23 can be used to complete the gear ratio selection.

b) When P04-05≠0，

electronic gear ratio $\frac{B}{A}=\frac{\text{encoder resolution}}{P04-05}$

At this time, the two sets of electronic gear ratio and electronic gear switching function are invalid.

c) The electronic gear ratio setting range is $\frac{0.001\times\text{encoder resolution}}{10000}<\frac{B}{A}<\frac{4000\times\text{encoder resolution}}{10000}$, otherwise, fault Er.49 (electronic gear ratio setting fault) will occur.

4.2.2.3 Position deviation clear function

Position deviation = (position command - position feedback) (encoder unit)

The position deviation clearing function means that the drive can clear the position deviation to zero when certain conditions are met (P04.22).

☆Associated function code:

Code	Name	Setting range	Unit	Function	Type	Effective time	Default setting
P04.22	Position deviation clear function	0-- Clear position deviation pulse when fault occurs or servo OFF 1-- Clear the position deviation pulse only when a fault occurs 2-- Cleared by DI input function (PERR-CLR)	-	Setting the conditions for clearing position deviation.	Setting after motor stop	Effective immediately	0

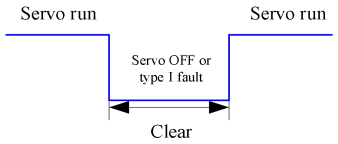
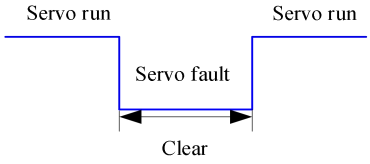
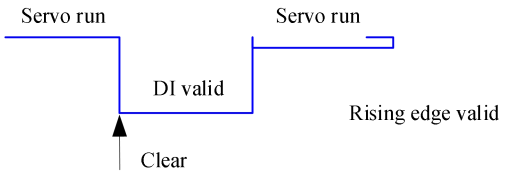
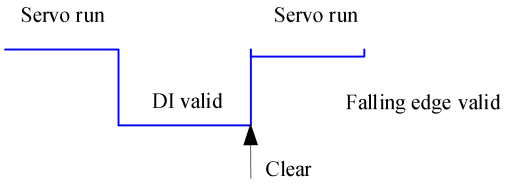
When P04.22=2, one DI terminal of the servo drive should be configured as function 5 (FunIN.5: PERR_CLR, clear position deviation), and determine the valid logic of the DI terminal.

☆ Associated function code:

Code	Name	Function name	Function
FunIN.5	Pulse deviation clear function	Clear position deviation	Valid, clear position deviation; Invalid, no clear operation.

The setting method is as follows.

Table 4.4 Position deviation clear setting

Setting value	Clear condition	Clear time
P04.22=0	Clear the position deviation when the servo is OFF or occuring the type 1 fault.	
P04.22=1	Clear position deviation pulses only when a fault occurs.	
P04.22=2	Clear the position deviation by the signal input by DI (function 5).	 

4.2.2.4 Positioning complete/approach function

The positioning completion function means that the position deviation meets the conditions set by the user (P04.24), and it can be considered that the positioning is completed in the position control mode. At this time, the servo drive can output a positioning completion (COIN) signal, and the host computer can confirm that the positioning of the servo drive is

completed after receiving this signal.

Its functional principle is shown in the figure below:

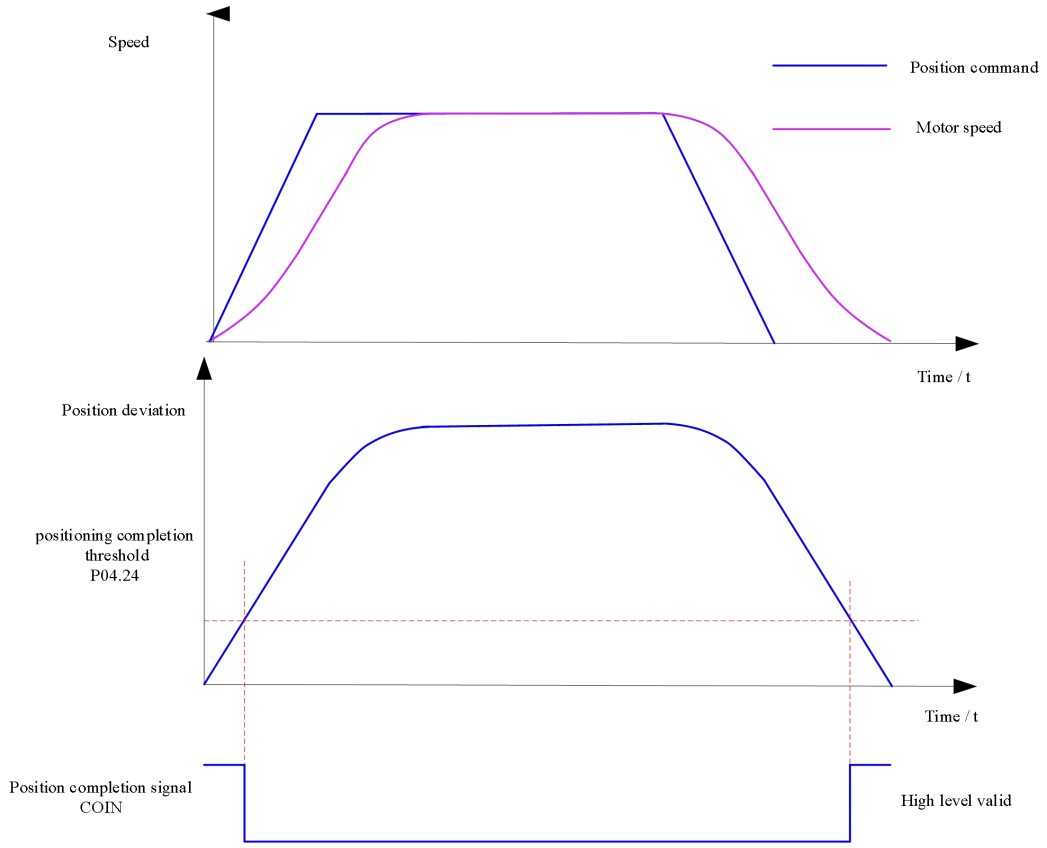


Figure 4.9 Positioning completion function description

When the position deviation satisfies the condition (P04.23), the servo driver can also output the positioning approach (NEAR) signal. Usually, the host computer can receive the positioning approach signal before confirming that the positioning is completed, so as to prepare for the positioning completion operation.

Before using the positioning completion/approach function, set the output conditions and thresholds for positioning completion/approach.

☆ Associated function code:

Code	Name	Setting range	Unit	Function	Type	Effective time	Default setting
P04.23	Condition for positioning completed	0-- Output when the absolute	-	Set the conditions for	Setting after motor	Effective	0

	/proximity output	position deviation value is smaller than the threshold of positioning completed / proximity; 1-- Output when the absolute position deviation value is smaller than the threshold of positioning completed / proximity and the filtered position command is 0; 2--Output when the absolute position deviation value is smaller than the threshold of positioning completed / proximity and position command is 0.		positioning completed (COIN)/ proximity (NEAR) to be valid.	stop	immediately	
P04.24	Threshold of positioning completed	1~65535	1P	Set the threshold of the absolute value of the position deviation when the positioning completed (COIN) is valid	Setting after motor stop	Effective immediately	According to encode type
P04.25	Threshold of positioning proximity	1~65535	1P	Set the threshold of the absolute value of the position deviation when the positioning proximity (NEAR) is valid	Setting after motor stop	Effective immediately	65535

Note:

• The threshold of positioning proximity (P04.25) generally needs to be greater than the positioning completed threshold (P04.24).

• The positioning completion threshold (P04.24) only reflects the threshold of the absolute value of the position deviation when the positioning completed is valid, and has nothing to do with the positioning accuracy.

• If the speed feedforward gain (P07.20) is set too large or runs at low speed, the absolute value of the position deviation will be small. If the P04.24 is set too large, the positioning completed will always be valid. Therefore, in order to improve the effectiveness of positioning completed, please reduce the setting value of P04.24

• When the positioning completed threshold (P04.24) is small and the position deviation is small, the output conditions of the positioning completed/ proximity signal can be changed by setting P04.23.

• When the servo enable (S-ON) is invalid, the output of the positioning completed signal (COIN) and positioning proximity signal (NEAR) is invalid.

When using the positioning completed and positioning proximity functions, the two DO terminals of the servo driver should be configured as DO function 7 (FunOUT.7: COIN, positioning completed) and DO function 8 (FunOUT.8: NEAR, positioning proximity), and determine the valid logic of the corresponding DO terminal.

☆Associated function code:

Code	Name	Function Name	Function
FunOut.7	COIN	Position completed	**Valid,** in the position control mode, the absolute value of the position deviation meets the setting conditions of P04.24, indicating that the servo positioning is completed. **Invalid,** in position control mode, the servo is in the process of completing positioning.
FunOut.8	NEAR	Position proximity	**Valid,** in the position control mode, the absolute value of the position deviation meets the setting conditions of P04.25, indicating that the servo positioning is close. **Invalid,** in the position control mode, the servo is in the process of positioning and proximity.

4.2.2.5 Interrupt Positioning

1）Description

In the position control mode, if interrupt positioning is triggered when the S-ON signal is active, the servo drive aborts current operation and turns to executing the preset position reference, which is to say, after interrupt positioning is triggered, the motor executes the position reference triggered by interrupt positioning, with motor direction unchanged.

When interrupt positioning is in progress, the servo drive does not respond to any other internal/ external position references (including another interrupt positioning command). In this case, the input position reference counter (P18.15) counts the interrupt positioning reference only. After interrupt positioning is done, the servo drive may or may not respond to other position references depending on the setpoint of P16.06 (Interrupt positioning cancel signal). The position references received during interrupt positioning will be abandoned.

After interrupt positioning is done, the servo drive outputs the interrupt positioning completed (FunOUT.18：XINT_DONE) signal and positioning completed (FunOUT.7：COIN) signal, while the host controller, upon receiving XINT_DONE signal, acknowledges interrupt positioning is done. The XINT_DONE signal output is not related to the S-ON signal or the logic of DI9.

Interrupt positioning is effective only when the following conditions are met:

●The motor speed is higher than or equal to 10 RPM before interrupt positioning is triggered, or the setpoints of P16.03 (Constant operating speed in interrupt positioning) and P16.01 (Displacement of interrupt positioning) are not 0.

●The DI assigned with FunIN.31 (Interrupt positioning inhibited) is not used or the logic of this DI is inactive.

2）Parameters setting

☆ Related parameters：

Code	Name	Setting range	Unit	Function	Type	Effective time	Default
P16.00	Interrupt positioning selection	0--Disable 1--Enable	-	Used to enable/disable interrupt positioning.	Setting after motor stop	Re-power	0

P16.01	Displacement of interrupt positioning	0~1073741824	Reference unit	Defines the displacement of interrupt positioning	Set when running	Effective immediately	10000
P16.03	Constant operating speed in interrupt positioning	0~6000	rpm	Defines the maximum motor speed in interrupt positioning, which is unrelated to the electronic gear ratio.	Set when running	Effective immediately	200
P16.04	Acceleration time of interrupt positioning	0~1000	ms	Defines the time for the motor to accelerate from 0 RPM to 1000 RPM.	Set when running	Effective immediately	10
P16.05	Deceleration time of interrupt positioning	0~1000	ms	Defines the time for the motor to accelerate from 0 RPM to 1000 RPM	Set when running	Effective immediately	10

P16.06	Interrupt positioning cancel signal Enable	0—Disable 1—Enable	-	Defines the condition for responding to other position references after interrupt positioning is done. When P16.06 is set to 1, interrupt positioning must be canceled by a DI assigned with FunIN.30 (Interrupt positioning cancel.	Set when running	Effective immediately	1

☆ Related function code：

Code	Name	Function name	Function
FunIN.30	XINT_FINISH	Interrupt positioning cancel	**Active:** The interrupt positioning state is unlocked, which means the servo drive can respond to other position references. **Inactive:** The interrupt positioning state is locked, which means the servo drive cannot respond to other position references.
FunIN.31	XINT_DISABLE	Interrupt positioning inhibit	**Active:** Interrupt positioning inhibited; **Inactive:** Interrupt positioning permitted
FunOut.18	XINT_DONE	Interrupt positioning completed	**Active:** Interrupt positioning completed during position control. **Inactive:** Interrupt positioning not completed during position control.

The constant operating speed for interrupt positioning is shown in the following figure.

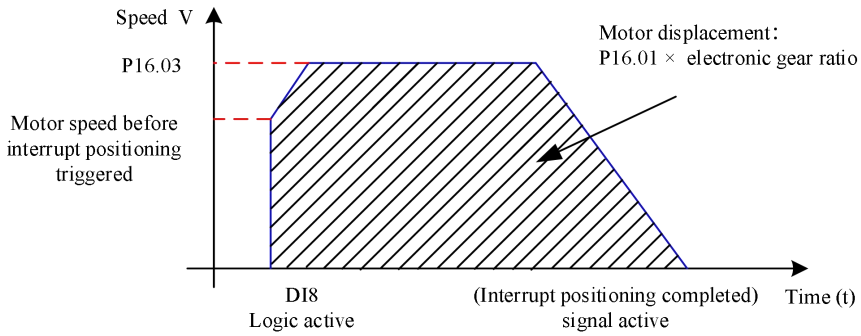


Figure 4.10 Motor operating curve during interrupt positioning

Table 4.4 Active logic of DI8 during interrupt positioning

P16.03	Motor Speed before Triggering Interrupt Positioning	Interrupt Positioning	Constant Operating Speed in Interrupt Positioning
0	<10	Inactive	-
	≥10	Active	Motor Speed before Triggering Interrupt Positioning
1~6000	-	Active	P16.03

4.2.2.6 Homing Function

Note:

The homing trigger signal is disabled when interrupt positioning or multi-position reference is in progress.

1）Description of the homing function

Home (or mechanical home): Indicates the position of the home switch or motor Z signal depending on the setting of P16-09 (Homing mode).

Zero position (or target point): Indicates the sum of the home plus the offset (defined by P16-14). When P16-14 (Home offset) is set to 0, zero position shares the same position with the home.

In the position control mode, when homing is triggered after the S-ON signal is activated, the motor starts searching for the zero point. When homing is in progress, the servo drive does not respond to other position references (including another homing trigger signal) until homing is done.

Homing: The servo drive, upon receiving the homing trigger signal, locates the relative position of the motor shaft and the mechanical home based on the pre-set mechanical home and starts searching for the home first. Then the servo drive executes the offset based on the home, after which it reaches the zero point. The homing mode usually applies in initial search of the zero point.

After the homing function is done executing, the motor position absolute value (P18-07) is consistent with the home offset (P16-14).

After homing is done, the servo drive outputs the homing completed signal (FunOUT.19：HomeAttain) . The host controller, upon receiving the signal, acknowledges the homing is done. HomeAttain is independent of the control mode or the operating state of the servo drive

Table 4.5 Homing trigger mode

Mode	Homing trigger mode (P16-08)	Homing Direction, Deceleration Point, Home	Trigger Signal	Total Motor Displacement
Homing	0: Disabled	-	-	-

	1: Homing enabled through the HomingStart signal inputted from DI	Determined by P16-09	HomingStart signal	Determined by the home coordinate and offset displacement.
	2: Homing enabled immediately upon power-on		S-ON	
	3: Homing executed immediately		S-ON	
	4: Current position as the home		S-ON	
	5: DI-triggered position as the home		S-ON	
	6: Host controller homing		S-ON	

Note:

The electronic gearing, the moving average filter and the low-pass filter are inactive during homing.

2）Homing

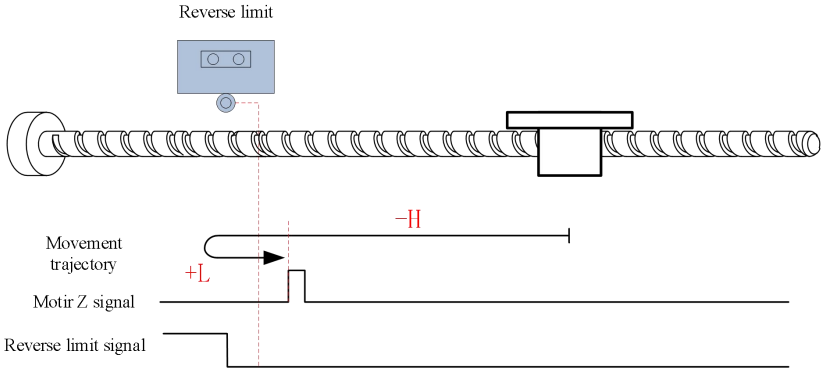
Note:

• Set mechanical limit switches before enabling homing. For homing upon hit-and-stop, set the offset to a value within the travel range to prevent the machine from collision due to high-speed operation during homing.

• When the motor hits the limit switch during homing, the servo drive reports Er.86 (Forward overtravel) or Er.87 (Reverse overtravel), and then it returns automatically and continues to search for the mechanical home. See the following examples for the homing mode:

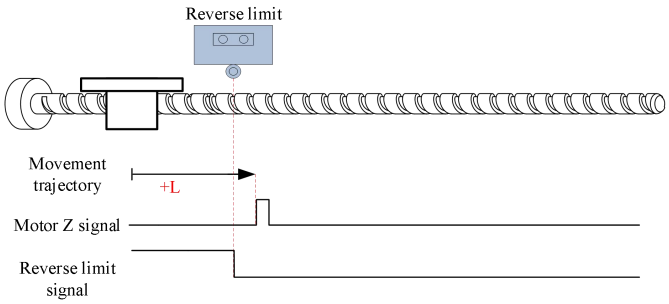
Homing Mode 1: Homing by reverse limit switch and zero position pulse signal（0x6098 =1）

1） During homing, the reverse limit switch signal is inactive.



Description: When homing, the reverse limit switch signal is invalid. The motor searches for the reverse limit switch signal at a higher speed until the rising edge of the limit switch signal is detected, then decelerates, reverses, and runs forward at a lower speed. When the falling edge of the reverse limit signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

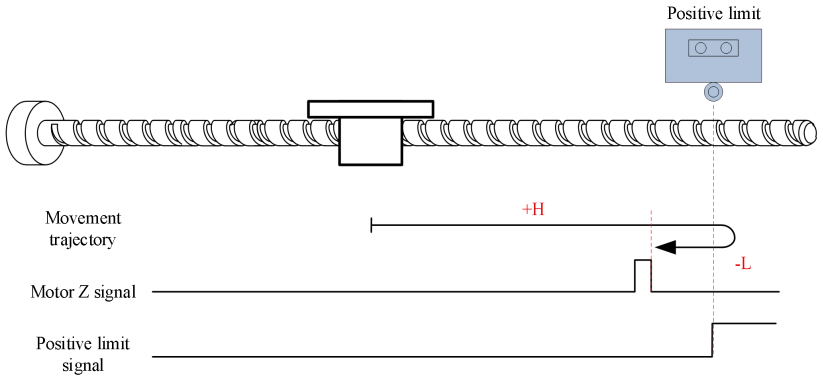
2）When homing, the reverse limit switch signal is valid



Description: When homing, the reverse limit switch signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the reverse limit signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

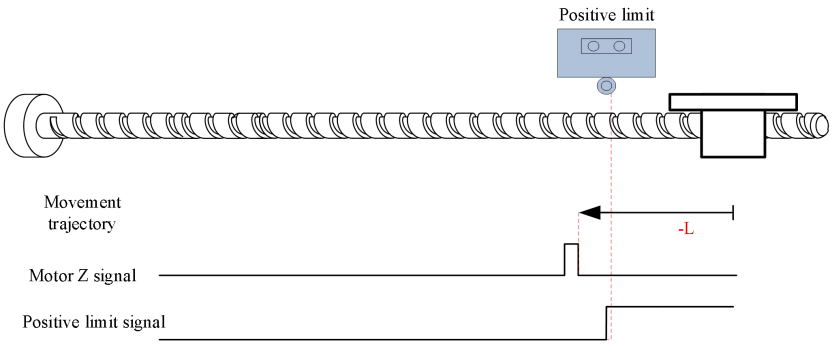
Homing mode 2: hoimg with positive limit switch and zero pulse signal (0x6098 = 2)

1）When homing, the positive limit switch signal is invalid



Description: When homing, the positive limit switch signal is invalid. The motor searches for the positive limit switch signal at a higher speed in the forward direction until the rising edge of the limit switch signal is detected, then decelerates, reverses, and runs in the reverse direction at a lower speed. When the falling edge of the positive limit signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

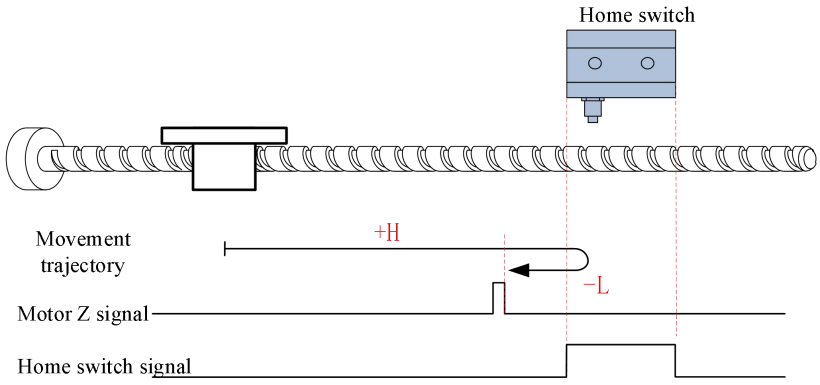
2）When homing, the positive limit switch signal is valid



Description: When homing, the positive limit switch signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the positive limit signal is detected, the motor starts to search for the Z signal of the motor, and stops running after the Z signal is found.

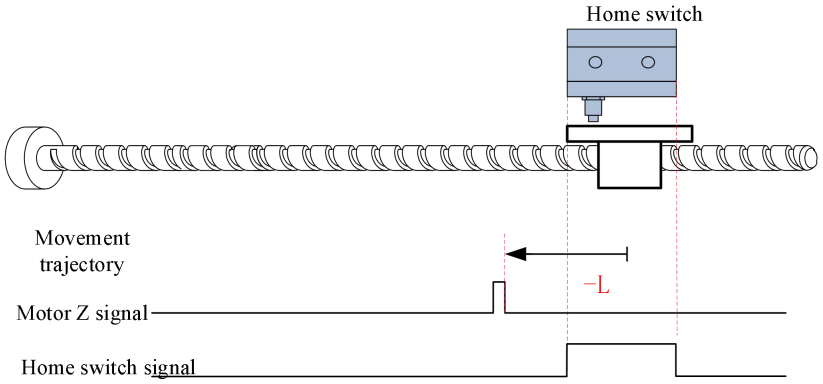
Homing mode 3: homing with Home switch and zero pulse signal (0x6098 = 3)

1）When homing, the Home switch signal is invalid



Description: When homing, the Home switch signal is invalid. The motor searches for the Home switch signal at a higher speed in the forward direction until the rising edge of the Home switch signal is detected, then decelerates, reverses, and runs in the reverse direction at a lower speed. When the falling edge of the Home signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

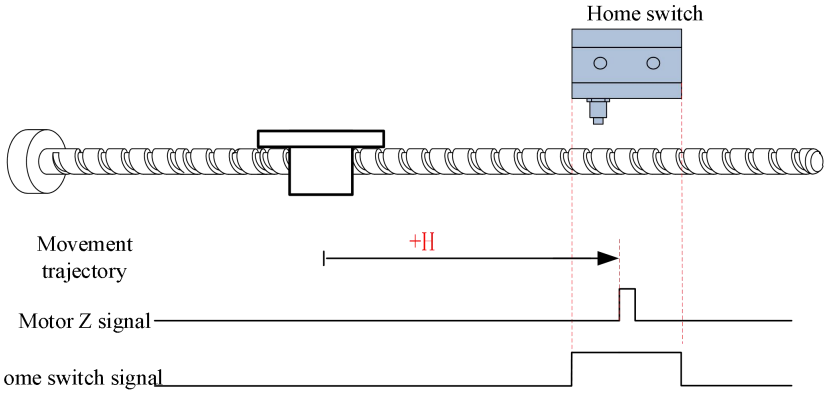
2）When homing,the home switch signal is valid



Description: When homing, the Home switch signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the Home switch signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

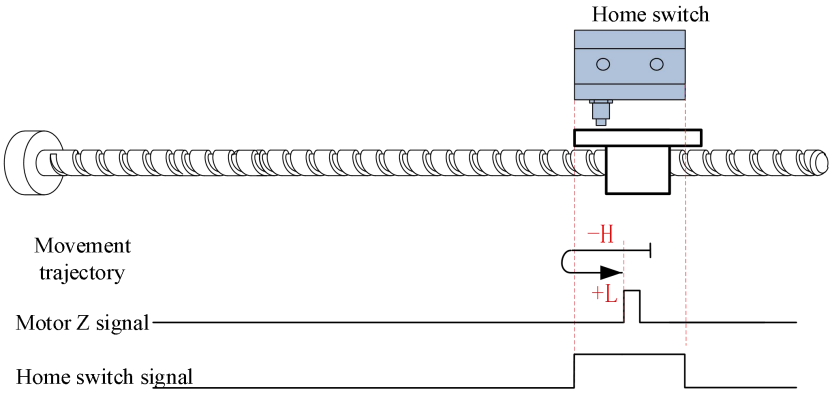
Homing mode 4: homing with Home switch and zero pulse signal (0x6098 = 4)

1）When homing, the Home switch signal is invalid



Description: When homing, the Home switch signal is invalid. The motor searches for the Home switch signal in the forward direction at a lower speed until the rising edge of the Home switch signal is detected. Then it starts to search for the motor Z signal and stops running after the Z signal is found.

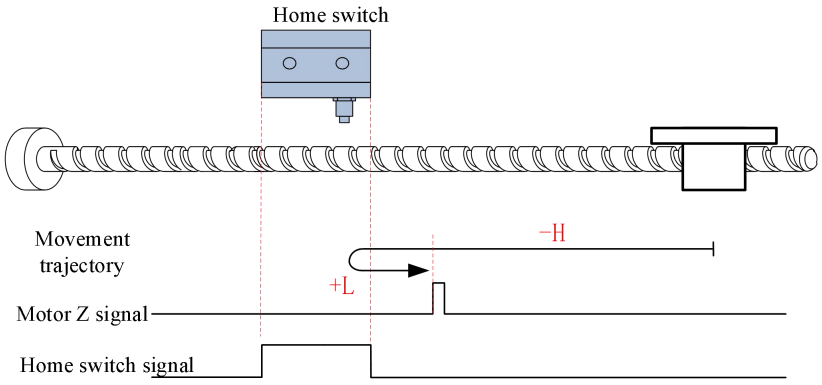
2）When homing, the Home switch signal is valid



Description: When homing, the Home switch signal is valid. The motor directly runs in the reverse direction at a higher set speed. When the falling edge of the Home switch signal is detected, it decelerates, reverses, and runs forward at a lower speed. When the rising edge of the Home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

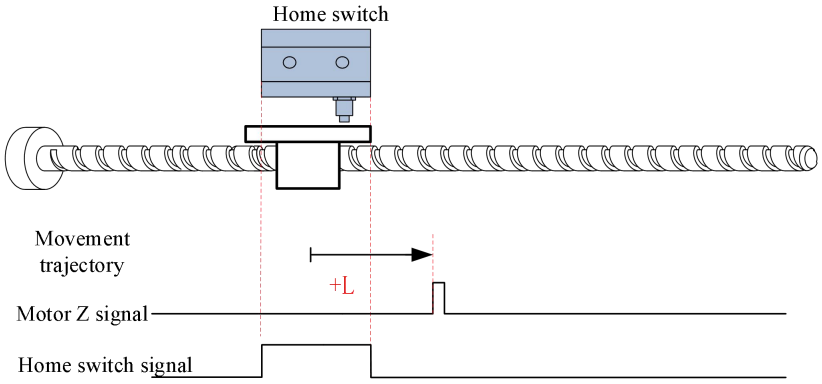
Homing mode 5: homing with Home switch and zero pulse signal (0x6098 = 5)

1）When homing, the Home switch signal is invalid



Description: When homing, the Home switch signal is invalid. The motor searches for the Home switch signal in reverse at a higher speed. When the rising edge of the Home switch signal is detected, it decelerates, reverses, and runs forward at a lower set speed. When the falling edge of the Home switch signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

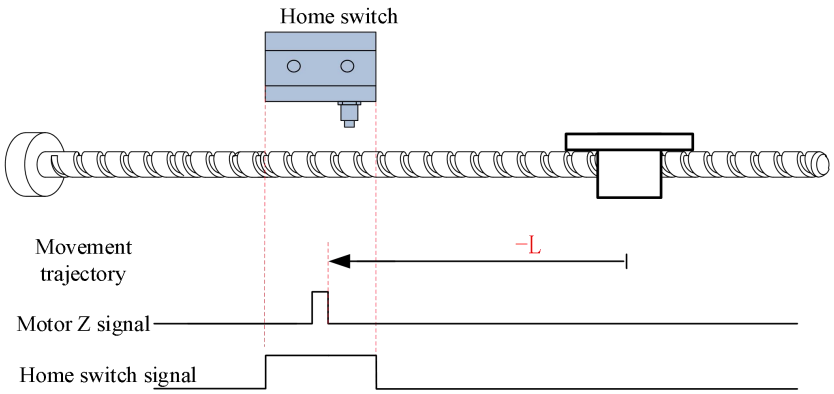
2）When homing, the Home switch signal is valid



Description: When homing, the Home switch signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the Home switch signal is detected, the motor starts to search for the Z signal, and stops running after the Z signal is found.

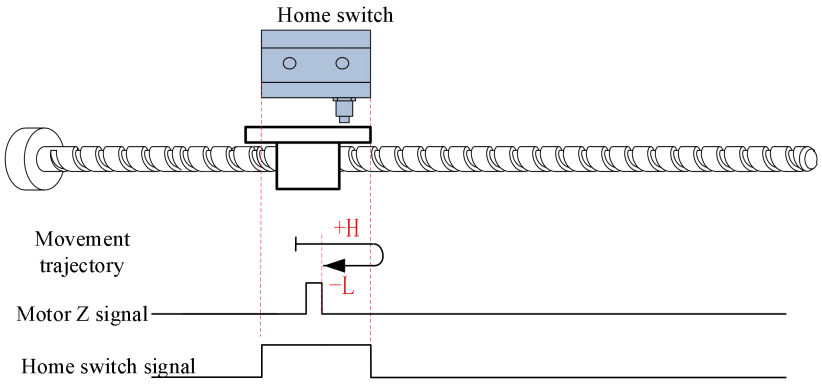
Homing mode 6: homing with Home switch and zero pulse signal (0x6098 = 6)

1）When homing, the Home switch signal is invalid



Description: When homing, the Home switch signal is invalid. The motor searches for the Home switch signal in reverse at a lower speed until the rising edge of the Home switch signal is detected. Then it starts searching for the motor Z signal and stops running after the Z signal is found.

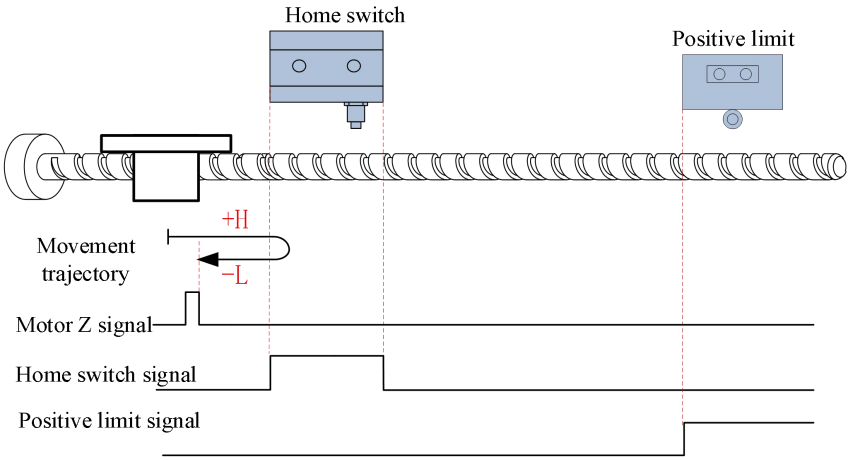
2）When homing, the Home switch signal is valid



Description: When homing, the Home switch signal is valid. The motor directly runs forward at a higher set speed. When the falling edge of the Home switch signal is detected, it decelerates, reverses, and runs in the reverse direction at a lower speed. When the rising edge of the Home signal is detected, it starts to search for the motor Z signal (falling edge), and stops running after the Z signal is found.

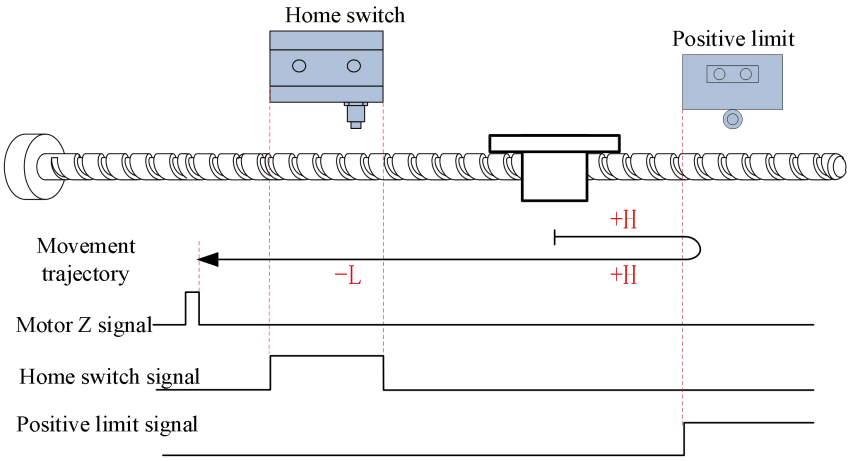
Homing mode 7: homing with Home switch, positive limit signal and zero pulse signal (0x6098 = 7)

1) The Home signal is invalid when starting the homing, and the positive limit switch is not encountered.



Description: When starting homing, the Home signal is invalid. The motor runs forward at a higher speed and does not encounter the forward limit signal. When the rising edge of the Home signal is detected, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the Home signal is encountered, it starts to search for the motor Z signal and stops running after the Z signal is found.

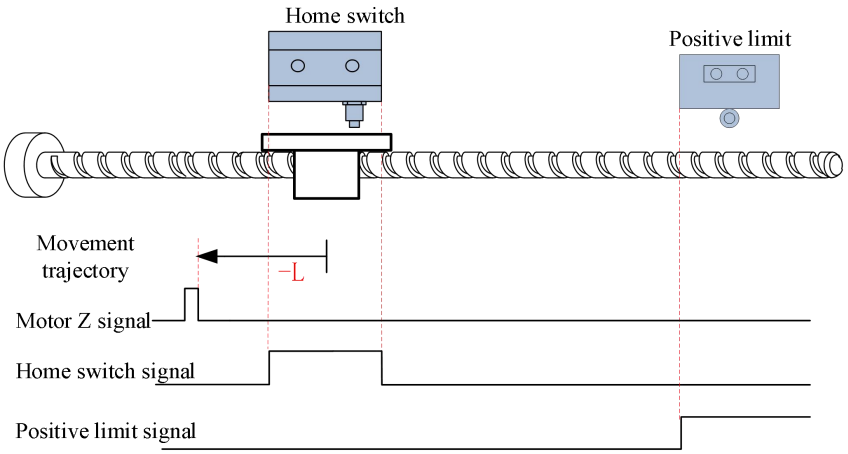
2) The Home signal is invalid when starting the homing, and the positive limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a high speed. When encountering the forward limit signal, it decelerates and runs in the reverse direction at a high speed. After detecting the rising edge of the home signal, it decelerates and continues to run in the reverse direction at a lower set speed. When encountering the falling edge of the home signal, it starts to

search for the motor Z signal and stops running after finding the Z signal.

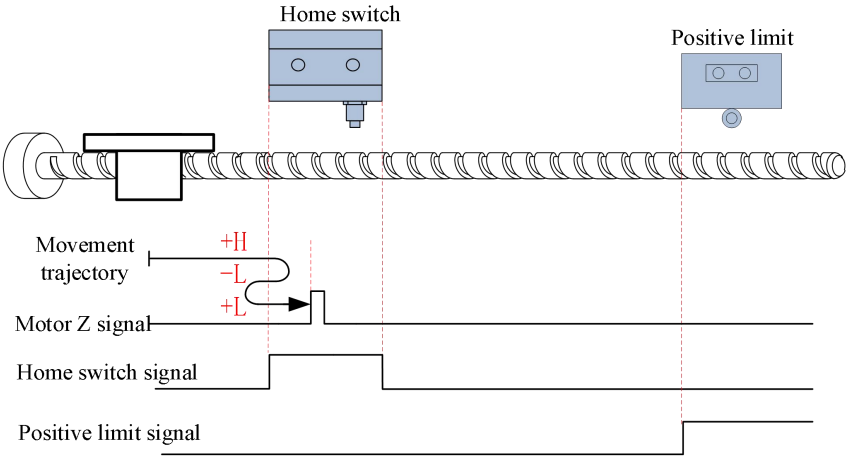
3) The home signal is valid when homing



Description: When starting the homing, the home signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

Homing mode 8: homing with home switch, positive limit signal and zero pulse signal (0x6098 = 8)

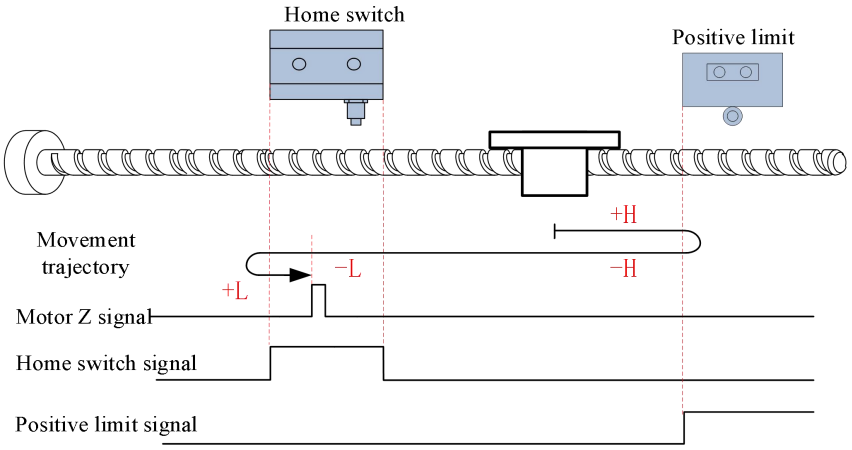
1) The home signal is invalid when starting the homing, and the positive limit switch is not encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a higher speed and does not encounter the forward limit signal. When the rising edge of the home signal is detected, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the home signal is encountered, it reverses and runs forward at a lower set speed. When the rising edge

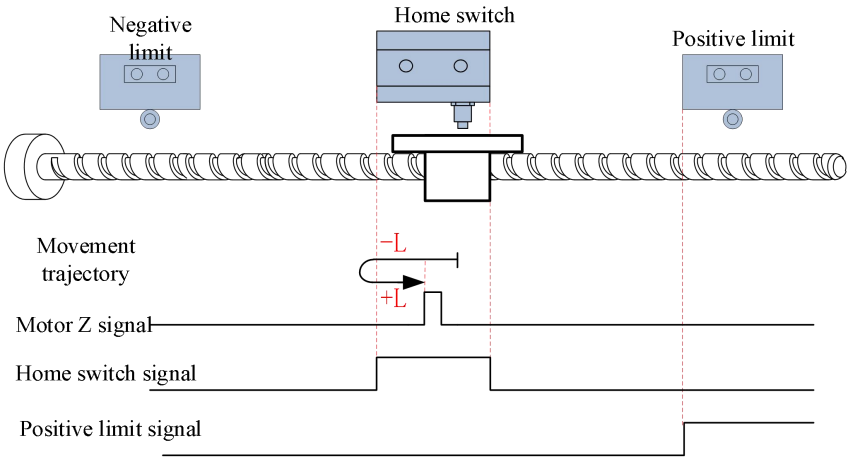
of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

2) The home signal is invalid when starting the homing, and the positive limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a high speed. When encountering the forward limit signal, it decelerates and runs in the reverse direction at a high speed. After detecting the rising edge of the home signal, it decelerates and continues to run in the reverse direction at a lower set speed. When encountering the falling edge of the home signal, it reverses and runs forward at a lower set speed. When detecting the rising edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

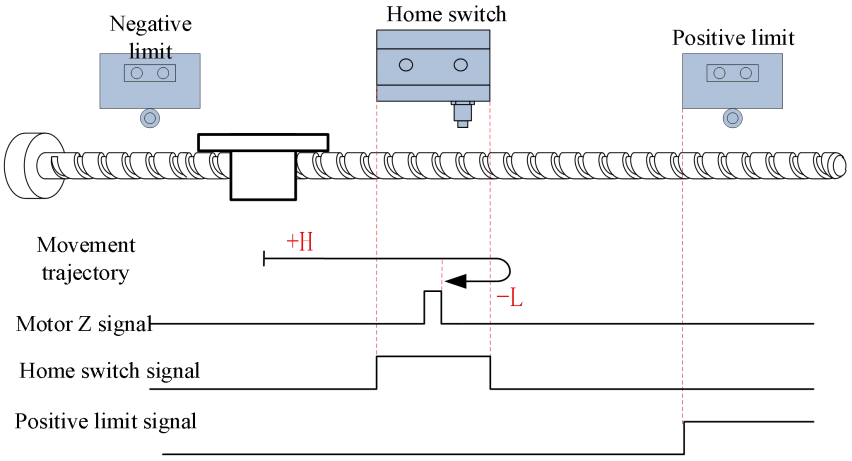
3) The home signal is valid when homing



Description: When starting the homing, the home signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it reverses and runs forward at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

Homing mode 9: homing with home switch, positive limit signal and zero pulse signal (0x6098 = 9)

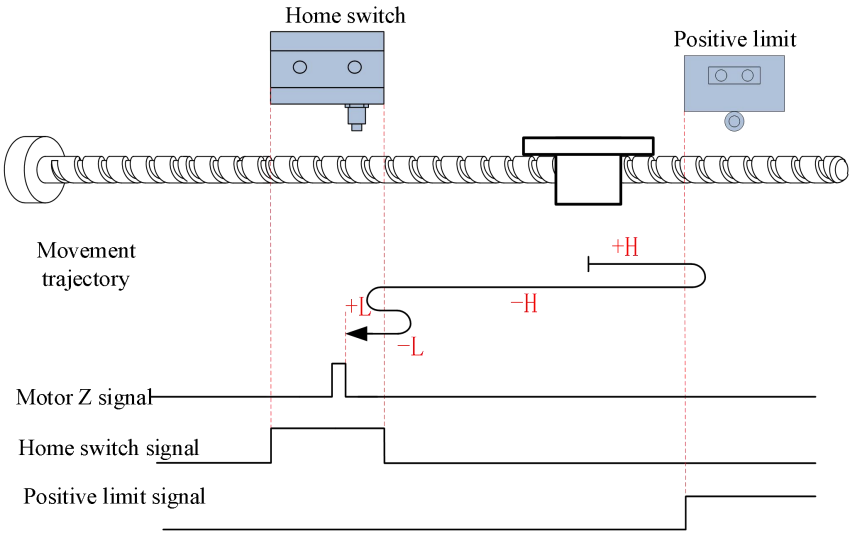
1) The home signal is invalid when starting the homing, and the positive limit switch is not encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a higher speed and does not encounter the forward limit signal. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal

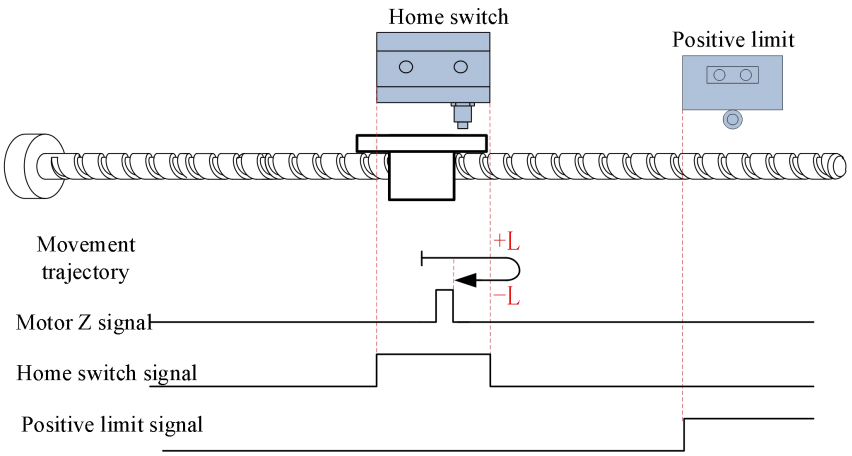
is encountered, it reverses and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

2)The home signal is invalid when starting the homing, and the positive limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a higher speed. When encountering the forward limit signal, it decelerates and runs in the reverse direction at a high speed. After detecting the rising edge of the home signal, it decelerates, reverses, and runs forward at a lower set speed. When encountering the falling edge of the home signal, it reverses and continues to run in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after finding the Z signal.

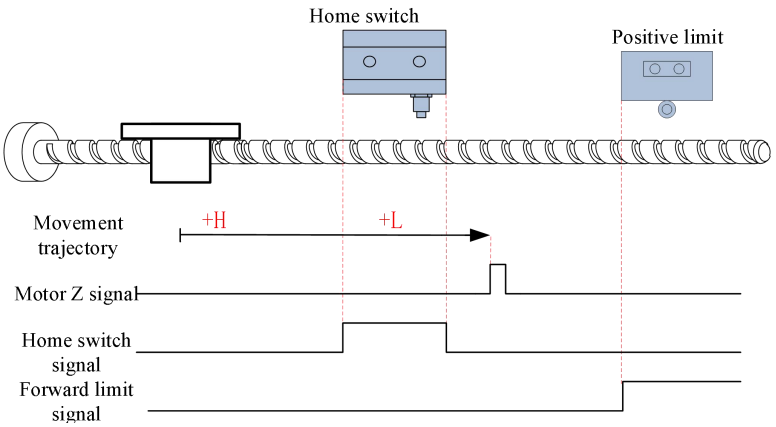
3) The home signal is valid when homing



Description: When starting the homing, the home signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the home signal is detected, it reverses and continues to run in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

Homing mode 10: homing with home switch, positive limit signal and zero pulse signal (0x6098 = 10)

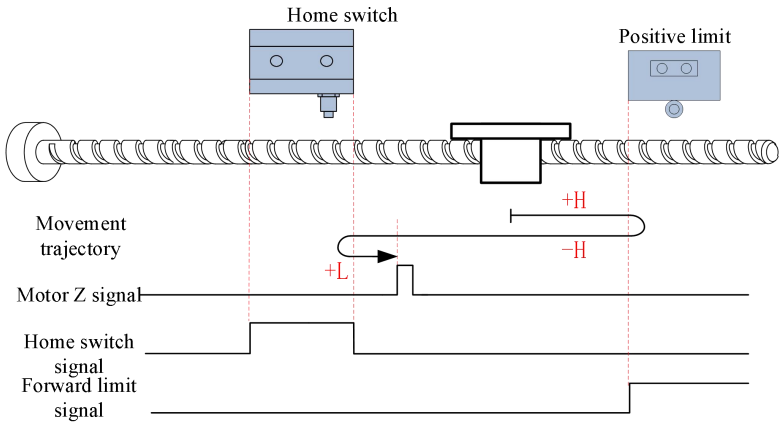
1）The home signal is invalid when starting the homing, and the positive limit switch is not encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a higher speed and does not encounter the forward limit signal. When the rising edge of the home signal is detected, it

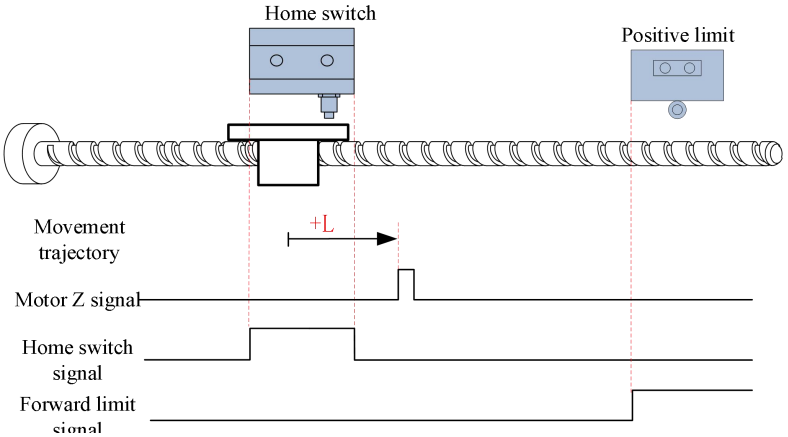
decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is encountered, it starts to search for the motor Z signal and stops running after the Z signal is found.

2) The home signal is invalid when starting the homing, and the positive limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs forward at a high speed. When encountering the forward limit signal, it decelerates and runs in the reverse direction at a high speed. After detecting the rising edge of the home signal, it decelerates, reverses, and continues to run forward at a lower set speed. When encountering the falling edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

3) The home signal is valid when homing

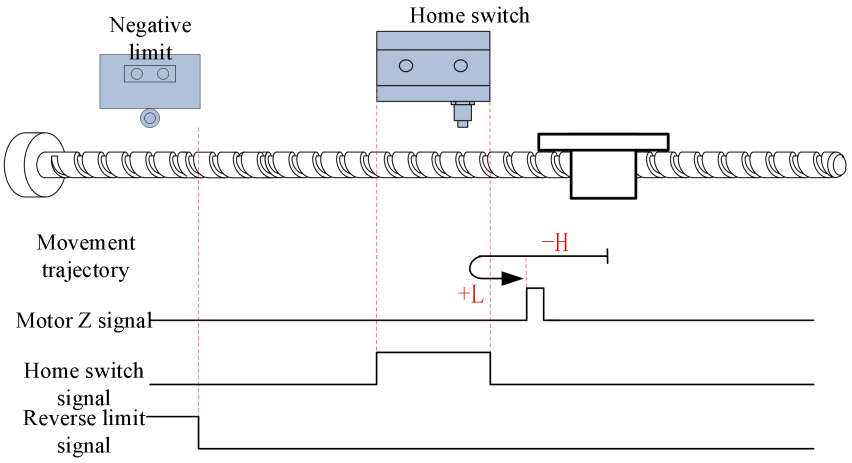


Description: When starting the homing, the home signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the home signal is detected, it starts to search for the motor

Z signal and stops running after the Z signal is found.

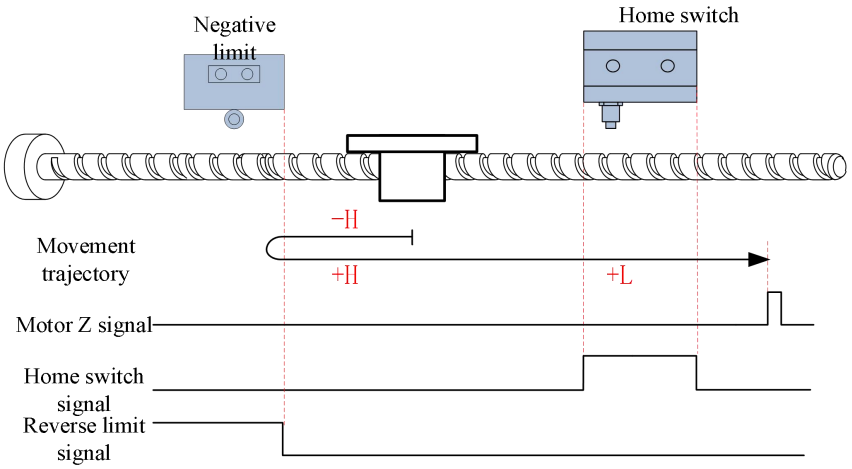
Homing mode 11: homing with home switch, reverse limit signal and zero pulse signal (0x6098 = 11)

1) The home signal is invalid when starting the homing, and no reverse limit switch is encountered.



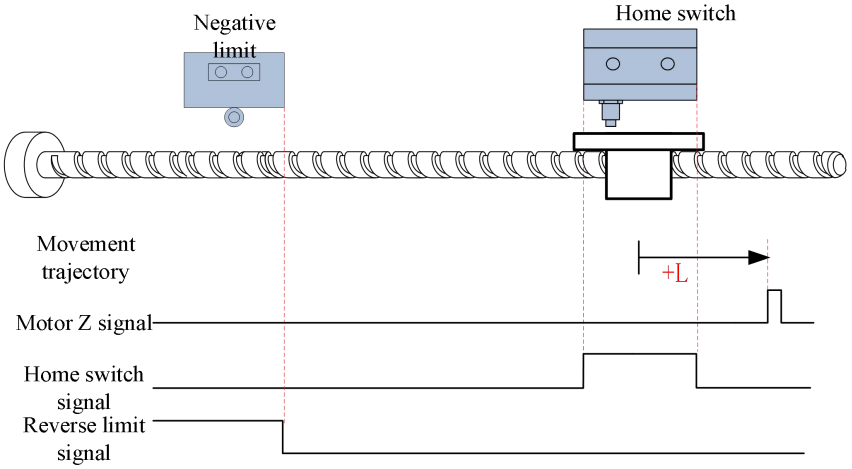
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed and does not encounter the reverse limit signal. When the rising edge of the home signal is detected, it decelerates, reverses, and runs forward at a lower set speed. When the falling edge of the home signal is encountered, it starts to search for the motor Z signal and stops running after the Z signal is found.

2）The home signal is invalid when homing, and the reverse limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed. When encountering the reverse limit signal, it decelerates and reverses, and runs in the forward direction at a higher set speed. After detecting the rising edge of the home signal, it decelerates and continues to run in the forward direction at a lower set speed. When encountering the falling edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

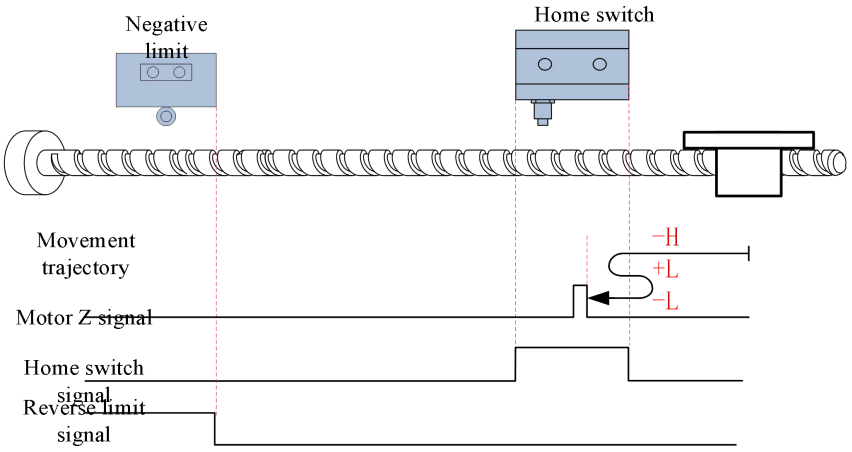
3) The home signal is valid when homing



Description: When starting the homing, the home signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

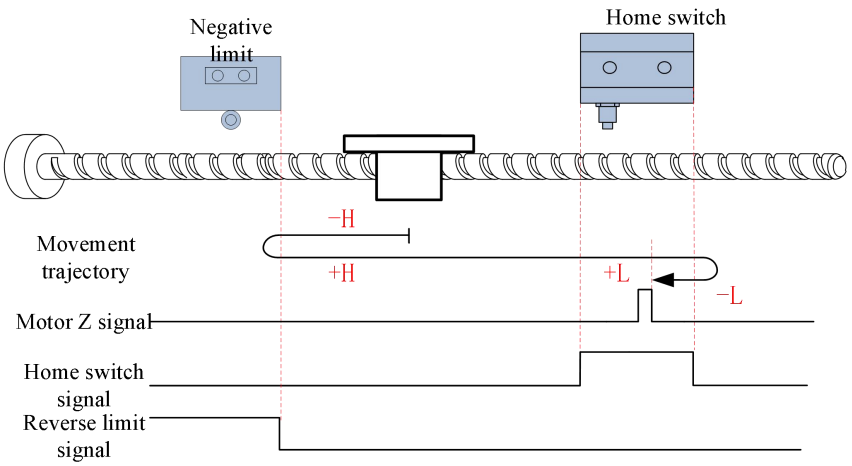
Homing mode 12: homing with home switch, reverse limit signal and zero pulse signal (0x6098 = 12)

1) The home signal is invalid when starting the homing, and no reverse limit switch is encountered.



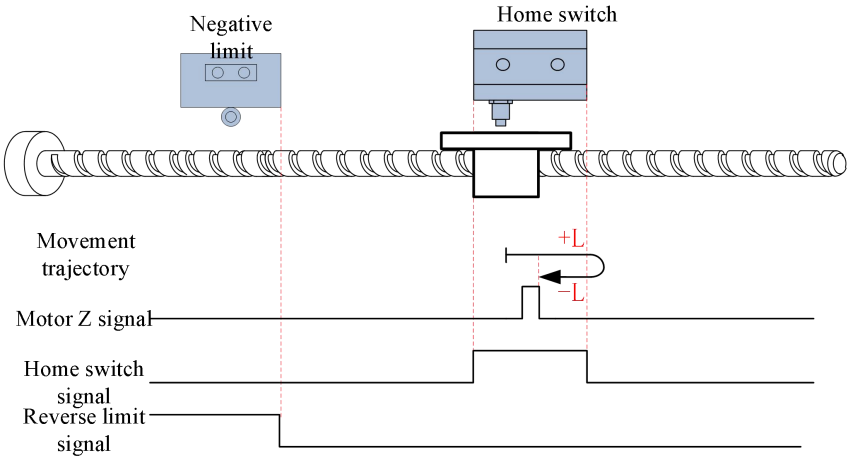
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed and does not encounter the reverse limit signal. When the rising edge of the home signal is detected, it decelerates, reverses, and runs forward at a lower set speed. When the falling edge of the home signal is encountered, it reverses again and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

2) The home signal is invalid when homing, and the reverse limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed. When encountering the reverse limit signal, it decelerates, reverses, and runs forward at a higher speed. After detecting the rising edge of the home signal, it decelerates and continues to run forward at a lower set speed. When encountering the falling edge of the home signal, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When detecting the rising edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

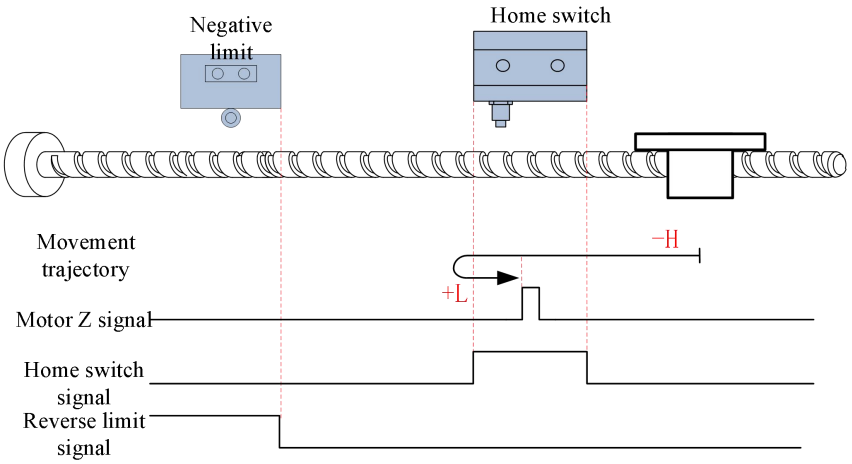
3) The home signal is valid when homing



Description: When starting from zero, the home signal is valid. The motor directly runs forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

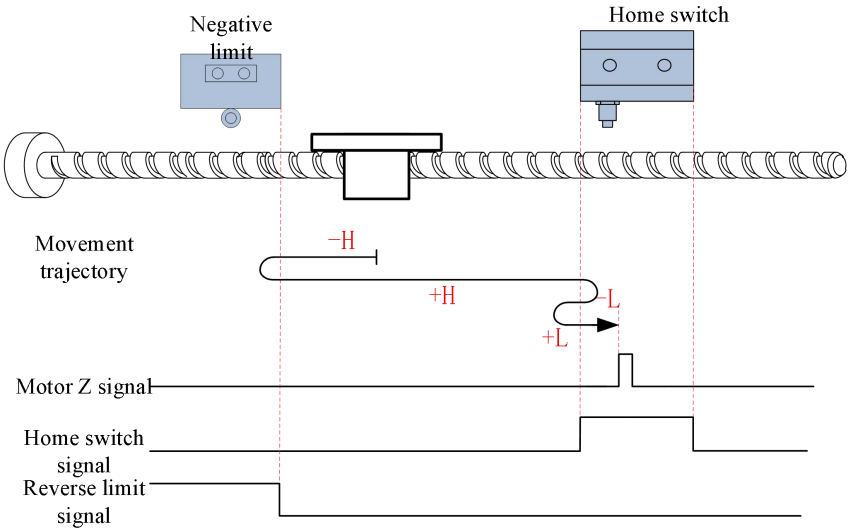
Homing mode 13: homing with home switch, reverse limit signal and zero pulse signal (0x6098 = 13)

1) The home signal is invalid when starting the homing, and no reverse limit switch is encountered.



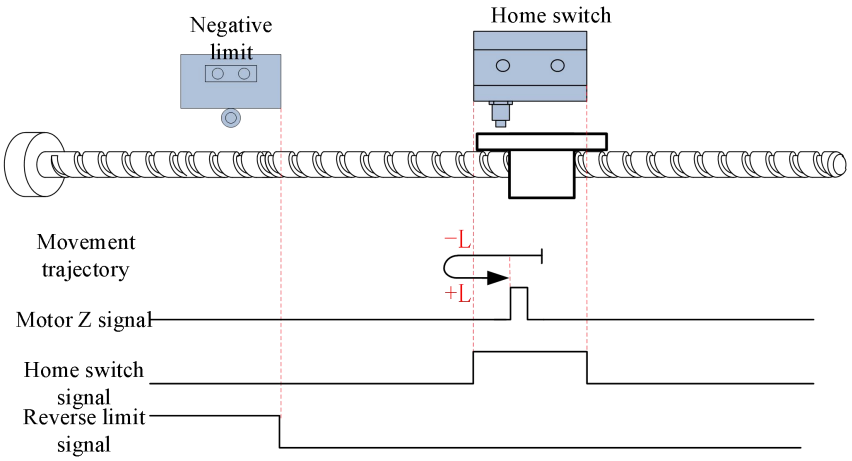
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed and does not encounter the reverse limit signal. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is encountered, it reverses and runs forward at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

2) The home signal is invalid when homing, and the reverse limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed. When encountering the reverse limit signal, it decelerates, reverses, and runs forward at a higher set speed. After detecting the rising edge of the home signal, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When encountering the falling edge of the home signal, it decelerates again and reverses, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

3) The home signal is valid when homing

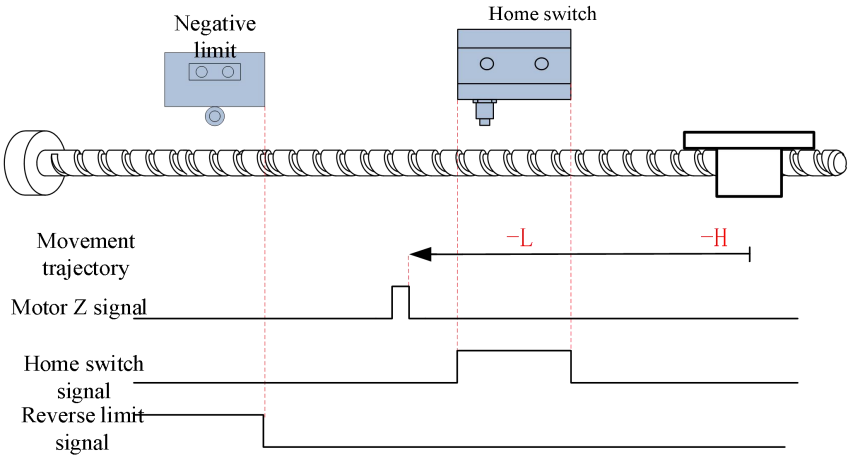


Description: When starting the homing, the home signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the rising edge of the home

signal is detected, it starts to search for the motor Z signal, and stops running after the Z signal is found.

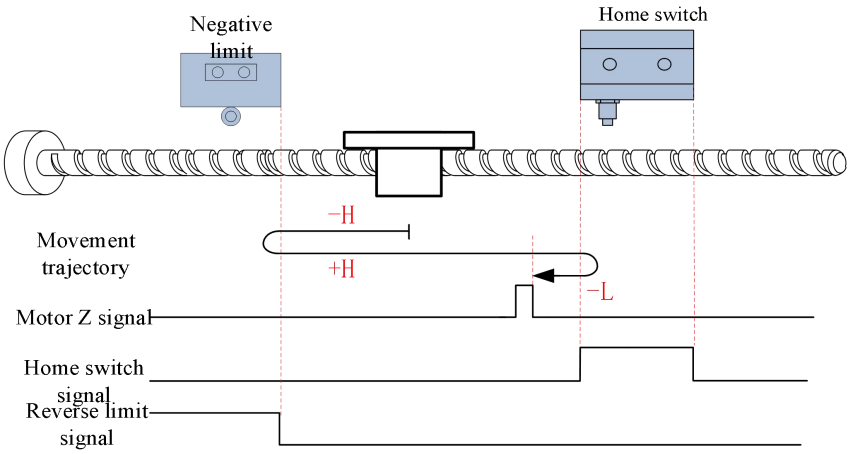
Homing mode 14: homing with home switch, reverse limit signal and zero pulse signal (0x6098 = 14)

1) The home signal is invalid when starting the homing, and no reverse limit switch is encountered.



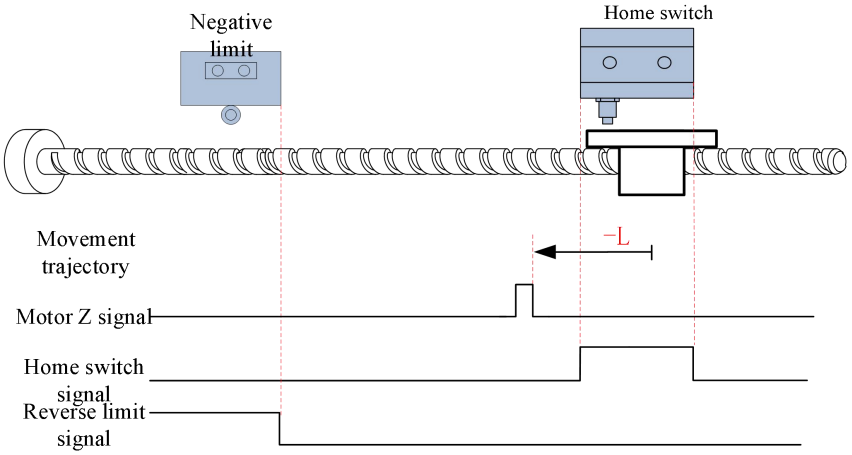
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed and does not encounter the reverse limit signal. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is encountered, it starts to search for the motor Z signal and stops running after the Z signal is found.

2) The home signal is invalid when homing, and the reverse limit switch is encountered.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher speed. When encountering the reverse limit signal, it decelerates, reverses, and runs forward at a higher set speed. After detecting the rising edge of the home signal, it decelerates, reverses, and runs in the reverse direction at a lower set speed. When encountering the falling edge of the home signal, it starts to search for the motor Z signal and stops running after finding the Z signal.

3) The home signal is valid when homing



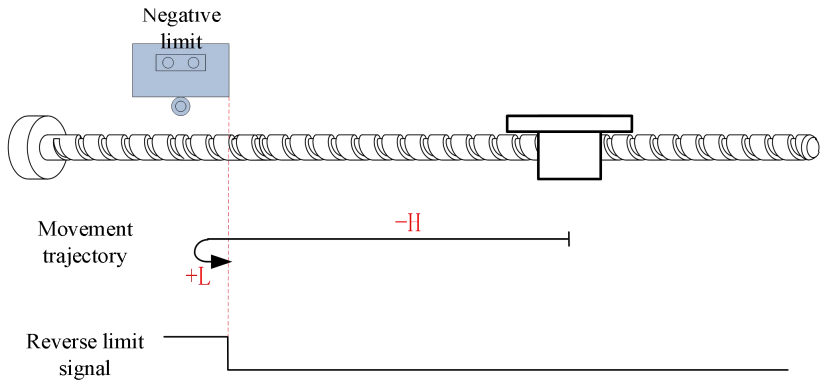
Description: When starting the homing, the home signal is valid. The motor directly runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it starts to search for the motor Z signal and stops running after the Z signal is found.

Homing method 15,16:

Reserve

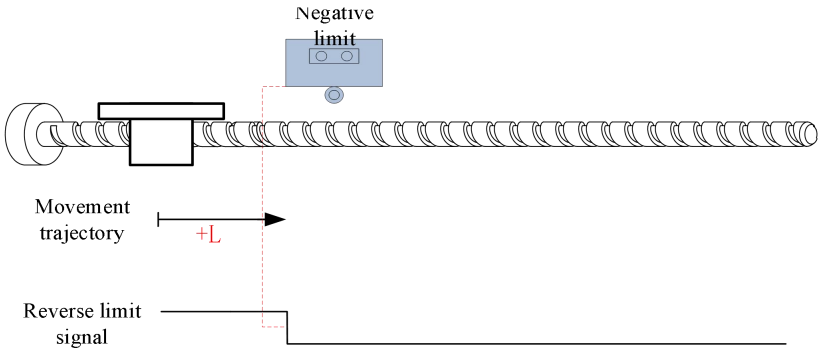
Homing mode 17: Homing with reverse limit signal (6098=17)

1) When homing, the reverse limit signal is invalid.



Description: When starting from zero, the reverse limit signal is invalid. The motor runs in the reverse direction at a higher set speed. When the rising edge of the reverse limit signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the reverse limit signal is detected, it stops running.

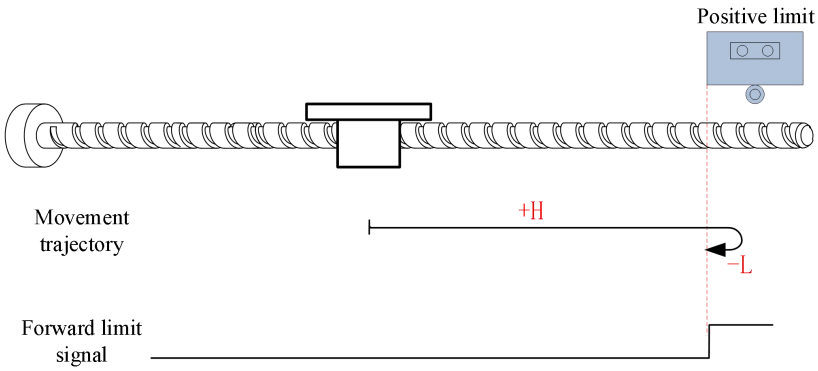
2) When homing and starting, the reverse limit signal is valid



Description: When starting from zero, the reverse limit signal is valid. The motor directly runs forward at a lower set speed and stops when the falling edge of the reverse limit signal is detected.

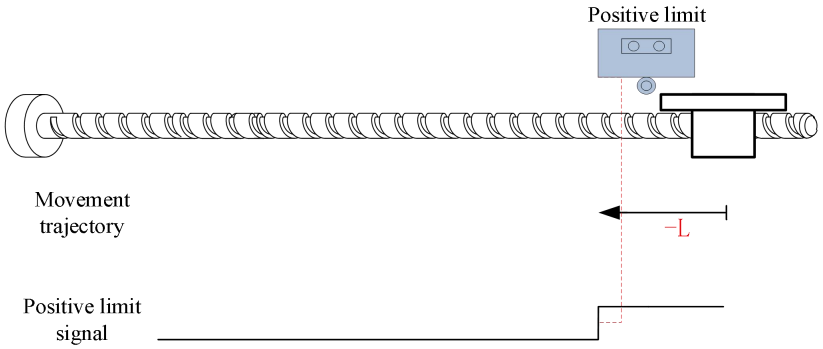
Homing mode 18: Homing with positive limit signal (6098=18)

1) When homing, the positive limit signal is invalid



Description: When starting from zero, the positive limit signal is invalid. The motor runs forward at a higher set speed. When the rising edge of the positive limit signal is detected, it decelerates and reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the positive limit signal is detected, it stops running.

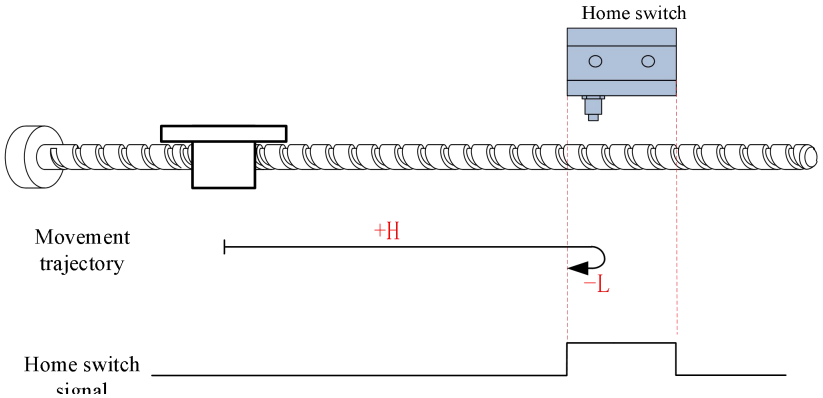
2) When homing, the positive limit signal is valid



Description: When starting from zero, the positive limit signal is valid. The motor directly runs in the reverse direction at a lower set speed, and stops when the falling edge of the positive limit signal is detected.

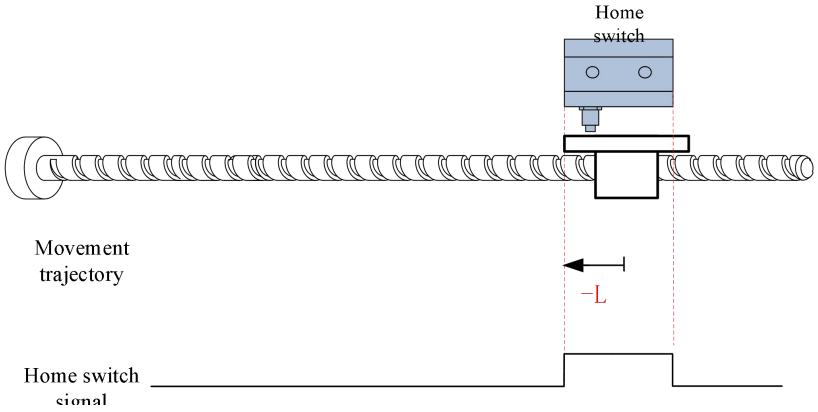
Homing method 19: homing with home signal (6098=19)

1) When homing, the home signal is invalid



Description: When starting from homing, the home signal is invalid. The motor runs forward at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it stops running.

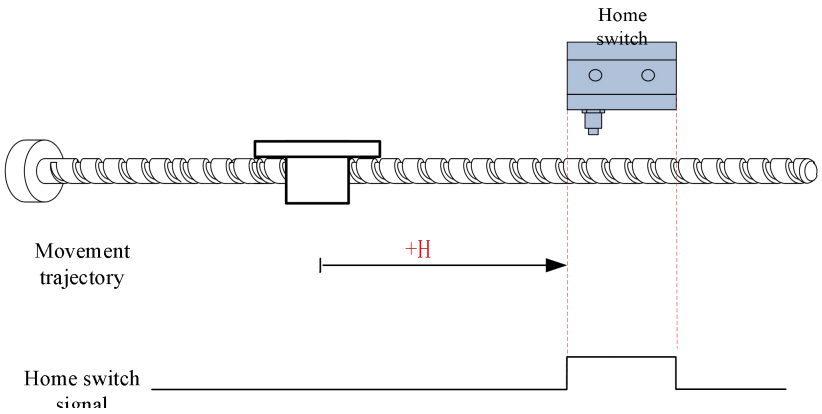
2) When homing, the home signal is valid



Description: When starting the homing, the home signal is valid. The motor directly runs in the reverse direction at a lower set speed, and stops when the falling edge of the home signal is detected.

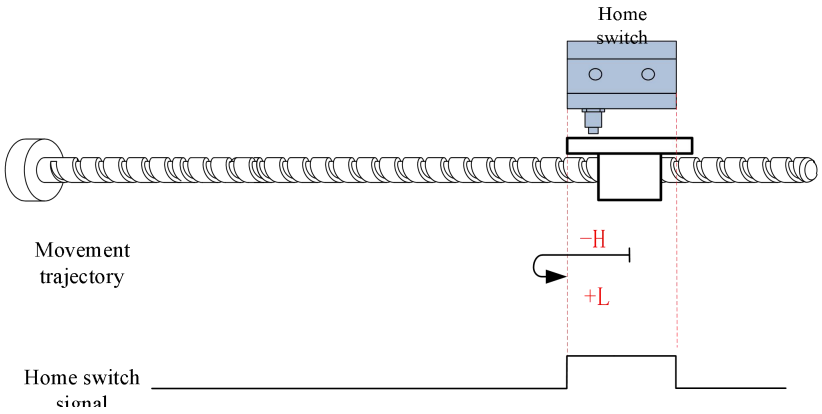
Homing method 20: homing with home signal (6098=20)

1) When homing, the home signal is invalid



Description: When starting the homing, the home signal is invalid. The motor runs forward at a lower speed and stops when the rising edge of the home signal is detected.

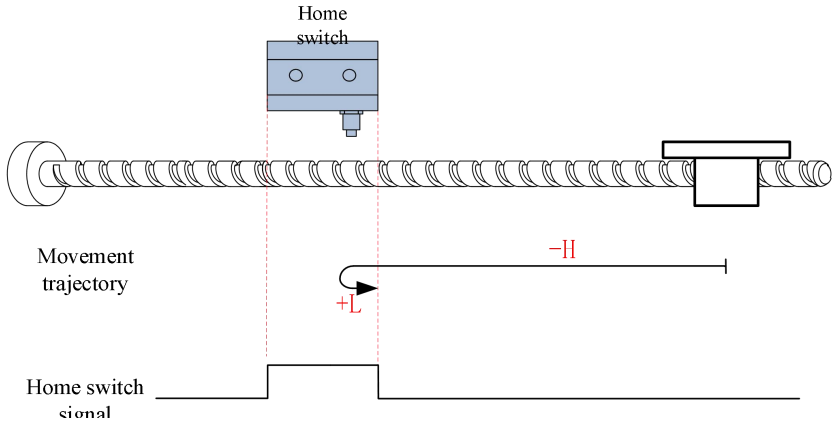
2) When homing, the home signal is valid



Description: When starting from zero, the home signal is valid. The motor first runs in the reverse direction at a higher set speed. When the falling edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops.

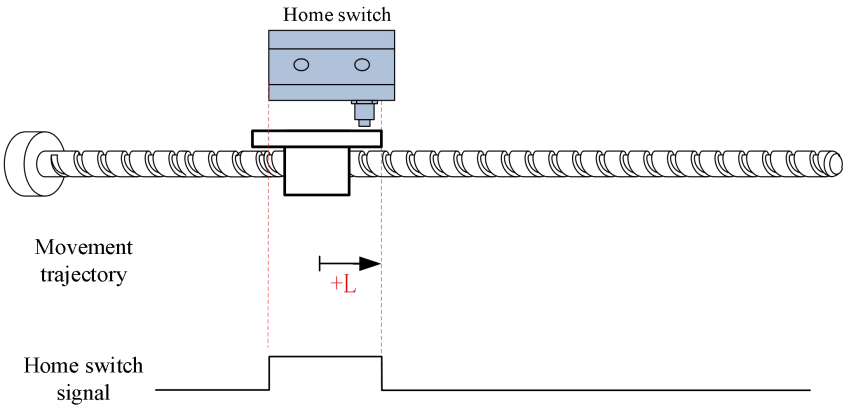
Homing method 21: homing with home signal (6098=21)

1) When homing, the home signal is invalid



Description: When starting from homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the home signal is detected, it stops.

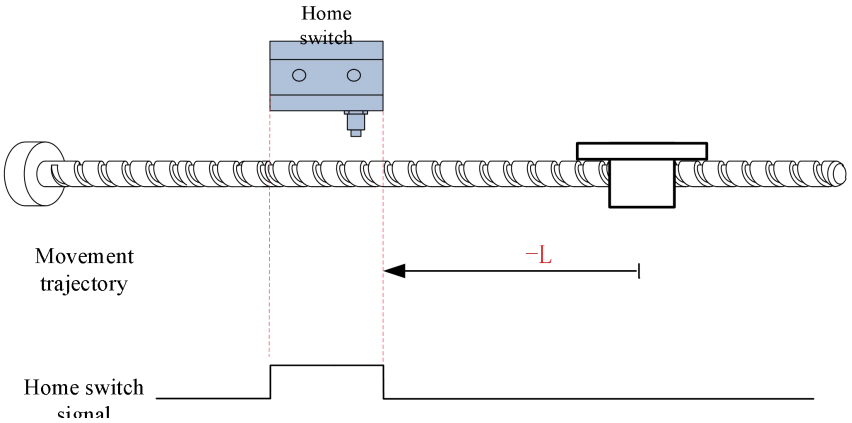
2) When homing, the home signal is valid



Description: When starting from zero, the home signal is valid. The motor directly runs forward at a lower set speed and stops when the falling edge of the home signal is detected.

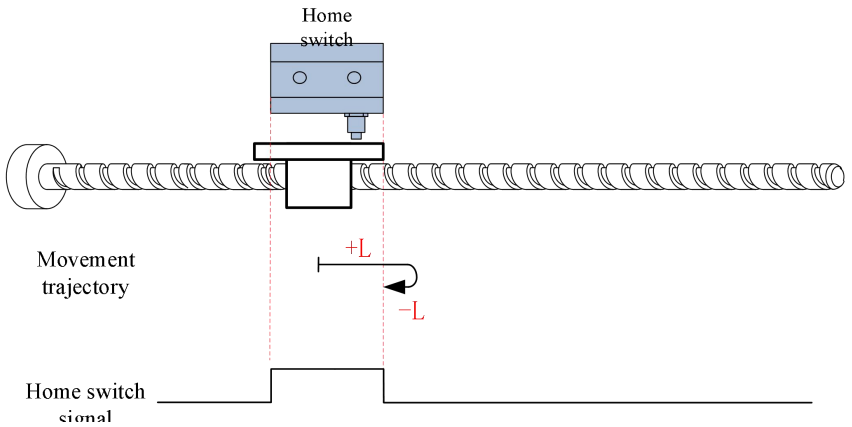
Homing method 22: Homing with home signal (6098=22)

1) When homing, the home signal is invalid



Description: When starting the homing, the home signal is invalid. The motor runs in the reverse direction at a lower speed and stops when the rising edge of the home signal is detected.

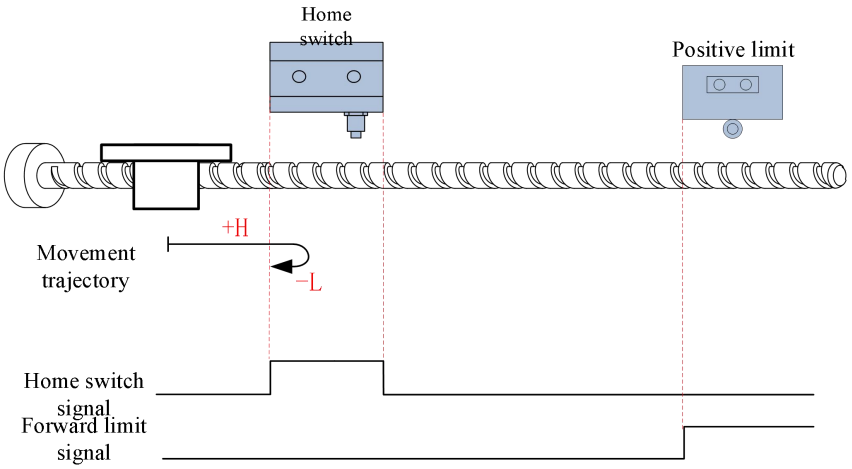
2) When homing, the home signal is valid



Description: When starting from homing, the home signal is valid. The motor first runs forward at a higher set speed. When the falling edge of the home signal is detected, it decelerates and reverses, running in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

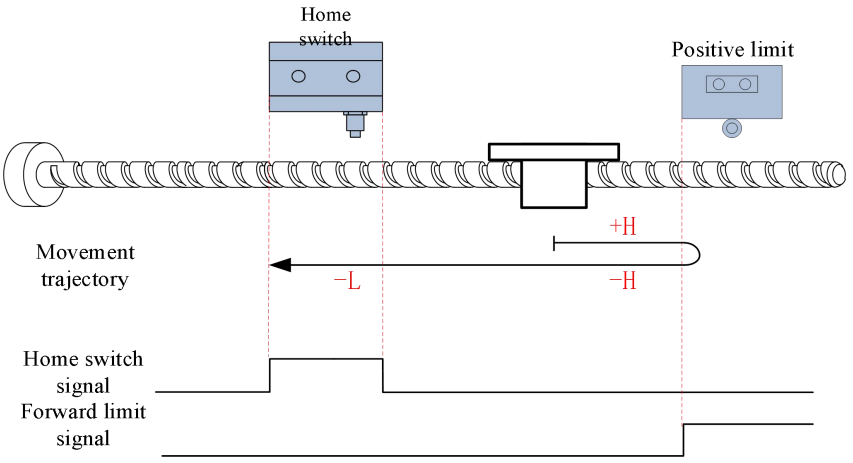
Homing mode 23: homing with home signal and positive limit signal (6098=23)

1) When homing, the home signal is invalid and no positive limit switch is encountered



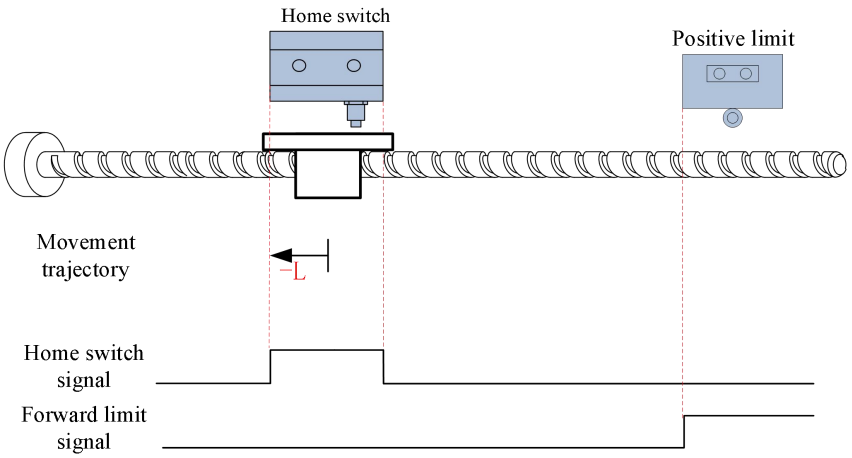
Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed and does not encounter the forward limit switch. When the rising edge of the home signal is detected, it decelerates and reverses, runs in the reverse direction at a lower set speed, and stops when the falling edge of the home signal is detected.

2) When homing, the home signal is invalid and encounters the positive limit switch.



Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed. When it encounters the forward limit switch, it decelerates and reverses, and runs in the reverse direction at a higher set speed. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it stops running.

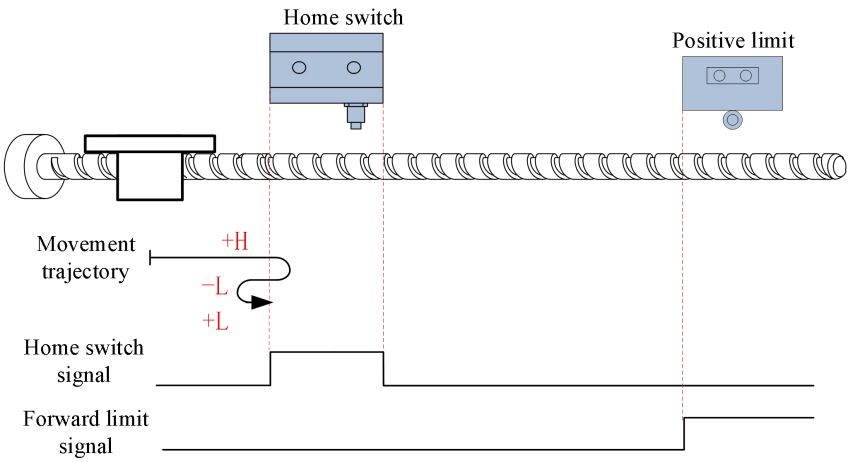
3) When homing, the home signal is valid



Description: When starting the homing, the home signal is valid. The motor runs in the reverse direction at a lower set speed and stops when the falling edge of the home signal is detected.

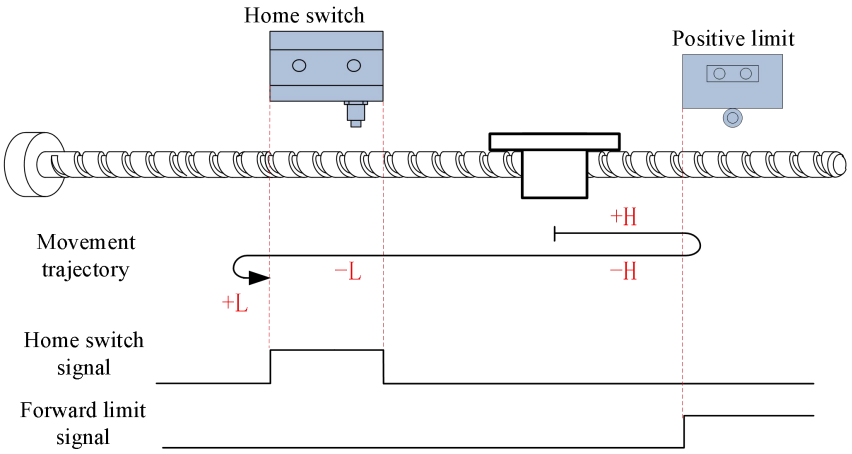
Homing mode 24: homing with home signal and positive limit signal (6098=24)

1）When homing, the home signal is invalid and no positive limit switch is encountered



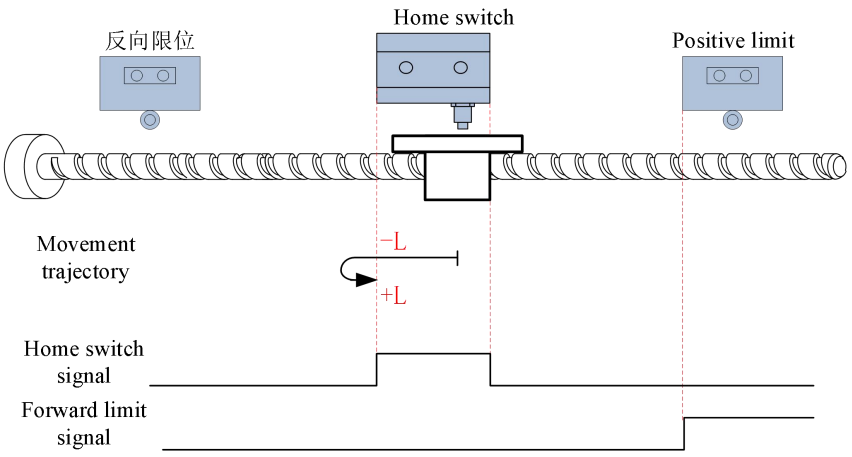
Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed and does not encounter the forward limit switch. When the rising edge of the home signal is detected, it decelerates and reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters the positive limit switch.



Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed. When encountering the forward limit switch, it decelerates and reverses, and runs in the reverse direction at a higher set speed. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops running.

3）When homing, the home signal is valid

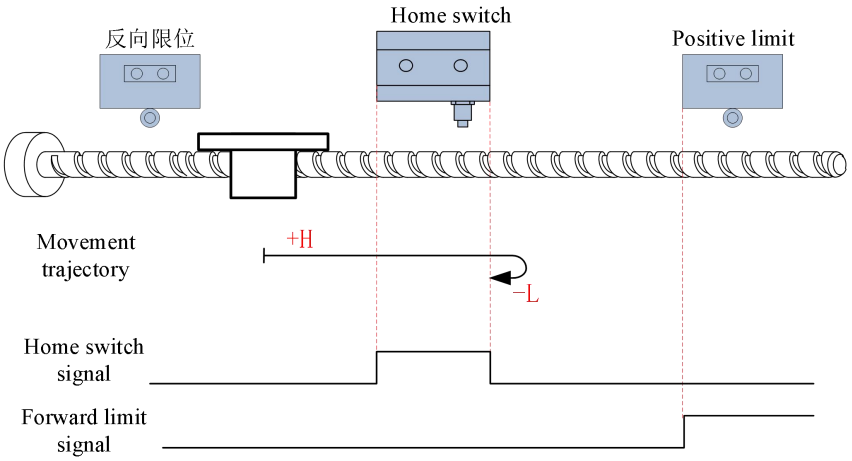


Description: When homing, the home signal is valid. The motor runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again,

and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops running.

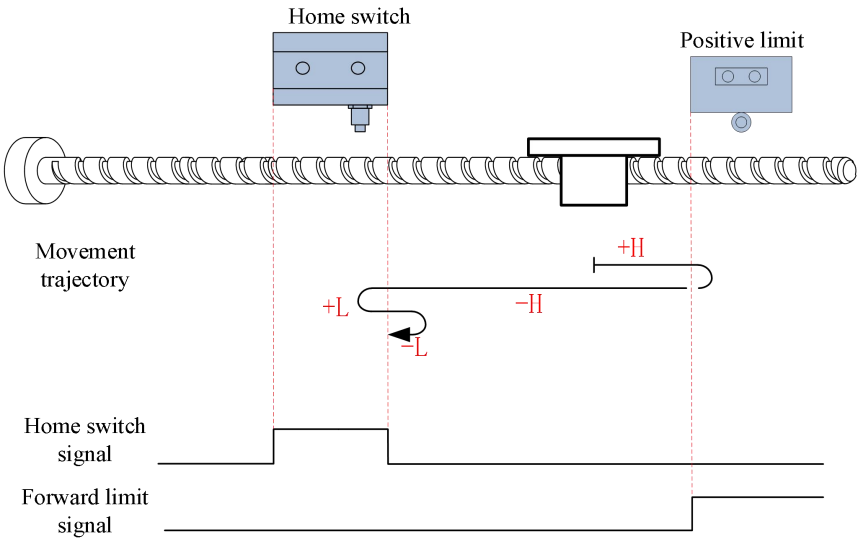
Homing mode 25: homing with home signal and positive limit signal (6098=25)

1）When homing, the home signal is invalid and no positive limit switch is encountered



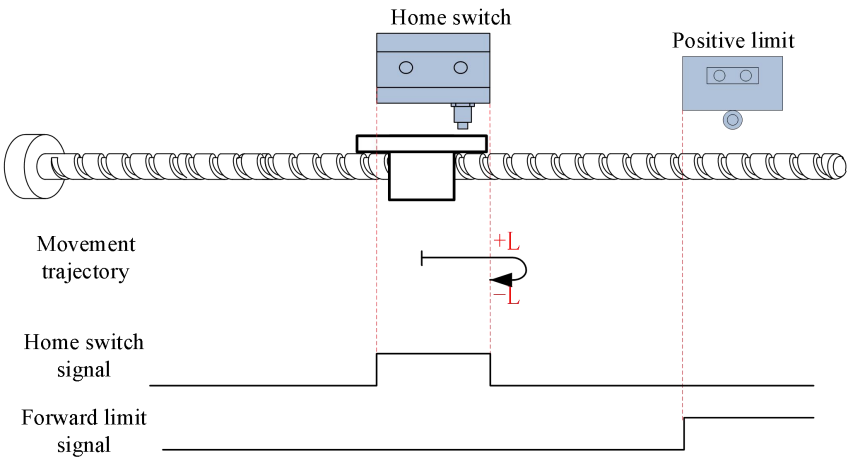
Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed and does not encounter the forward limit switch. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates again and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters the positive limit switch.



Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed. When encountering the forward limit switch, it decelerates and reverses, and runs in the reverse direction at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

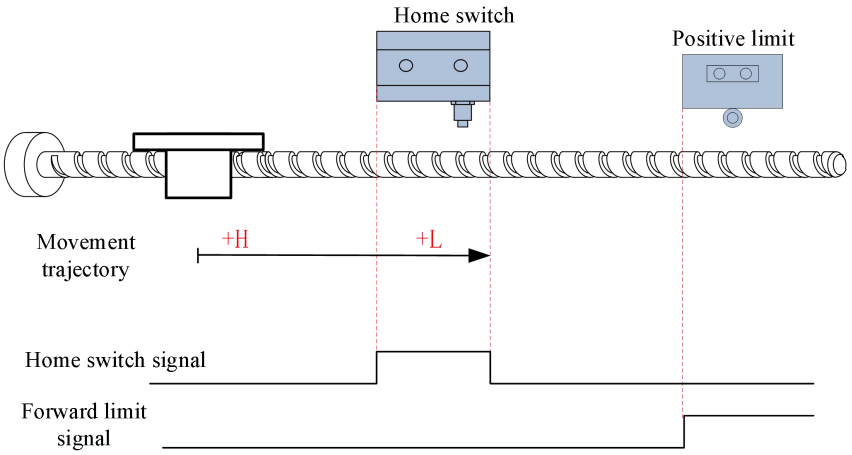
3）When homing, the home signal is valid



Description: When homing, the home signal is valid. The motor runs forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses, running in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

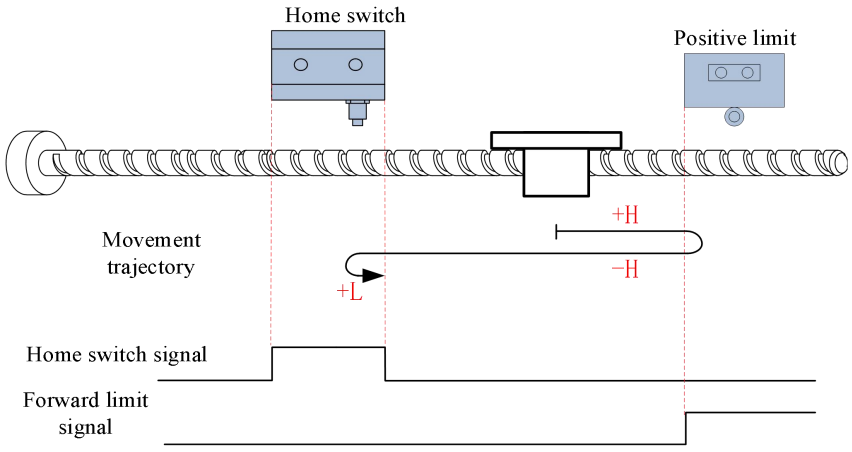
Homing mode 26: homing with home signal and positive limit signal (6098=26)

1）When homing, the home signal is invalid and no positive limit switch is encountered



Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed and does not encounter the forward limit switch. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is detected, it stops running.

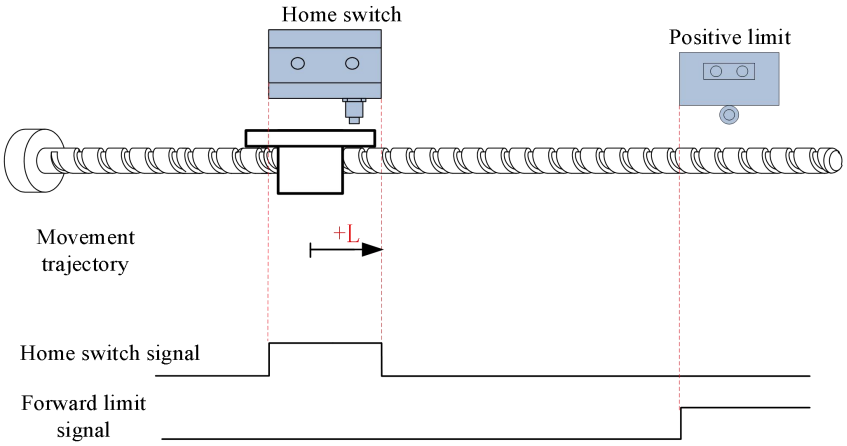
2）When homing, the home signal is invalid and encounters the positive limit switch.



Description: When homing, the home signal is invalid. The motor runs forward at a higher set speed. When encountering the forward limit switch, it decelerates and reverses, and runs in the reverse direction at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the home signal is detected, it

stops running.

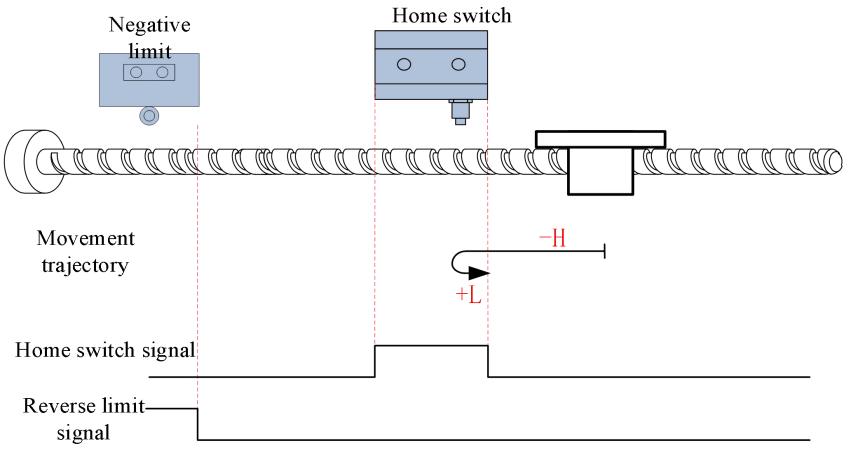
3）When homing, the home signal is valid



Description: When homing, the home signal is valid. The motor runs forward at a lower set speed and stops when the falling edge of the home signal is detected.

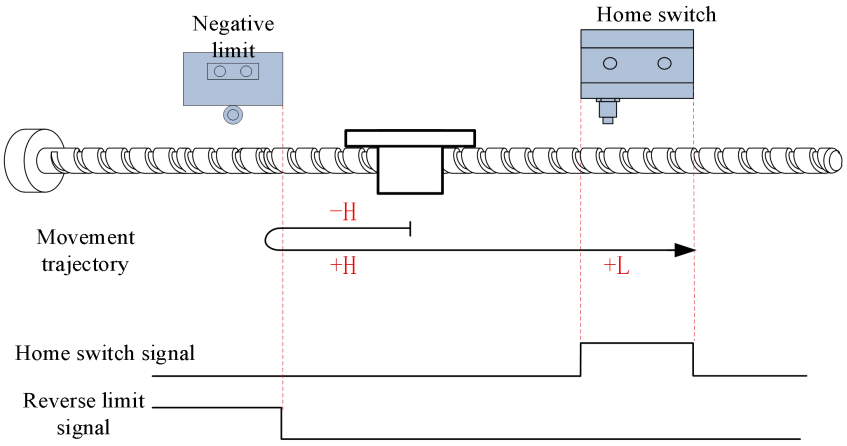
Homing mode 27: homing with home signal and reverse limit signal (6098=27)

1）When homing, the home signal is invalid and no reverse limit switch is encountered



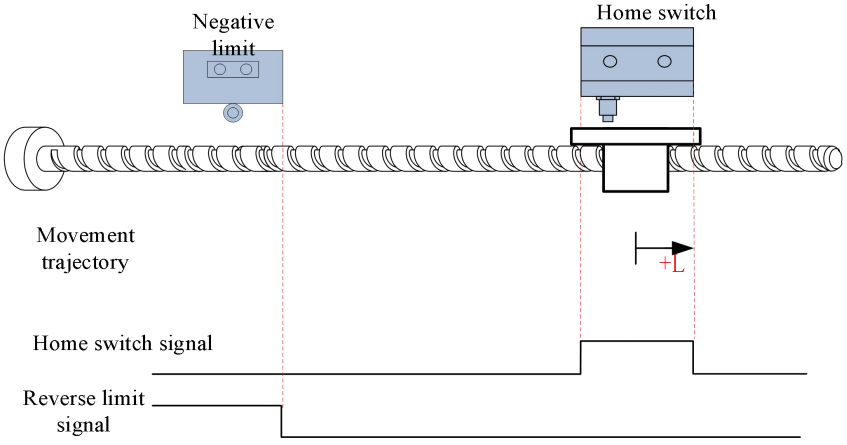
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed and does not encounter the reverse limit switch. When the rising edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters a reverse limit switch



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed. When it encounters the reverse limit switch, it decelerates and reverses, and runs forward at a higher set speed. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is detected, it stops running.

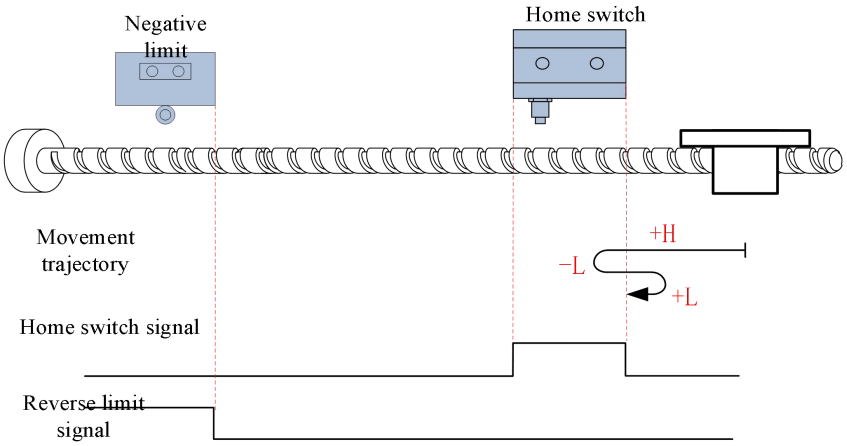
3）When homing, the home signal is valid



Description: When homing, the home signal is valid. The motor runs forward at a lower set speed and stops when the falling edge of the home signal is detected.

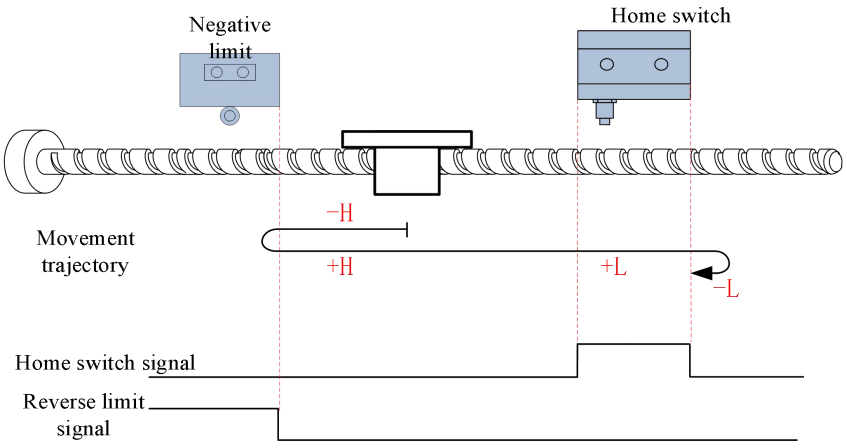
Homing mode 28: homing with home signal and reverse limit signal (6098=28)

1）When homing, the home signal is invalid and no reverse limit switch is encountered



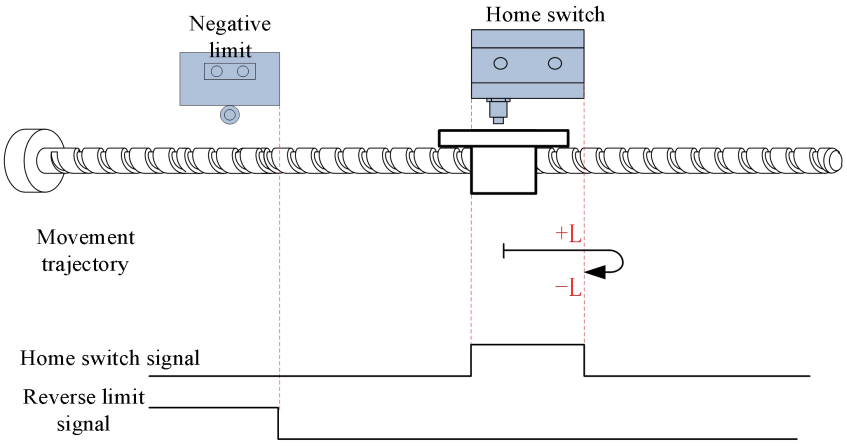
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed and does not encounter the reverse limit switch. When the rising edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters a reverse limit switch



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed. When encountering the reverse limit switch, it decelerates and reverses, and runs forward at a higher set speed. When the rising edge of the home signal is detected, it decelerates and continues to run forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

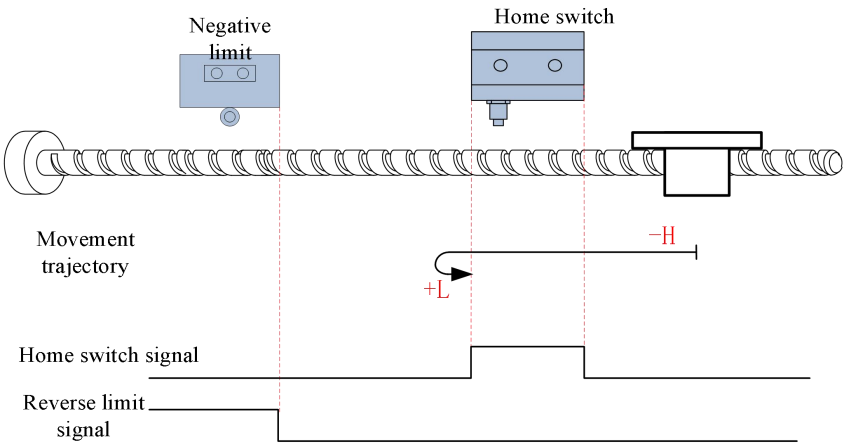
3）When homing, the home signal is valid



Description: When homing, the home signal is valid. The motor runs forward at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses, running in the reverse direction at a lower set speed. When the rising edge of the home signal is detected, it stops running.

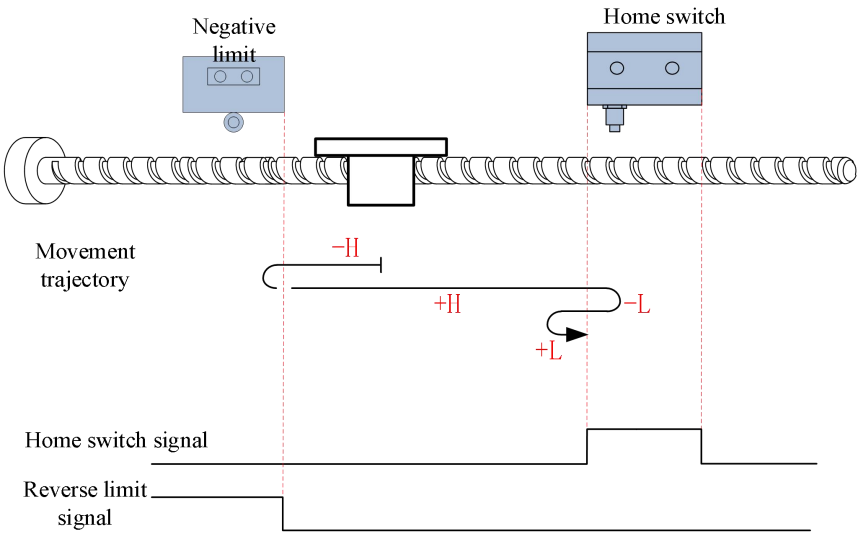
Homing mode 29: homing with home signal and reverse limit signal (6098=29)

1）When homing, the home signal is invalid and no reverse limit switch is encountered



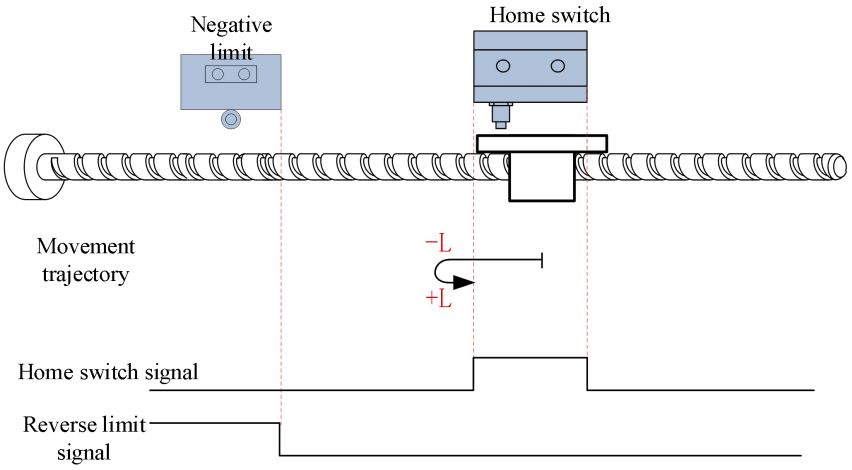
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed and does not encounter the reverse limit switch. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, runs forward at a lower set speed, and stops when the rising edge of the home signal is detected.

2）When homing, the home signal is invalid and encounters the positive limit switch.



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed. When encountering the reverse limit switch, it decelerates and reverses, and runs forward at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses again, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops running.

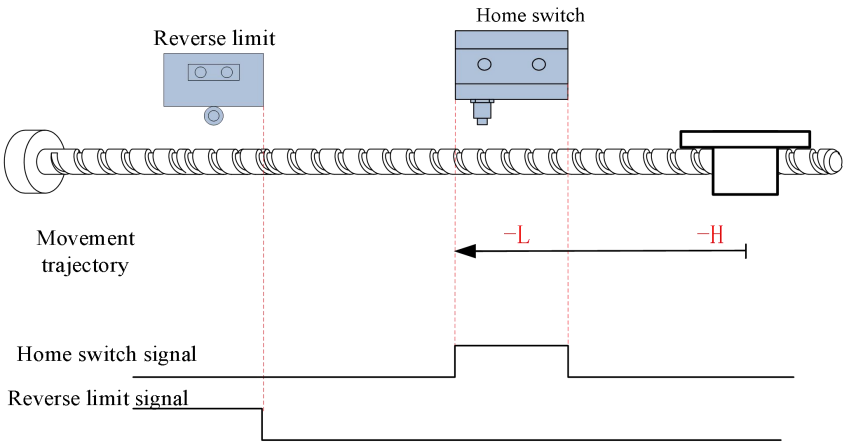
3）When homing, the home signal is valid



Description: When homing, the home signal is valid. The motor runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it decelerates and reverses, and runs forward at a lower set speed. When the rising edge of the home signal is detected, it stops.

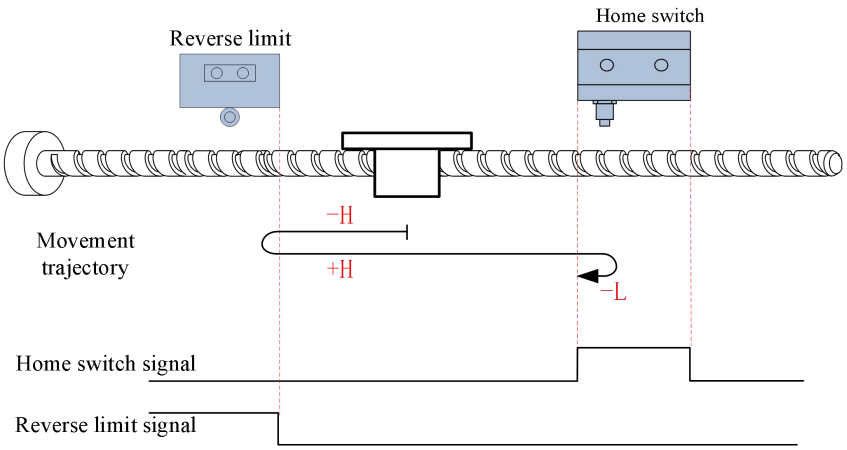
Homing mode 30: homing with home signal and reverse limit signal (6098=30)

1）When homing, the home signal is invalid and no reverse limit switch is encountered



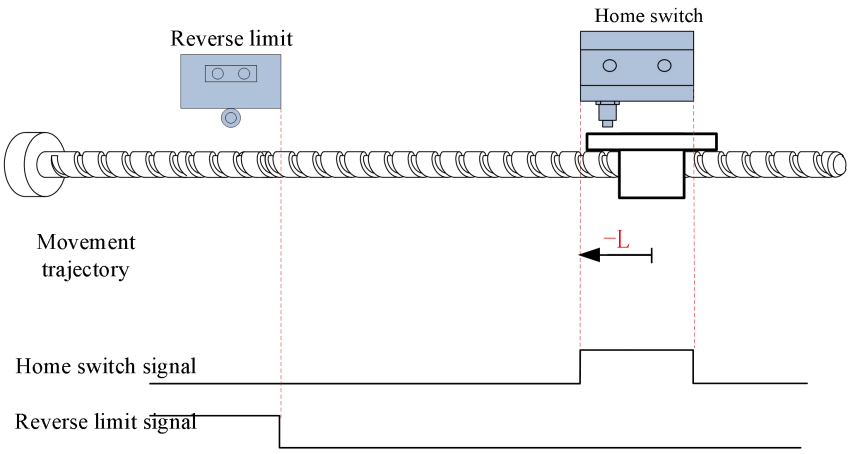
Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed and does not encounter the reverse limit switch. When the rising edge of the home signal is detected, it decelerates and continues to run in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it stops running.

2）When homing, the home signal is invalid and encounters a reverse limit switch



Description: When homing, the home signal is invalid. The motor runs in the reverse direction at a higher set speed. When it encounters the reverse limit switch, it decelerates and reverses, and runs forward at a higher set speed. When the rising edge of the home signal is detected, it decelerates and reverses, and runs in the reverse direction at a lower set speed. When the falling edge of the home signal is detected, it stops running.

3）When homing, the home signal is valid



Description: When starting the homing, the home signal is valid. The motor runs in the reverse direction at a lower set speed and stops when the falling edge of the home signal is detected.

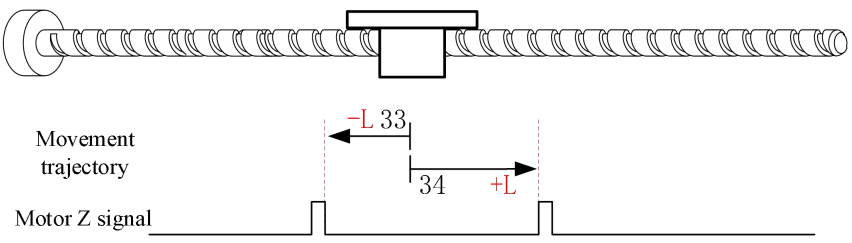
Homing method 31: (used in conjunction with homing method 32)

Description: When the motor uses an absolute position encoder, perform absolute position homing (the absolute position recorded by homing method 32).

Homing method 32: (used in conjunction with homing method 31)

Description: When the motor uses an absolute position encoder, the current absolute position is recorded as the zero point for absolute position return.

Homing mode 33,34: Homing with motor zero position pulse signal (6098=33,34)



Description:

Homing mode 33: The motor runs in the reverse direction at a set lower speed and stops when it encounters the motor Z signal;

Homing mode 34: The motor runs in the forward direction at a set lower speed and stops when it encounters the motor Z signal;

Homing method 35: current point homing

Description: With bit 4 of control word 6040 as the trigger signal, the motor does not move and directly sets the homing completion state.

4.2.2.7 Internal multi-position function

The multi-segment position operation function means that the servo driver stores 16 segments of position instructions. The displacement, maximum operating speed, acceleration and deceleration time of each segment can be set separately. The waiting time and connection method between each segment can also be selected according to actual needs. The setting process is as follows:

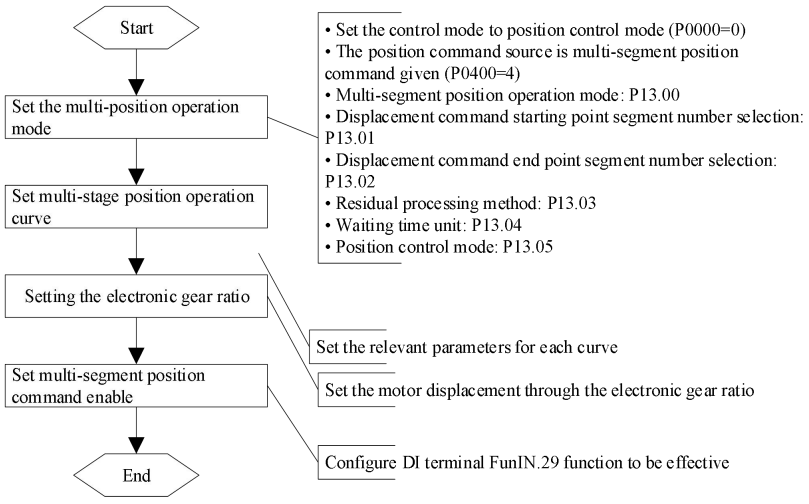


Figure 4.11 Multi-segment position command source setting process

1) Set the multi-position operation mode

☆Related function code:

Function code	Name	Setting range	Function	Setting method	Effective time	Factory setting
P13.00	Multi-stage position operation mode	0- Single run ends and stops 1- Cycle run 2- DI switch run	Set the connection between paragraphs	Shutdown setting	Effective immediately	1
P13.01	Specify the starting segment	1 ~ 16	Set the starting point of multiple position commands	Shutdown setting	Effective immediately	1
P13.02	Specify the end segment	1 ~ 16	Set the end segment of multiple position instructions	Shut down setting	Effective immediately	16
P13.03	Residue processing method	0- Continue running the unfinished segment 1- Restart running from the first	Set the servo enable to ON, and the starting segment number when	Shut down setting	Effective immediately	0

		segment	the multi-segment position operation is interrupted and resumed			
P13.04	Waiting time unit	0-ms 1-s	Set the waiting time unit	Shut down setting	Effective immediately	0
P13.05	Position control method	0- Relative position command 1- Absolute position command	Set the displacement instruction type	Shut down setting	Effective immediately	0

a) Stop at the end of a single run

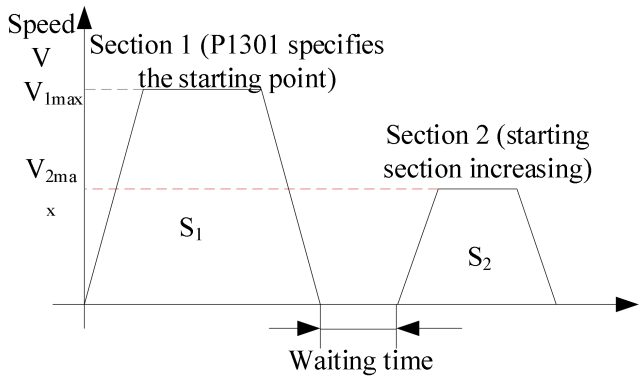


Figure 4.12 Operation curve of shutdown at the end of a single operation

V_{1max}、V_{2max}：The maximum operating speed of the first segment (the starting segment specified by P1301) and the maximum operating speed of the second segment (the starting segment increasing segment).

S1、S2：The first segment (the starting segment specified by P1301) is displaced, and the second segment (the starting segment and the increasing segment) is displaced.

Mode Description:

◆The first segment of the operation is the starting segment specified by P1301, and the last segment of the operation is the ending segment specified by P1302.

◆The segment number automatically increases and switches.

◆Waiting time can be set between each segment.

◆The internal multi-segment position enable (FunIN.29: PSEC_EN) signal is level-valid.

◆After each segment of operation is completed, the positioning completion signal is valid.

◆During operation, the internal multi-stage position enable is OFF, the servo abandons the unfinished displacement of this stage and stops. After the stop, the positioning completion signal is valid.

◆Turn on the internal multi-stage position enable again, and the servo will run in the corresponding stage according to P13.03 setting.

◆During a certain period of operation, the servo is enabled and turned off. The motor stops in the servo OFF mode. After the stop, the positioning is invalid.

◆During a certain operation, the reverse DI (FunIN.24: POS_DIR) logic switching of the position command has no effect on the operation direction of this section.

b) Cycle operation (P13.00=1)

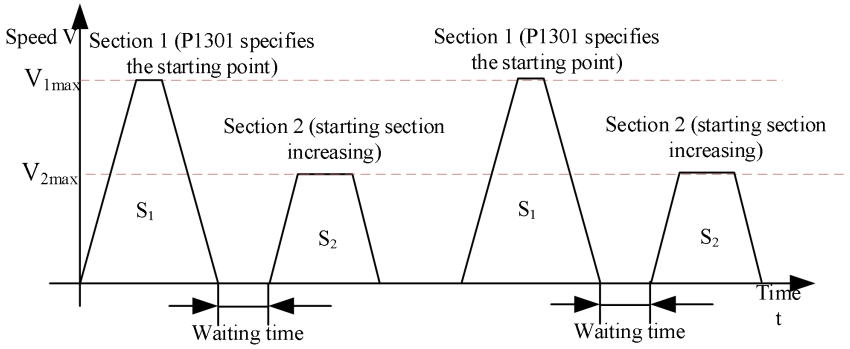


Figure 4.13 Cyclic operation curve

V1max、V2max：The maximum operating speed of the first segment (the starting segment specified by P1301) and the maximum operating speed of the second segment (the starting segment increasing segment).

S1、S2：The first segment (the starting segment specified by P1301) is displaced, and the second segment (the starting segment and the increasing segment) is displaced.

Mode Description:

◆Cyclic operation, the starting segment number of each round is set by parameter P1301;

◆The segment number automatically increases and switches;

◆Waiting time can be set between each segment;

◆The internal multi-segment position enable (FunIN.29: PSEC_EN) signal is level-valid; if valid, the loop operation state is maintained.

◆After each segment is completed, the positioning completion signal is valid;

◆During operation, turn off the multi-stage position command enable, the servo abandons the unfinished displacement of this stage and stops. After the stop, the positioning completion signal is valid;

◆Turn on the multi-segment position command again, and the servo will run in the corresponding segment according to P13.03 setting;

◆During a certain period of operation, the servo is enabled and turned off. The motor stops in the servo OFF mode. After the stop, the positioning is invalid.

◆During a certain operation, the reverse DI (FunIN.24: POS_DIR) logic switching of the position command has no effect on the operation direction of this section.

c) DI switch operation (H11-00=2)

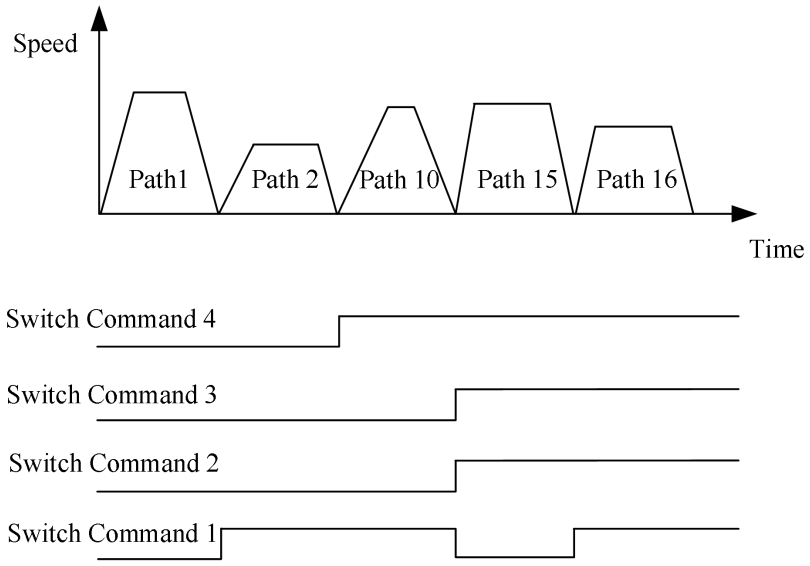


Figure 4.14 Multi-stage position timing diagram

When the multi-segment position operation mode is set to DI switching operation, please configure the DI terminal of the servo drive to function 6~9 (FunIN.6: CMD1~FunIN.9: CMD4, multi-segment operation command switching) and confirm the effective logic of the DI terminal.

☆Related function code:

<table>
<tr><th>Code</th><th>Name</th><th>Function name</th><th>Function</th></tr>
<tr><td>FunIN.6</td><td>CMD1</td><td>Multi-stage operation command switching 1</td><td rowspan="4">The multi-segment number is a 4-bit binary number. The corresponding relationship between CMD1~CMD4 and the segment number is as follows: (DI terminal logic is level-effective, when the input level is effective, the CMD value is 1, otherwise it is 0)
<table>
<tr><th>CMD4</th><th>CMD3</th><th>CMD2</th><th>CMD1</th><th>段号</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>1</td><td>2</td></tr>
<tr><td colspan="5">……</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>0</td><td>15</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>16</td></tr>
</table></td></tr>
<tr><td>FunIN.7</td><td>CMD2</td><td>Multi-stage operation command switching 2</td></tr>
<tr><td>FunIN.8</td><td>CMD3</td><td>Multi-stage operation command switching3</td></tr>
<tr><td>FunIN.9</td><td>CMD4</td><td>Multi-stage operation command switching 4</td></tr>
</table>

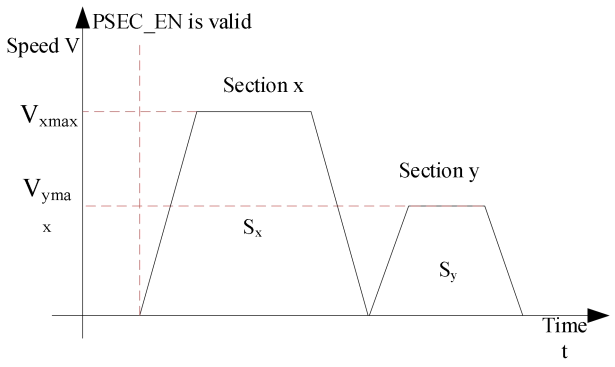


Figure 4.15 Operation curve of DI switching operation

Vxmax、Vymax：Maximum operating speed of segment x and segment y;

Sx、Sy：displacement of segment x and segment y;

Mode Description:

◆When running the current segment number, you can set the next running segment number. After completing the position command set for the current segment number, the motor stops; then immediately runs the next running segment number;

◆The segment number is determined by the DI terminal logic;

◆There is no waiting time between each segment, and the interval time is determined by the host computer command delay;

◆The internal multi-segment position enable (FunIN.29: PSEC_EN) signal is level-valid; if valid, the loop operation state is maintained;

◆After each segment is completed, the positioning completion signal is valid;

◆During operation, turn off the multi-stage position command enable, the servo abandons the unfinished displacement of this stage and stops. After the stop, the positioning completion signal is valid;

◆If the servo is enabled OFF during a certain period of operation, the motor stops in the servo OFF stop mode. After the stop is completed, the positioning completion is invalid;

◆During a certain operation, the reverse DI (FunIN.24: POS_DIR) logic switching of the position command has no effect on the operation direction of this section.

2) Position control method

a) Incremental position control

The relative displacement of each execution of the incremental position control method is the position increment of the target position relative to the current position of the motor.

b) Absolute position control

The absolute position control mode is based on the successful motor zeroing. If the motor zeroing is not successful, that is, the motor home is not zeroed before running the absolute position function, the multi-segment position function of the motor will not be executed, and the motor will enter the lock axis waiting state.

Absolute displacement is the position increment of the target position relative to the motor home. For example, if the mechanical home offset (P1614) after the motor returns to zero is set to 1000, then the absolute position counter (P1807) when the motor stops after the homing is also 1000. After the absolute position offset is executed, the value of the motor absolute position counter (P1807) is the motor mechanical home offset (P1614) plus the absolute displacement increment of the corresponding segment (for example, the first segment P1308 is 20000), so the absolute position of the motor after execution is 20000+1000.

3) Multi-stage position operation curve setting

The multi-stage position operation function can set 16 different position commands. The displacement, maximum operating speed, acceleration and deceleration time of each stage and the waiting time between each stage can be set separately. Take the first stage as an example:

☆Related function code:

Function code	Name	Setting range	Unit	Function	Setting method	Effective time	Factory setting
P13.09	The first stage of movement	-1073741824 ~1073741824	Command unit	Set the total position command of the first segment	Running setting	Effective immediately	10000
P13.10	Maximum operating speed of the first displacement	1~6000	rpm	Set the maximum running speed of the first stage	Running setting	Effective immediatel	200
P13.11	The first stage of displacement acceleration and deceleration time	0~65535	ms(s)	Set the time for the motor to change speed from 0rpm to 1000rpm in the first stage of multi-stage position.	Running setting	Effective immediately	10
P13.12	Waiting time after the first displacement is completed	0~10000	ms(s)	Set the waiting time after the first segment positioning is completed	Running setting	Effective immediately	10

According to the above settings, the actual operation curve of the motor is shown in the figure below:

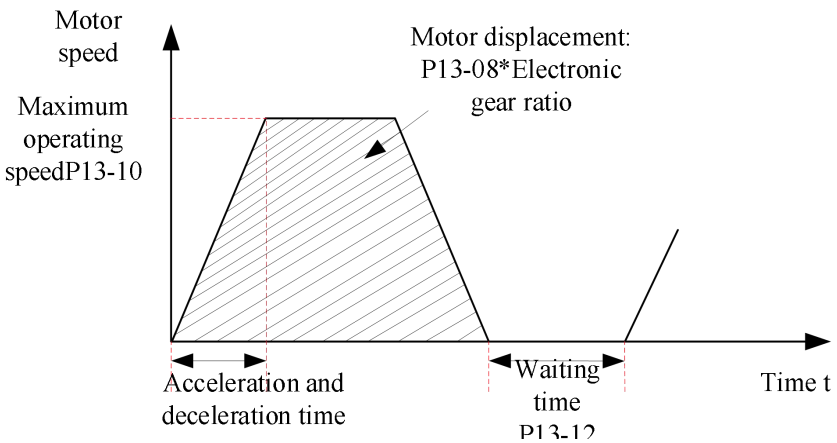


Figure 4.16 The first section of the motor operation curve

Therefore, the actual acceleration time t to P13.10 (maximum running speed of the first displacement segment) is:

$$t = \frac{(P13.10)}{1000} \times (P13.11)$$

4) Internal multi-segment position command enable

When selecting the internal multi-segment position command as the position command source, please configure one DI terminal of the servo drive as function 29 (FunIN.29: internal multi-segment position enable) and confirm the effective logic of the DI terminal.

☆Related function code:

Code	Name	Function name	Function
FunIN.29	PSEC_EN	Internal multi-stage position enable	If valid, the servo motor runs multiple position commands; If invalid, the servo motor is in a locked state;

4.3 Speed control mode

4.3.1 Speed control diagram

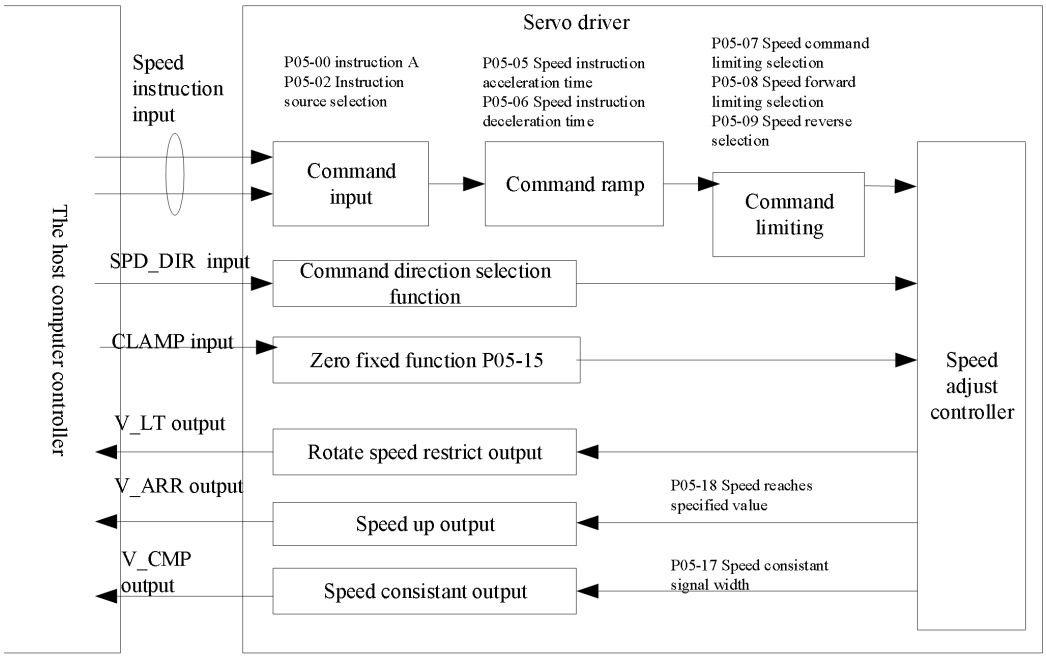


Figure 4.17 Speed control diagram

The main use steps of speed control mode are as follows:

1. Correctly connect the power supply of servo main circuit and control circuit, as well as the motor power cable and encoder cable.

2. Disconnect the motor and load and perform the servo JOG test run through the host computer to comfirm wether the motor can run as normal.

3. Please refer to speed mode wiring description to connect the necessary DI/DO signal and analog instruction in DIDO terminal.

4. make relatesettings of speed mode.

5. Enable servo, firstly make the motor rotate at a low speed and judge wether the rotation direction of motor is normal, then adjust gain.

4.3.2 Internal multi-segment speed function

The servo driver has multi-segemnt speed running function. It refers the internal of servo drive storage 16 segments speed instruction, the maximum speed of each segment and running time can be set respectively. And equipped with 4 sets of acceleration and deceleration times to choose from. The setting process are as follows:

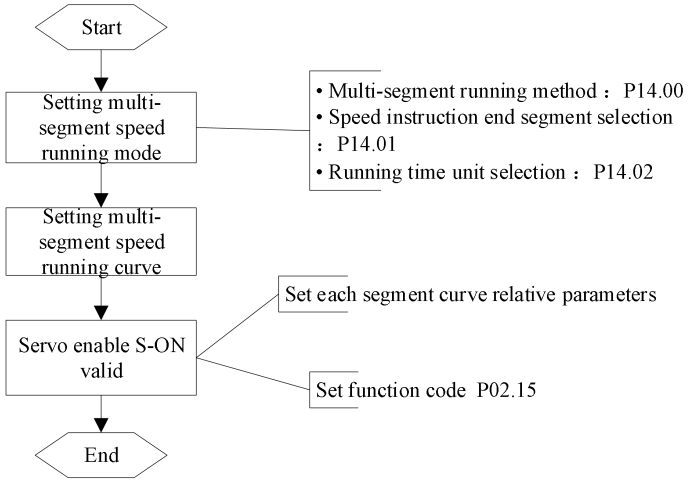


Figure 4.18 Multi-segment setting process diagram

①Setting multi-segment running mode

☆Relative function code：

Functioncode	Name	Setting arrange	Unit	Function	Setting method	Effective time	Factory settings
P14.00	Multi-segment speed running method	0- Stop at the end of signal run (P14.01 Segments selection) 1- Loop operation (P14.01 Segment selection) 2- Switching through external DI	-	Setting multi-segments speed instruction operation method	Shutdown settings	Effective immediately	1
P14.01	Speed command end segment selection	1~16	-	Setting the number of segments for multi-segment speed command	Shutdown settings	Effective immediately	1
P14.02	Running time	0-Sec	-	Selecting time unit for the	Operation	Effective	1

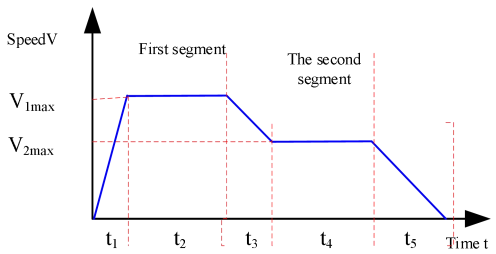
	unit selection	1-Min		multi-segment speed command operation	settings	immediately	

Take P14.01=2 as example to illustrate each mode.

●Signal operation end shutdown (P14.00=0)

Setting function code P14.00 to 0, select signal running shutdown method. After setting function code P14.01 and P14.02 according to execute total segments and implement time unit individually, and setting parameters such as corresponding command value, operation time, accelerate and decelerate time according to requirement. The driver will running as the method of from the first segment to the N segment according to segment code, until finish the last segment and then shutdown.

Table 4.6 The signal running end up shutdown description

Mode description	Operation curve
◆Running one round ◆Segment number automatic increment switching	 ◆V1max、V2max：First segment, second segment command speed ◆t1：First segment actual accelerate and decelerate time ◆t3、t5：The second time accelerate, decelerate time The running time of a certain segment: The speed changing time from the last segment speed command switching to this segment speed command+ this segment uniform running time(such as: in this figure, the first segment running time is t1+t2, the second running time is t3+t4, and so on.) Do not set a certain segment running time to 0, driver will skip this speed command, execute next segment; The actual rotate speed of motor reaches up to the setting maximum running speed of this segment,

	the speed reaches signal is valid.

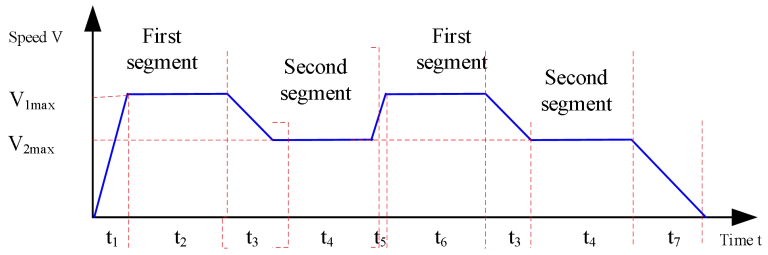
★Glossary：

The drive runs completely once the multi-segment speed command total segments setting by P14.01 is called complete a round operation.

- **Cyclic operation (P14.00=1)**

Setting function code P14.00 to 1, and select the loop operation method. After setting function code P14.01,P14.02 respectively according to the execution total segment number and execution time unit, and setting parameters command value, running time, acceleration and deceleration time corresponding segment according to requirements. The moduel will set the running time and acceleration and deceleration time of each segment, the drive will run from 1st segment to Nth segment according to segment code. After running the last segment, it will automatically jump to the 1st segment for cyclic operation.

Table 4.7 Cyclic operation description

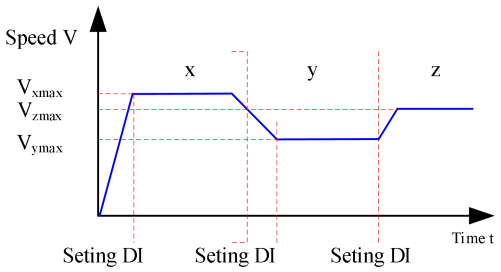
Mode description	Operation curve
◆Cyclic operation, The starting segment number of each round is 1; ◆Segment number automatic increment switching; ◆The Servo enable isvalid and always keep the cyclic operation status.	 ◆ V1max、V2max：The maximum running speed of first segment, second segment; ◆A certain segment running time: The speed changing time of previous speed command switching to this segment speed command + this segment uniform running time(For example: in this figure, the first segment running time is t1+t2, the second running time is t3+t4, and so on.) ◆Do not set a certain segment running time to 0, driver will skip this speed command, execute next segment; ◆The actual rotate speed of motor reaches the maximum running speed setting of this segment,

	the speed reaches signal is valid.

● **DI Switching operation (P14.00=2)**

Setting the function code P14.00 to 2, select the external DI switching method. Set corresponding segment command value according to requirements, the driver will select run corresponding segment speed instruction according to the combination of ON/OFF of external DI(CMDx).

Table 4.8 DI Switching operation description

Mode description	Running curve
◆As long as the segment number is updated, it can continue to run. ◆The segment number is determined by DI terminal logic; ◆The interval between segments is determined by the host computer command delay time; ◆Multi-segment position enable is valid for edge changing.	 x，y：Segment number, the logic relationship between segment number and DI terminal is description as follows; The running time of a certain section is not affected by the function code setting value, during a certain section speed instruction running, if the segment number is changed, it will switch to a new segment number for running at once; The actual rotate speed reaches this segment setting maximum running speed, the speed reaches signal is valid.

When setting the multi-segment speed running method to DI switching running, must configure the Servo drive DI terminal as function 6~9(FunIN.6 ~FunIN.9 Multi-segment operation command switching), and comfirm that DI terminal valid logic.

☆Relative function code：

<table>
<tr><th>Code</th><th>Name</th><th>Function Name</th><th>Function</th></tr>
<tr><td>FunIN.6</td><td>CMD1</td><td>Multi-segment running instruction switching</td><td rowspan="4">Multi-segment number is 4-bit binary number, the relationship between CMD1~CMD4 and segment number are as follows:
<table><tr><th>CMD4</th><th>CMD3</th><th>CMD2</th><th>CMD1</th><th>Segment Number</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>2</td></tr><tr><td colspan="5">……</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>16</td></tr></table>DI CMD value is 1 when terminal input level is valid, or is 0.</td></tr>
<tr><td>FunIN.7</td><td>CMD2</td><td>Multi-segment running instruction switching</td></tr>
<tr><td>FunIN.8</td><td>CMD3</td><td>Multi-segment running instruction switching</td></tr>
<tr><td>FunIN.9</td><td>CMD4</td><td>Multi-segment running instruction switching</td></tr>
</table>

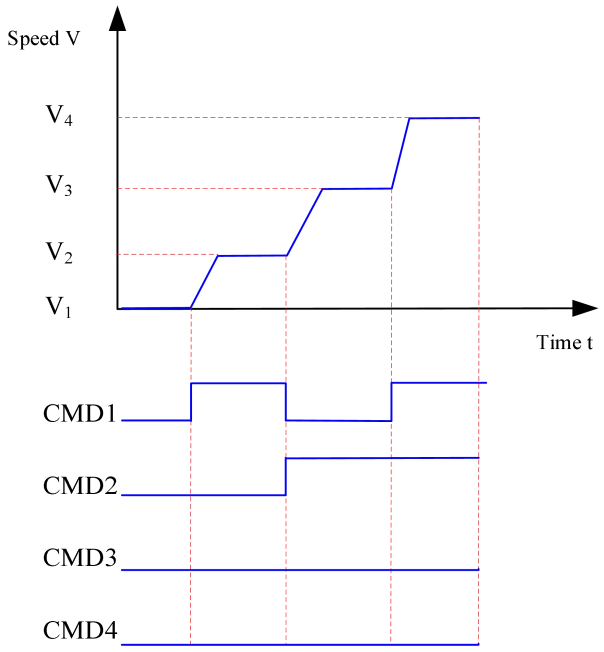


Figure 4.19 Multi-segment speed curve examples

②Multi-segment Speed Running Curve Settings

Take the first segment speed instruction as examples, the relative function code is as below:

☆Relative function code：

Function code	Name	Setting range	Unit	Function	Setting method	Effective time	Factory settings
P14.03	Accelerate and decelerate time	0~65535	ms	Setting Accelerate and decelerate time	Shutdown settings	Effective immediately	0
P14.04	Accelerate and decelerate time	0~65535	ms		Shutdown settings	Effective immediately	0
P14.05	Accelerate and	0~65535	ms		Shutdown	Effective	0

	decelerate time				settings	immediately	
P14.06	Accelerate and decelerate time	0~65535	ms		Shutdown settings	Effective immediately	0
P14.07	The first segment speed command	-6000~6000	rpm	Setting the first segment speed instruction value	Shutdown settings	Effective immediately	0
P14.08	The first segment speed command	0~65535	0.1s (min)	Setting the first segment instruction running time	Shutdown settings	Effective immediately	50
P14.09	The first segment accelerate and decelerate time	0- No accelerate and decelerate time 1- Accelerate and decelerate time 2- Accelerate and decelerate time 3- Accelerate and decelerate time 4- Accelerate and decelerate time	-	Select the first segment accelerate and decelerate method	Shutdown settings	Effective immediately	0

Among the multi-segment speed instruction parameters, except 1~16 segments command value

and command running time, there are 4 sets acceleration and deceleration time to choose from, the default method is no accelerate and decelerate time. Take multi-segment P14.01=1 signal running end as example, to illustrate the actual accelerate and decelerate time and running time:

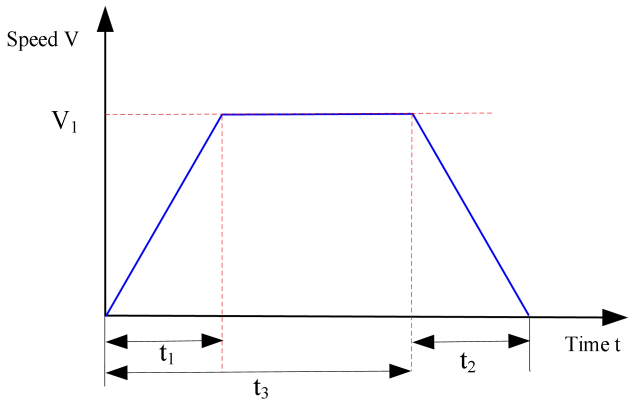


Figure 4.20 Multi-segment speed curve example

As shown above, this segment speed instruction is V1, actual the accelerate time t_1 is:

$$t_1 = \frac{V_1}{1000} \times \text{The acceleration time setting of this segment speed}$$

Actual deceleration time:

$$t_2 = \frac{V_1}{1000} \times \text{The deceleration time setting of this segment speed}$$

Running time: The speed changing time from previous speed instruction switching to this segment speed command+ this segment uniform running time, shown as the figure t3.

4.4 Torque control mode

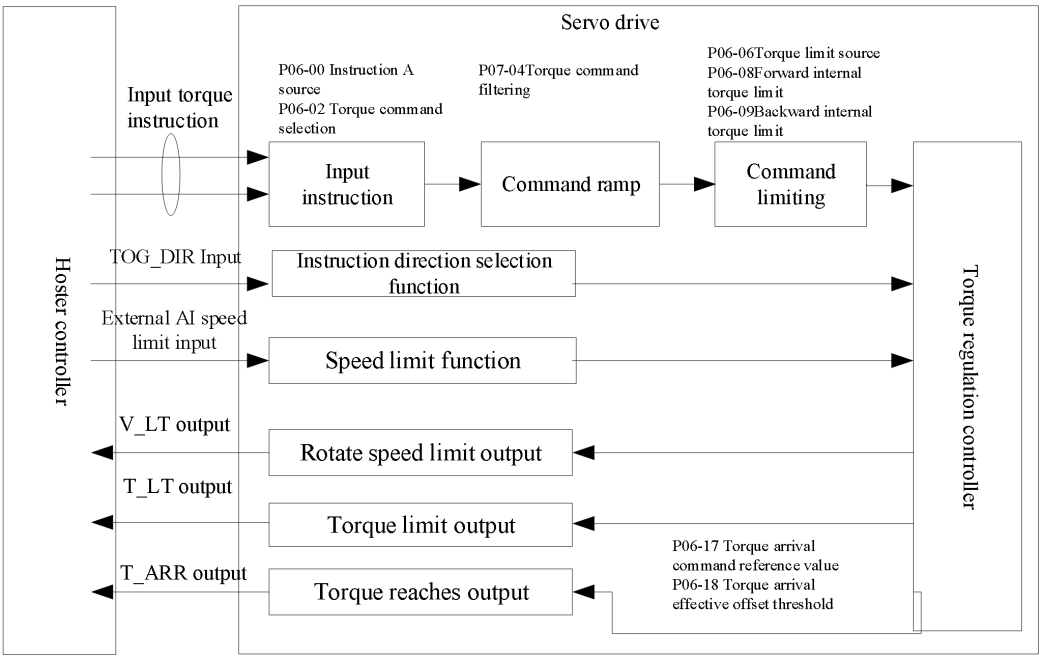


Figure 4.21 Torque control diagra

The main using steps of torque control mode are as follows:

1. Correctly connect the power supply of the servo main circuit and control circuit, as well as the motor power cable and encoder cable.

2. Disconnect the motor from the load, and conduct a servo JOG test run through the host computer to confirm whether the motor can run normally.

3. Connect necessary DI/DO and torque command source, speed limit, etc.

4. Make related settings of torque mode.

5. Enable servo, set a lower speed limit value, impose a forward or reverse torque command to servo, and confirm wether the rotation direction is right, the rotation speed is limited correctly, if normal, you can start using it.

4.5 Mixed mode switching

The mixed control mode is refer to, when the servo enable is ON, the servo status is “run”, the

working mode of servo driver can be switching in different control mode.

The mixed control mode is divided into three types as below:

●Torque mode↔ speed mode

●Speed mode ↔ position mode

●Torque mode ↔ position mode

Setting function code P00.00 through panel or hoster, the servo driver will working in mixed control mode.

☆Relative function code：

Function code	Name	Setting range	Function	Setting method	Effective time	Factory settings
P00.00	Control mode selection	0-Position mode 1- Speed mode 2- Torque mode 3- Torque mode ↔ Speed mode 4- Speed mode ↔ Position mode 5- Torque mode ↔ Position mode	Setting servo driver control mode	Shutdown settings	Effective immediately	0

Please setting the servo driver's parameters individually in different control mode according to mechanical structure and index.

When P00.00=3/4/5, please configure one DI terminal of servo driver to function 10 (FunIN.10：MODE_SWITCH，Mode switching), and confirm the valid logic of DI terminal.

☆Relative function code：

<table>
<tr><th>Code</th><th>Name</th><th>Function name</th><th>Functios</th></tr>
<tr><td>FunIN.10</td><td>MODE_SWITCH</td><td>Mode switching</td><td>Applied in mixed control mode, when the servo status is “run”, setting the driver current control mode
<table>
<tr><th>P00.00</th><th>Terminal logic</th><th>Control mode</th></tr>
<tr><td rowspan="2">3</td><td>Invalid</td><td>Torque mode</td></tr>
<tr><td>Valid</td><td>Speed mode</td></tr>
<tr><td rowspan="2">4</td><td>Valid</td><td>Speed mode</td></tr>
<tr><td>Valid</td><td>Position mode</td></tr>
<tr><td rowspan="2">5</td><td>Invalid</td><td>Torque mode</td></tr>
<tr><td>Valid</td><td>Position mode</td></tr>
</table></td></tr>
</table>

Chapter 5 Operation performance adjustment

5.1 Overview

The Servo drive need to drive the motor as quickly and accurately as possible to track the instructions from host or internal settings. In order to meet the requirement, the servo gain must be reasonably adjustment.

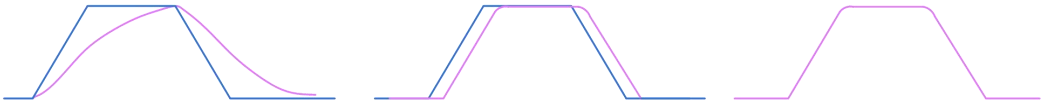


Figure 5.1 Gain setting example

Position loop gain: 40.0Hz Speed loop gain: 200.0Hz Speed loop gain: 200.0Hz

Speed loop gain: 200.0Hz Speed loop gain: 25.0Hz Speed loop gain: 25.0Hz

Speed loop integral time constant Speed loop integral time constant Speed loop integral time constant

: 100.00ms : 50.00ms : 50.00ms

Speed feedforward gain: 0 Speed feed forward gain: 0 Speed feedforward gain: 50.0%

Load inertia ratio: 30 Load inertia ratio: 30 Load inertia ratio: 30

The servo gain is setting by the combination of multiple parameters (Position loop, speed loop gain, filter, load rotate inertia ratio, etc.), which influence each other. Therefore, the setting of servo gain must considerate the balance between each parameter setting value.

Note: Before performing gain adjustment, it is recommended to perform a jog test run to confirm that the motor can operate normally.

The normal process of gain adjustment is shown as the figure below:

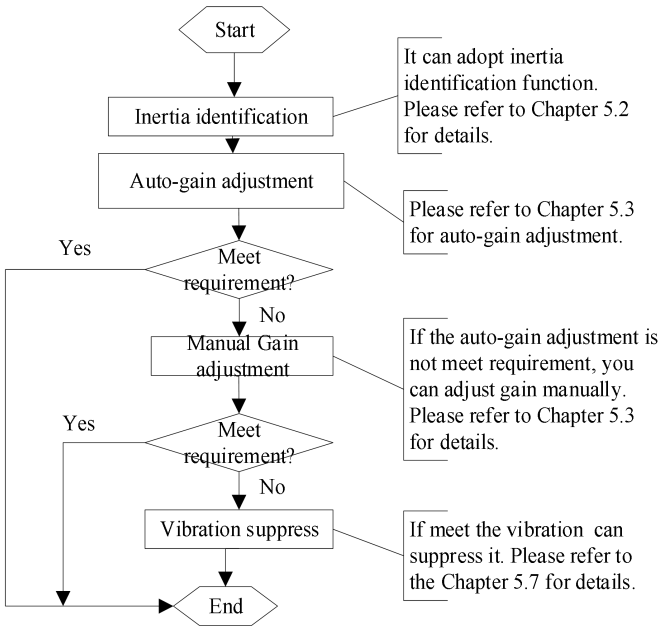


Figure 5.2 Gain adjustment process

5.2 Inertia Identification

Load inertia ratio= $\frac{\text{mechanical load ttotal rotation inertia}}{\text{Motor}-\text{self rotation inertia}}$

Load inertia ratio is the important parameter of servo system, correctly set load inertia ratio is benefit for quickly complete test. The load inertia ratio can be set manually or automatically identified through the inertia identification function of the servo drive.

Use the “Rotating inertia identification function (P11.03)” to realize inertia identification by operating the function code of servo drive host computer to rotate the motor.

Note:

To use the inertia identification function, in order to accurately calculate the load inertia ratio, the following conditions must be met:quirements:

• The actual motor maximum rotation speed is up to 200rpm.

• When the actual motor accelerates and decelerates, the acceleration is above 3000rpm/s.

• The load torque is relatively stable and cannot change strenuously.

• The actual load inertia ratio doea not exceed 120 times.

If the actual load inertia ratio is large and the drive gain is low, the motor will move slowly and cannot meet the motor's maximum speed and accerlation requirements. In this case, the speed loop gain (P07.01) can be increased and the inertia identification can be performed again.

If vibration occurs during the identification process, the inertia identification should be stopped immediately and the gain should be reduced. In addition, when the backlash of the transmission mechanism is large, the inertia identification may fail. Before performing the offline inertia identification, first comfirm the following:

1) The motor movable process should meet the requirement bellowing

There is a movable stroke of more than 1 turn in the forward and reverse directions between the mechanical limit switches: Before performing the offline inertia identification, please make sure that a limit switch is installed on the machine and sure the motor has a movable stroke of more than 1 turn in the forward and reverse directions to prevent overtravel during the inertia identification process and cause an accident!

2) Estimated load inertia ratio P00.05 value

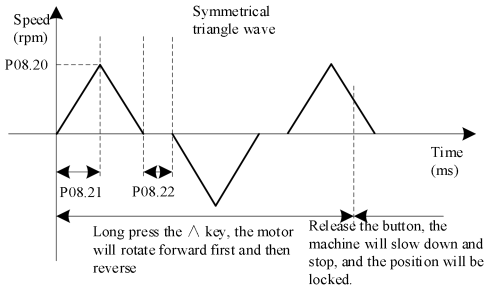
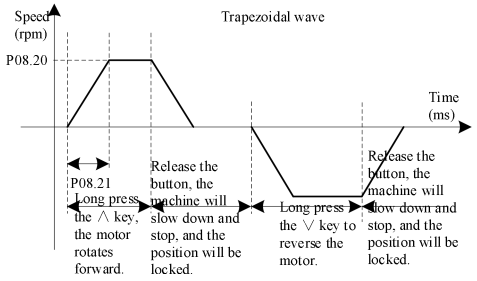
If P00.05 is the default value (1.00) and the actual load inertia ratio is greater than 30.00, the motor may move slowly causing identification failure. In this case, the following two measures can be taken:

a) The preset P00.05 is a larger initial value: it is recommended to use 5.00 times as the starting value for the preset value, and gradually increase it until the panel display value is updated during the identification process.

b) Appropriately increase the drive rigidity level (P00.04) so that the actual motor speed can reach the maximum speed of inertia identification (P08.20).

Inertia identification is divided into positive and negative triangle wave mode and JOG mode, and the motor rotation needs to be realized through the host computer function code.

Table5.1 Inertia identification method

Project	Positive and negative triangle wave form (P08.23=0)	JOG mode (P08.23=1)
Instruction form		
Max speed	P08.20	P08.20
Acceleration and deceleration time	P08.21	P08.21
Interval time	P08.22	The time interval between two key operations
Motor rotation times	View P08.24	Human Control
Button Description	Long press the inertia identification forward button: the motor first rotates forward and then reverses Long press the inertia identification reverse button: the motor will first reverse and then forward Release the button: stop at zero speed and keep the position locked	Long press the inertia identification forward button: the motor runs forward Long press the inertia identification reverse button: the motor reverses Release the button: zero speed stop, keep the position locked
Applicable occasions	When the motor stroke is short.	The motor has a long stroke and can be manually controlled.

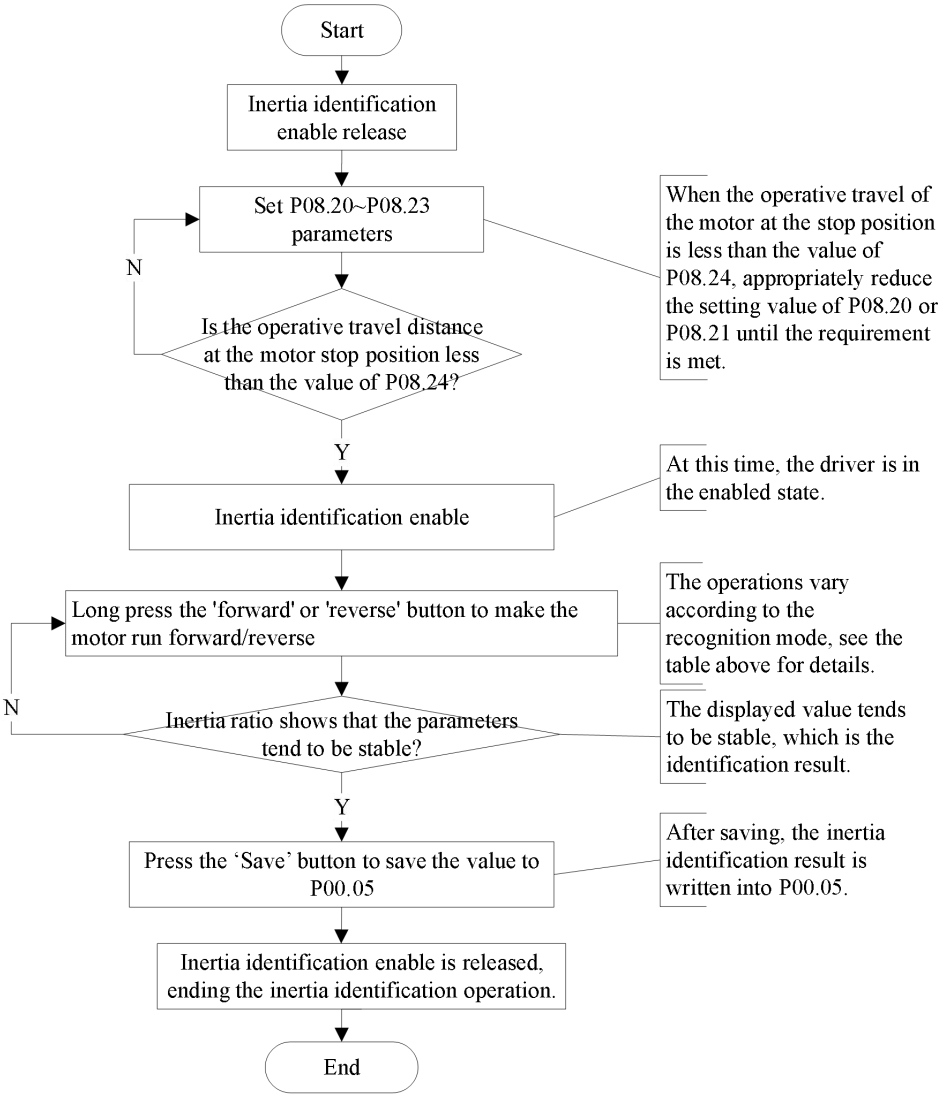


Figure 5.3 Inertia identification process

☆Related function code:

Function code	Name	Setting range	Minimum unit	Function	Factory setting	Effective time	Setting method
P08.20	Inertia identification maximum speed	200~1000	1rpm	Set the maximum speed command for inertia identification	500	Effective immediately	Shutdown setting
P08.21	Inertia identification acceleration and deceleration time	50~800	1 ms	Set the time for the motor to accelerate from 0rpm to the maximum speed of inertia identification (P08.20) during inertia identification.	100	Effective immediately	Shutdown setting

5.3 Gain adjustment

5.3.1 PI Parameters Adjustment PI

When the automatic gain can not achieve the expected result, you can manually minute adjust the gain. To optimize the result through more detailed adjustment. The servo system is composed of three control loops, from outside to inside, they are position loop, speed loop and current loop, the basic control block diagram is shown in figure 5.4.

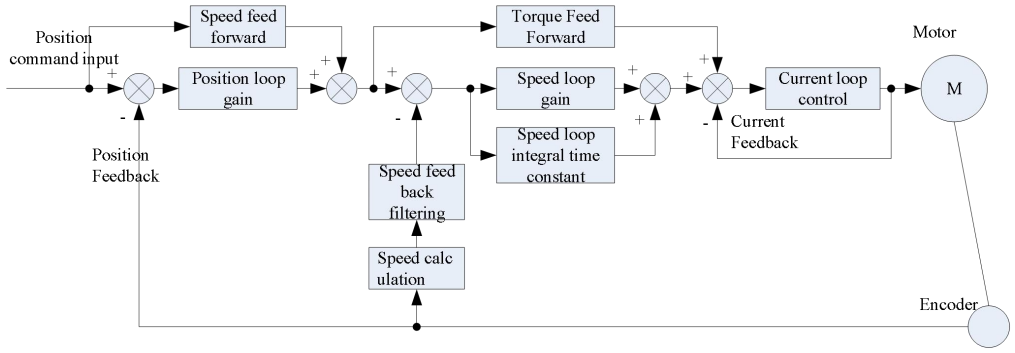
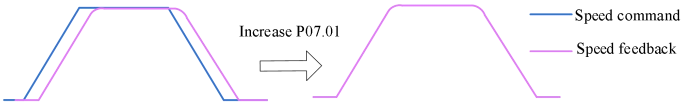
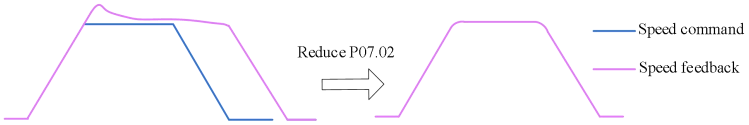
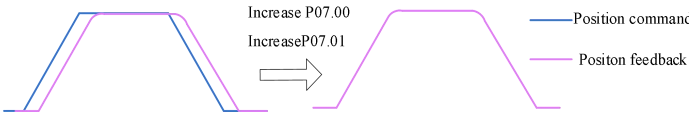


Figure 5.4 Manual Gain Basic Description Diagram

The default current loop gain of servo drive is already ensured the sufficient responsiveness, generally does not need to be adjusted, only position loop gain, speed loop gain and other auxiliary gain need to be adjusted. Therefore, when adjust gain in position control mode, in order to ensure system stability, while increasing the position loop gain, it is necessary to improve speed gain, and ensure that the response of position loop is lower than speed loop, the basic gain parameters adjustment method are as follows.

Table 5.2 Gain adjustment method

Steps	Function code	Name	Adjustment Description
1	P07.01	Speed loop gain	◆Parameters function: Determines the maximum frequency of changing speed commands that the speed loop can follow. Under the premise that the load inertia ratio (P00.05) is set correctly,it can be considered that the maximum follow frequencyof speed loop=P07.01

			 ◆Adjustment method: Within the range where noise and vibration do not occurs, increasing this parameter can accelerate positioning time and bring better speed stability and followability. If noise occurs, reduce the parameter setting value.
2	P07.02	Speed loop integral time	◆Parameters function: Eliminate speed loop deviation.  ◆Adjustment method: It is recommend to take the value from the following relationship: 500 ≤P07.01×P07.02 ≤ 1000 For example,when the speed loop gain P07.01=40.0Hz, the speed loop integrated time constant should meet: 12.50ms ≤P07.02 ≤ 25.00ms。 Decreasing the set value can strengthen the integral effect and speed up the positioning time, but a set value that is too small can easily cause mechanical vibration. If the setting value is too high, the speed loop deviation will never be returned to zero. When P07.02=512.00ms，integral is invalid.
3	P07.00	Position loop gain	◆Parameters function: Determines the maximum frequency of position instruction that the position loop can follow. Position loop maximum following frequency = P07.00  ◆Adjustment method: In order to ensure system stability, it should be ensured that the maximum following frequency of speed loop is 3-5 times the maximum following frequency of position loop. Therefore: $3 \le \frac{2\pi \cdot \mathrm{P07.01}}{\mathrm{P07.00}} \le 5$ For example, when the speed loop gain P07.01=40.Hz, positon loop gain should meet: 50.2Hz ≤ P07.00≤ 83.7Hz。 According the position time to adjust, increasing this parameter can accelerate positioning time and improve the ability of motor to resist external disturbance when it is stationary. If the setting value is too high may causing systemunstable and occurs vibration.

4	P07.04	Torque command filtering	◆Parameters function: Eliminate high-frequency noise and suppress mechanical resonance.  ◆Adjustment method: It should be ensured that the cutoff frequency of the torque instruction filter is higher than 4 times highest following frequency of speed loop, therefore: $$\frac{1000}{2\pi \cdot \mathrm{P07.04}} \geq (\mathrm{P07.01}) \cdot 4$$ For example, when the speed loop gain P07.01=40.0Hz, the torque instruction filter c=time constant should meet: P07.04 ≤ 1.00ms。 When vibration occurs when P07.01 is increased, the vibaration can be suppressed by P07.04; If the setting value is too high, it will cause the responsivemess decrease of current loop. The vibration in motor shutdown status need to be suppressed, you can try to increase P07.01, decrese P07.04; The vibration of the motor is too large when it is stopped. You can try to reduce the setting value of P07.04.

5.3.2 Speed Feedforward Gain

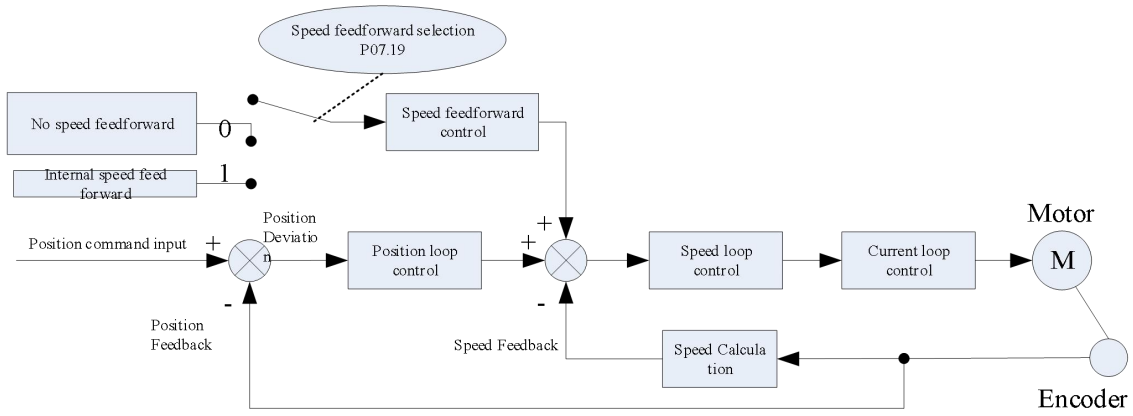


Figure 5.5 Speed feedforward control operation diagram

The velocity forward is only applicable to position control mode. Using velocity feedforward function can improve response of speed command, reduce the position deviation at a fixed speed.

The operation steps of speed feedforward:

a）Setting speed feedforward signal source

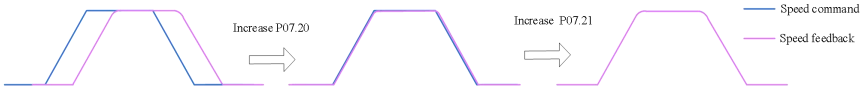
Set P07.19 (Speed feedforward control selection) to a non-0 value, the speed feedforward function takes effect and the corresponding signal source is selected.

Function code	Name	Setting value	Remark
P07.19	Velocity feedforward control selection	0- No speed feedforward	-
		1-Internal speed feedforward	Use the speed information corresponding to the position command (encoder unit) as the source of the speed feedforward signal.
		4- EtherCTASpeed feedforward EtherCTA	Use the 60B1 input from the master station as speed feedforward.

b）Set the speed feedforward parameters

Including the speed feedforward gain (P07.20) and speed feedforward filter time constant (P07.21).

Table 5.3 Speed feedforward adjustment

Function code	Name	Adjustment description
P07.20	Speed feefforward gain	 ◆Parameters function： Increasing P07.20 can improve response, but may generate speed overshoot during acceleration and deceleration;
P07.21	Speed feedforward filter time constant	Reducing P07.21, can suppress the speed overshoot occur in acceleration and deceleration; Increasing P07.21 can suppress the noise caused by the position command update cycle being longer than the drive control cycle, the pulse frequency of the position command being uneven, and suppressing the jitter of the positioning completion signal; ◆Adjustment method: First, setting P07.21 to a fixed value; then gradually increase P07.20 from zero start, until the speed feedforward takes effect at a certain setting value. When adjusting, P07.20 and P07.21 should be adjusted repeatedly the to find a setting with good balance.

5.4 Torque feedforward

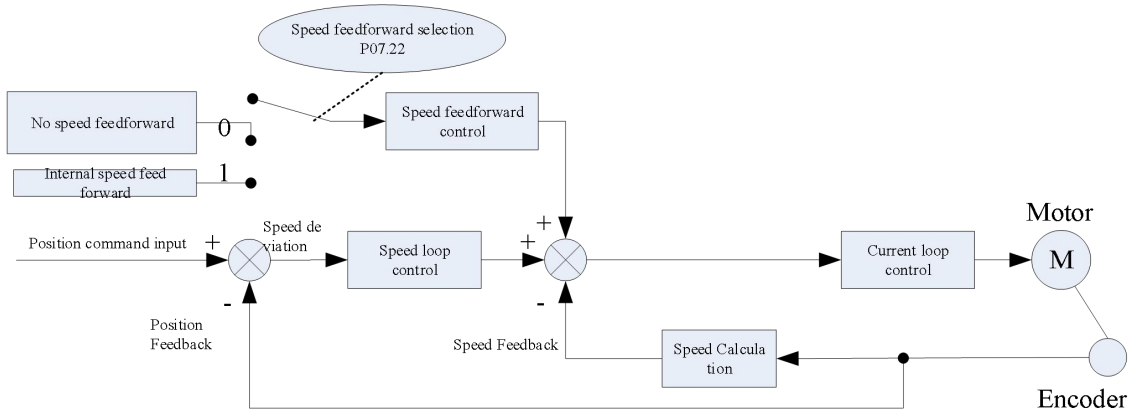


Figure 5.6 Torque feedforward control operation diagram

Only applicable to non-torque control mode situations. The position control mode uses torque feedforward, which can improve the torque command response and reduce the position deviation at fixed acceleration and deceleration; the speed control mode uses torque feedforward, which can improve the torque command response and reduce the speed deviation at fixed speed.

Torque feedforward function operation steps:

a）Setting torque feeforward signal source

Set P07.22 (Torque feedforward control selection) to a non-0 value, the torque feedforward function takes effect and the corresponding signal source is selected.

Function code	Name	Setting value	Remark
P07.22	Torque feedforward control selection	0- No torque feedforward	-
		1- Internal torque feedforward	Using the acceleration information correspondingto speed instruction as the source of the torque feedforward signal. In position control mode, the speed instructioncomes from the output of the position controller.

b）Setting torque feedforward parameters

Function code	Name	Adjustment description
P07.23	Torque feedforward gain	◆Parameters function: Increasing the P07.23 can improve the response, but may occurs overshoot when accelerate and decelerate; Reducing the P07.24 can suppress the overshoot occurs in accelerate and decelerate; Increasing P07.24 can suppress the noise; ◆Adjustment method: When adjusting, first keep P07.24 as the default value; then gradually increase the setting value of P07.23 from 0 until the torque feedforward takes effect at a certain setting value. When adjusting, P07.23 and P07.24 should be adjusted repeatedly to find a setting with good balance.
P07.24	Torque feedforwardfilter time constant	

5.5 Command filter adjustment

5.5.1 Position command smoothing filter

Setting the time constant of the once delay filter for the position command.

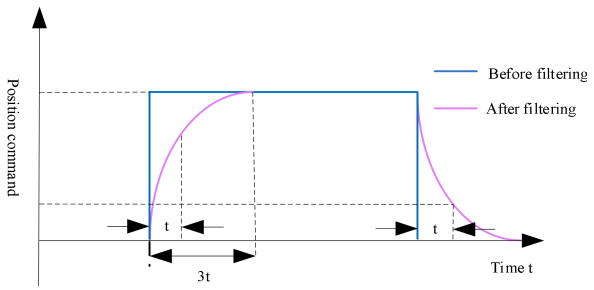


Figure 5.7 Position command smoothing filter diagram

☆Related function code:

Function code		Name	Setting range	Minimum unit	Factory setting	Effective time	Category	Mode
P04	03	Position command smoothing filter	0～65535	0.1ms	0	Effective immediately	Shutdown setting	P

5.5.2 Position command FIR filter

Setting the time constant of FIR filter for position command.

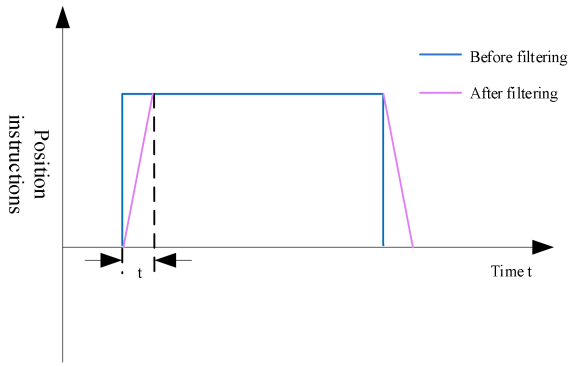


Figure 5.8 Position command FIR filter diagram

Function code		Name	Setting range	Minimum unit	Factory setting	Effective time	Category	Mode
P04	04	Position command FIR filter	0.～1280	0.1ms	0	Effective immediately	Shutdown setting	P

5.6 Adjustment parameters in different mode

Parameters adjustment under different control mode must follow the consequence of “Inertia identification” ==> “Automatic Gain Adjustment” ==> “Manual Gain Adjustment”.

5.6.1 Adjustment parameters in position mode

1）Through inertia identification, obtain load inertia ratio P00.05:

2）The gain parameter in position mode:

①The first gain:

Function code	Name	Function	Default value
P07.00	Position loop gain	Set position loop proportional gain	48.0Hz
P07.01	Speed loop gain	Set speed loop proportional gain	50.0Hz

P07.02	Speed loop integral time constant	Set the integral time constant of the speed loop	12.00ms
P07.04	Torque command filter time constant	Set the torque command filter time constant	1.26ms

②The second gain:

Function code	Name	Function	Default value
P07.05	Set the integral time constant of the speed loop	Set position loop proportional gain	38.0Hz
P07.06	The second speed loop gain	Set speed loop proportional gain	18.0Hz
P07.07	The second speed loop integral time constant	Set the integral time constant of the speed loop	512.00ms
P07.09	The second torque command filter time constant	Set the torque command filter time constant	1.26ms
P07.10	DI function GAIN-SWITCH switching action selection	Set the GAIN-SWITCH switching action selection	0
P07.11	Gain switching mode	Set the gain switching condition	0
P07.12	Gain switching delay	Set the delay time for gain switching	5.0ms
P07.13	Gain switching level	Sets the level at which gain switches	50
P07.14	Gain switching hysteresis	Set the hysteresis of gain switching	30
P07.15	Position gain switching time	Set the switching time of the position loop gain	3.0ms

Public Gain:

Function code	Name	Function	Default value
P07.03	Speed feedback filtering	Set the speed feedback filter time	0.00ms
P07.16	Pseudo-differential feedforward control coefficient	Setting the coefficients of the PDFF controller	100.0%
P07.20	Speed feed forward gain	Set the speed feedforward gain	0.0%
P07.21	Speed feedforward filter time constant	Set the filter time constant of the speed feedforward signal	0.50ms
P07.23	Torque feed forward gain	Setting the torque feedforward gain	0.0%
P07.24	Torque feedforward filter time constant	Set the filter time constant of the torque feedforward signal	0.50ms

3) By setting the rigidity level of P00.04 and automatic adjustment gain, obtain the first gain (or second gain). If the rigid connection equipment of the lead screw and rack encounters whistling when the rigidity level is increasing, can open the instantaneous speed observer (P08.39), in order to improve rigidity level.

4）Mannually fine-tune the following gains, if there are problems such as mechanical whistling, mechanical shaking, and tip jitter, you can use notch and model compensation control to optimize:

Function code	Name	Function
P07.00	Position loop gain	Set the position loop proportional gain
P07.01	Speed loop gain	Set the speed loop proportional gain
P07.02	Speed loop integral time constant	Set the integral time constant of the speed loop
P07.04	Torque command filter time constant	Set the torque command filter time constant

P07.20	Speed feedforward gain	Set the speed feedforward gain
P07.21	Speed feedforward filter time	Set the speed feedforward command filter

5.6.2 Parameters adjustment in the speed mode

The parameters adjustment in speed control mode is same as that in position mode, except for the position loop gain (P07.00, P07.05), please adjust according to 5.6.1 “Parameters adjustment in position mode”.

5.6.3 Parameters adjustment in torque control mode

Parameters adjustment in torque control mode:

When the actual speed reaches the speed limit value, the adjustment method is the same as 6.7.2 "Parameter adjustment in speed mode";

When the actual speed do not reaches the speed limit value, except for speed loop gain(P07.01, P07.06) and speed loop integral time constant (P07.02, P07.07), the adjustment method is same as 6.5.2 “Parameter adjustment in speed mode”.

5.7 Mechanical resonance suppression

Mechanical system has a certain resonance frequency. When the servo gain is increased, resonance may occur near the mechanical resonance frequency, causing the gain to be unable to continue to increase. There are two ways to suppress mechanical resonance:

1）Torque command filter (P07.04，P07.09)

By setting filter time constant, the torque command is attenuated in high-frequency above the cutoff frequency to achieve the purpose of suppressing mechanical resonance.

2）Notch filter:

The notch filter can achieve the purpose of suppressing mechanical resonance by reducing the gain at specific frequencies. After correctly set notch filter, the vibration can get effectively suppress, and try to

increase servo gain continuously. The primciple of Notch filter is as shown below.

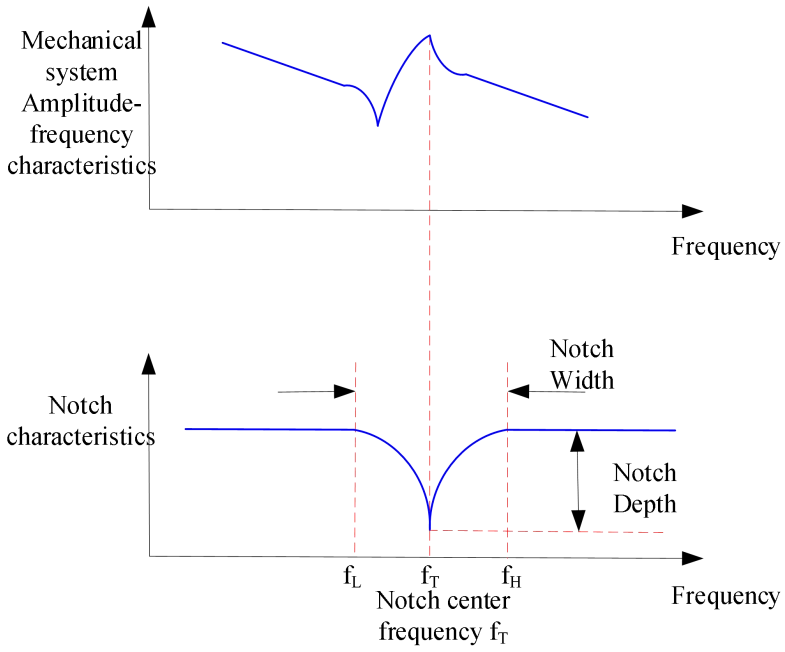


Figure 5.9 Notch filter principle

The servo drive has two sets of notches, each of them have 3 parameters, divided into Notch filter frequency, width level and depth level. This notch is a manual notch, and each parameter is set manually by the user.

Notch filter width and depth

The notch width is used to express the ratio of the notch width to notch center frequency:

$$\text{The notch width} = \frac{f_H - f_L}{f_T}$$

Among:

f_T：Notch center frequency, that is to say, mechanical resonance frequency

f_H-f_L：The notch width represents the frequency bandwidth with an amplitude attenuation rate of -3dB relative to the notch center frequency.

The notch depth level represents the relationship of the ratio of the input to output at center frequency. When the notch depth level is 0, the input is completely be suppressed at center frequency;

When the notch depth level is 100, the input is completely pass at center frequency. Therefore, the setting level of notch depth is smaller, the depth of notch is more deeper, the suppression to mechanical resonance is more stronger, but this may cause unstable system, should be attention when using.

Table 5.4 Parameters setting of notch

Notch width	Bandwidth/Center frequency		Notch depth	Ratio of input to output	[dB] Gain Represent
0	0.5		0	0	-∞
1	0.59		1	0.01	-40
2	0.71		2	0.02	-34
3	0.84		3	0.03	-30.5
4	1		4	0.04	-28
5	1.19		5	0.05	-26
6	1.41		6	0.06	-24.4
7	1.68		7	0.07	-23.1
8	2		8	0.08	-21.9
			9	0.09	-20.9
			10	0.1	-20
			15	0.15	-16.5
			20	0.2	-14
			25	0.25	-12
			30	0.3	-10.5

			35	0.35	-9.1
			40	0.4	-8
			45	0.45	-6.9
			50	0.5	-6
			60	0.6	-4.4
			70	0.7	-3.1
			80	0.8	-1.9
			90	0.9	-0.9
			100	1	0

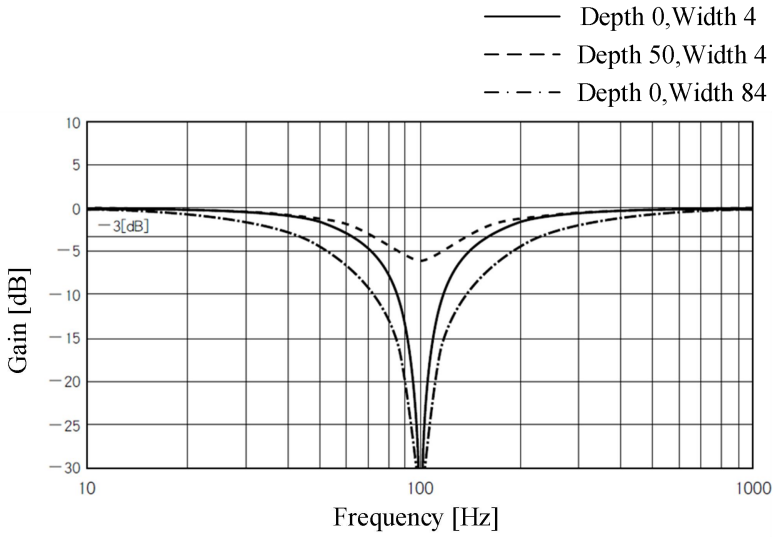


Figure 5.9 Notch frequency characteristic

Notch using steps

① Analyze resonance frequency;

When using a manual notch, you need to set the frenquency of the notch to the actual resonant

frequency that occurs. The method to obtain resonance frequency: By setting P08.00=3, the resonant frequency is automatically tested when the servo is running, and the test results are saved in P08.01. After the test is completed, be sure to set P08.00 to 0.

② Enter the of Enter the resonant frequency obtained in step ① into the notch parameters of the selected group, and enter the width level and depth level of the notch group.

③ If the resonance is suppressed, it means the notch is effective, you can continue to adjust the gain. After the gain is increased, if occours new resonance, you can repeat steps ① ~ ②;

④If the vibration can not be eliminated in a long time, please close the servo function in time.

☆Related function code:

Function code		Name	Setting range	Minimum unit	Factory setting	Effective time	Category	Mode
P08	00	Adaptive filter mode	0~5	1	0	Effective immediately	Running setting	PST
P08	01	Resonance frequency	-	1Hz	0	N/A	Display parampara	PST
P08	02	1st notch filter frequency (manual)	10~4000	1Hz	4000	Effective immediately	Running setting	PST
P08	03	1st notch filter width	0~8	1	2	Effective immediately	Running setting	PST
P08	04	1st notch depth	0~100	1	50	Effective immediately	Running setting	PST
P08	05	2nd notch filter frequency (manual)	10~4000	1Hz	4000	Effective immediately	Running setting	PST
P08	06	2nd notch filter width	0~8	1	2	Effective immediately	Running setting	PST
P08	07	2nd notch depth	0~100	1	50	Effective immediately	Running setting	PST

Chapter 6 Alarm and Processing

Servo drive alarm level is divided into two levels

Table 6-1 Alarm level

Alarm level	Name	Decription
Level 1	Fault	The servo drive has a serious alarm and cannot work normally. It needs to be shut down to cheak. DO terminal outputs ALM singal.
Level 2	Warning	The servo drive warns that the device will not be damaged for the time being, but if it is not handled correctly in time, it may cause a high level fault output. DO terminal outputs WARN singal.

6.1 Fault diagnosis and treatment

Faults can be divided into:

a) Non-resettable NO.1 fault;

b) Resettable NO.1 fault;

c) Resettable NO.2 fault.

Among them, “Resettable” means that after troubleshooting, P11-01 can be set to 1or the DI function can be configured FunIN.2：ALM-RST alarm reset to clear the servo fault state.

“Non-resettable” means that after the fault is handled, drive needs to be re-powered.

The NO.1 and NO.2 faults have different stop modes. The NO.1 fault is free to stop and keeps running freely. The NO.2 fault stop mode is set by P00-12.

The reset method of NO.1, NO2 resettable fault: first turn off the servo enable signal (S-ON is set to OFF), then set P11-01=1or use DI function 2.

6.1.1 Non-resettable NO.1 fault

Table 6-2 Non-resettable NO.1 fault

Fault number Er_	Fault name	Cause of fault	Stop method when fault occurs	Whether the fault can be reset
1	System parameter exception	Abnormal data of servo internal parameters	No.1	NO
2	Product model selection failure	Invalid drive model	No.1	NO
3	Motor model selection failure	P01.00 Function code Set an invalid motor code	No.1	NO
4	Fault in parameter storage	1. Parameter storage device failure 2. Parameters read and write too frequently 3. Control power supply instability 4. Drive failure	No.1	NO
5	FPGA fault	1. FPGA initialization exception 2. FPGA logic version is abnormal 3. FPGA detected an exception	No.1	NO
6	Encoder matching fault	In absolute value system mode (P00.06 ≠0), an incremental or single-turn absolute value encoder motor is used	No.1	NO
7	Control power supply undervoltage	Control circuit power supply undervoltage	No.1	NO
9	Overcurrent A	1. The drive or motor parameters are incorrect 2. UVW phase-to-phase short circuit 3. Motor burned out 4. Motor is shorted to ground 5. Drive failure	NO.1	NO
10	Overcurrent B	1. Servo motor wiring is not normal 2. Software detects the overcurrent of power transistor	No.1	NO
11	Encoder disconnection	Incremental encoder A/B/Z three-phase line break	No.1	NO
13	Encoder check exception	1.The bus encoder data check is abnormal 2.The incremental encoder zero check is abnormal	No.1	NO
14	Motor initial angle detection abnormal	The encoder signal interferes when the incremental encoder is powered on	No.1	NO
15	Motor out of control	1. Motor UVW phase sequence setting error 2. UVW wiring error 3. Initial position or encoder parameter setting error	No.1	NO
16	Current sampling fault	The current A/D sampling circuit is faulty	No.1	NO
18	Motor data verification failure	1. No motor parameters are written into the encoder EEPROM 2. Motor parameter verification error	No.1	NO

6.1.2 Resettable NO.1 fault

Table 6-3 Resettable NO.1 fault

Fault number Er_	Fault name	Cause of fault	Stop method when fault occurs	Whether the fault can be reset
20	Overvoltage	The main circuit DC voltage abnormally high.	No.1	Yes
21	Undervoltage	The main circuit DC voltage low.	No.1	Yes
22	Overspeed	1. Speed command exceeds the maximum speed value 2. UVW phase sequence error 3. Speed response too high, severely exceed adjustment range 4. Drive failure	No.1	Yes
25	Abnormal motor rotation during bootstrap charging	Abnormal motor rotation during bootstrap charging	No.1	Yes
27	DI terminal parameter setting fault	Different DI are assigned with same function repeatedly.	No.1	Yes
28	DO terminal parameter setting fault	Different DO are assigned with same output.	No.1	Yes
30	Reference position failure	When using the PTP function, P04.00=5 is not set.	No.1	Yes

6.1.3 Resettable NO.2 fault

Table 6-4 Resettable NO.2 fault

Fault number Er_	Fault name	Cause of fault	Stop method when fault occurs	Whether the fault can be reset
43	Position deviation is too large	In the servo ON state, the position deviation exceeds the position deviation value (P09.09)	No.2	Yes
46	Drive overload	1.The load operation exceeds the inverse time curve of the drive. 2.The UVW output may be lack of phase or phase sequence connected incorrectly.	No.2	Yes
47	Motor overload	1. The load operation exceeds the inverse time curve of the motor. 2.The UVW output may be lack of phase or phase sequence connected incorrectly.	N0.2	Yes
48	Motor stall	1. The mechanical position is stuck, causing the motor current to continuously increase abnormally.	N0.2	Yes

Fault number Er_	Fault name	Cause of fault	Stop method when fault occurs	Whether the fault can be reset
		2. Inconsistent response of dual-drive motors with gantry structure.		
49	Electronic gear setting error	Electronic gear ratio exceeds specification range [0.001, 4000]	No.2	Yes
50	Heat sink (fan) overheating	Servo unit heat sink (fan) exceeds the set fault value	No.2	Yes
51	Encoder battery failure	No battery or battery voltage below 2.6V	No.2	Yes
52	Encoder multi-turn count error	Absolute encoder multi-turn count error	No.2	Yes
53	Encoder multi-turn count overflow	Absolute encoder multi-turn count overflow	No.2	Yes
54	Soft limit setting error	1. The position between positive/negative limit is too short 2. When the motor runs in the counterclockwise direction as the positive direction, the positive limit value is smaller than the negative limit value 3. When the motor runs in the clockwise direction as the positive direction, the negative limit value is smaller than the positive limit value	No.2	Yes

6.2 Reasons for warning and treatment

Table 6-5 Reasons for warning and treatment

Fault number Er_	Warning name	Cause of warning
81	Drive overload warning	Warning when 80% of the drive overload limit value is reached.
82	Motor overload warning	The warning before the motor is about to fail, the warning value is determined by P09-05.
83	Changed parameters need to be re-powered to take effect	Changed parameters that need to be re-powered.
84	Reset encoder warning	In enabled state, resets encoder when host computer sending commands.
86	Forward overtravel warning	Positive overtravel switch POT terminal is valid. The motor position exceeds the maximum soft limit.
87	Negative overtravel warning	Negative overtravel switch NOT terminal is valid. The motor position exceeds the soft limit minimum.

88	Divided pulse output setting fault	The number of encoder divided pulses does not meet set condition or range.
89	AI1\ AI2 zero drift too large	AI1\ AI2 zero drift too large.
90	External regenerative resistor overload	External regenerative resistor power is too small.
91	External regenerative resistor value is too small	The external regenerative resistor is less than the minimum requirement by the drive; Or parameter setting error.
94	DI emergency brake	External emergency brake E_STOP terminal trigger.
95	Absolute encoder battery low	Battery voltage is lower than 3.2V.
96	Homing timeout	1. Home switch failure. 2. Time limit to find home is too short. 3. The high speed for searching home signal is too small.
97	Mechanical home point offset error	1. When the homing mode parameter P16-09=6 or P16-09=8 or P16-09=14, the mechanical home offset parameter P16-14 is set to a value greater than 0. 2.When the homing mode parameter P16-09=7 or P16-09=9 or P16-09=15, the mechanical home offset parameter P16-14 is set to less than 0.
99	Multi-segment position reference position fault	1. In the multi-segment absolute position operation mode, the system does not perform the homing operation or the drive is not set to the absolute value system. 2. In the multi-segment incremental position operation mode, when the function code of P13.07 is set to 1, the system does not perform the zero return operation.
100	The homeing mode is incorrectly set	The homeing mode is set 0, Or the mode is not supported.

Note: The reset method of warning: set P11-01=1 or use DI function 2.

Chapter 7 Parameter List

7.1 Parameter group number

Parameter group number	Parameter group description
P00	Basic control parameters
P01	Servo motor parameters
P02	Digital input and output parameters
P04	Position control parameters
P05	Speed control parameters
P06	Torque control parameters
P07	Gain parameters
P08	Advanced adjustment parameters
P09	Fault and protection parameters
P10	Communication parameters
P11	Auxiliary function parameters
P12	Keyboard display parameters
P13	Multi-segment position function parameters
P14	Multi-speed function parameters
P16	Special function parameters
P17	Drive parameters
P18	Display parameters

7.2 Each group of parameters

P00 group: basic control parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P00	00	Control mode selection	0-position mode 1-speed mode 2-torque mode	1	0	Effectiveimmediately	Setting after motor stop	PST
P00	01	Direction of rotation	0-CCW direction is the forward direction (counterclockwise) 1-CW direction is the forward direction (clockwise)	1	0	Re-power	Setting after motor stop	PST
P00	02	Pulse output positive direction definition	0-CCW direction is forward direction (the corresponding motor rotation direction when the pulse output OA leads the OB) 1- CW direction is forward direction (reverse mode, OA lag OB)	1	0	Re-power	Setting after motor stop	PST
P00	04	Rigidity level setting	0~31	1	11	Effective immediately	Set when running	PST
P00	05	Inertia ratio	0~3000	0.01	100	Effective immediately	Set when running	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P00	06	Absolute value system selection	0-incremental position mode 1-Absolute position linear mode （If homing completed when the absolute value encoder is used, power on the encoder immediately. If the value of 6064 is not 0 after the encoder is powered on, set the absolute value mode to this mode.） 2- Absolute value position rotation mode （Mask multi-circle data overflow errors, In the multi-turn data -32768~32767 has been running cycle.）	1	0	Re-power	Setting after motor stop	PST
P00	07	System maximum speed	0~10000	1rpm	4500	Effective immediately	Setting after motor stop	PST
P00	10	Servo OFF stop mode	0- Freewheel stop, keep free running state 1- Stop at zero speed (deceleration time is controlled by P05.06), keep free running state	1	1	Effective immediately	Setting after motor stop	PST
P00	11	Fault No.1 stop mode selection	0- Freewheel stop, keep free running state 1-Resaved	1	0	Effective immediately	Setting after motor stop	PST
P00	12	Fault No.2 stop mode selection	0- Freewheel stop, keep free running state 1- Stop at zero speed (deceleration time is controlled by P05.06), keep free running state	1	1	Effective immediately	Setting after motor stop	PST
P00	13	Stop mode when overtravel	0-free running stop. 1- Decelerate the motor with the set torque of the emergency stop torque as the maximum value, and then enter the servo lock state. 2- Decelerate the motor with the set torque of the emergency stop torque as the maximum value, and then enter free running state.	1	1	Effective immediately	Setting after motor stop	PST
P00	14	Brake output ON to command receiving delay	0～10000	1ms	200	Effective immediately	Set when running	PST
P00	15	In the rotation state, No.2 fault occurs, or the servo enable is OFF, the brake output OFF delay; In the static state,brake output OFF to the motor not energized delay.	10～10000	1ms	200	Effective immediately	Set when running	PST
P00	16	Rotating state, the speed threshold when the brake output is OFF	0～1000	1rpm	50	Effective immediately	Set when running	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P00	17	NO.1 fault occurs in rotating state , delay between servo enable OFF and brake output OFF	0～10000	1ms	500	Effective immediately	Set when running	PST
P00	18	Energy consumption resistance setting	0-Use built-in power consumption resistor 1-Use external power-consuming resistors and cool naturally 2-Use external power-consuming resistors and force air-cooled 3- No energy consumption resistor, relying on capacitance Absorption (The brake tube is always closed at this time)	1	0	Effective immediately	Setting after motor stop	PST
P00	19	External resistor power capacity	1～65535	1W	Model parameters	Effective immediately	Setting after motor stop	PST
P00	20	External resistance value	1～1000	1Ω	Model parameters	Effective immediately	Setting after motor stop	PST
P00	21	External resistance heating time constant	1000~65535	1ms	Model parameters	Effective immediately	Setting after motor stop	PST
P00	22	Energy consumption braking start voltage	0~1000	1V	Model parameters	Effective immediately	Set when running	PST
P00	37	Pulse increment threshold	0～200	1	1	Effective immediately	Set when running	PS
P00	38	Continuous pulseless reception cycle number	1～200	1	3	Effective immediately	Set when running	PS

P01 group: Servo motor parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P01	00	Motor model code	0~65535	1	101	Re-power	Setting after motor stop	PST
P01	01	Motor power line phase sequence direction	0：CCW (The opposite electromotive force U is ahead of V) 0：CCW (The opposite electromotive force V is ahead of U)	1	0	Re-power	Setting after motor stop	PST
P01	02	Rated voltage	1-1000	1	48	Re-power	Setting after motor stop	PST
P01	03	Rated power	0~65535	0.01Kw		Re-power	Setting after motor stop	PST
P01	04	Rated current	1～10000	0.01A		Re-power	Setting after motor stop	PST
P01	05	Rated torque	0~65535	0.01Nm		Re-power	Setting after motor stop	PST
P01	08	Max speed	0~9000	1rpm		Re-power	Setting after motor stop	PST
P01	09	Rotor inertia	0~10000	0.01 $kgcm^2$		Re-power	Setting after motor stop	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P01	10	Number of pole pairs of Permanent magnet synchronous motor	1~50	1 pole pair		Re-power	Setting after motor stop	PST
P01	11	stator resistance Rs	1~65535	0.001Ω		Re-power	Setting after motor stop	PST
P01	12	Q-axis inductance Lq	1~65535	0.01mH		Re-power	Setting after motor stop	PST
P01	13	D-axis inductance Ld	1~65535	0.01mH		Re-power	Setting after motor stop	PST
P01	14	Back EMF	1~65535	0.01 mV/rpm		Re-power	Setting after motor stop	PST
P01	15	Torque factor	1~65535	0.001 Nm/A		Re-power	Setting after motor stop	PST
P01	18	Encoder selection	0: 2500 lines encoder 1: 17bit incremental encoder 2: 17bit absolute encoder 3: 23bit incremental encoder 4: 23bit absolute encoder			Re-power	Setting after motor stop	PST
P01	20	Encoder resolution	1~1073741824			Re-power	Setting after motor stop	PST
P01	22	Z corresponding Electrical angle	0~3600	0.1°		Re-power	Setting after motor stop	PST
P01	23	U rising edge corresponding electrical angle	0~3600	0.1°		Re-power	Setting after motor stop	PST
P01	24	FPGA upload motor model	0～65535			Effective immediately	Only read parameter	PST

P02 group: digital terminal input and output parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P02	00	FunINL signal unassigned state (HEX)	0~0xFFFF Bit0- corresponds to FunIN.1; Bit1- corresponds to FunIN.2; ..... Bit15-corresponds to FunIN.16	1	0	Re-power	Set when running	PST
P02	01	DI1 terminal function selection	Input function code: 0, 1-45 0: no definition 1~45: FunIN.1 to 45 (refer to the DIDO basic function coding table)	1	1	Effective immediately	Set when running	PST
P02	02	DI2 terminal function selection	Input function code: 0, 1-45 0: no definition 1~45: FunIN.1 to 45 (refer to the DIDO basic function coding table)	1	2	Effective immediately	Set when running	PST
P02	03	DI3 terminal function selection	Input function code: 0, 1-45 0: no definition 1~45: FunIN.1 to 45 (refer to the DIDO basic function coding table)	1	10	Effective immediately	Set when running	PST
P02	10	FunINH signal unassigned state (HEX)	0~0xFFFF Bit0- corresponds to FunIN.17; Bit1- corresponds to FunIN.18; ..... Bit15-corresponds to FunIN.32	1	0	Re-power	Setting after motor stop	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P02	11	DI1 terminal logic selection	Input polarity: 0-4 0-low level effective 1-high level effective 2-rising edge is valid 3-falling edge is valid 4- rising and falling edges are valid	1	0	Effective immediately	Set when running	PST
P02	12	DI2 terminal logic selection	Input polarity: 0-4 0-low level effective 1-high level effective 2-rising edge is valid 3-falling edge is valid 4- rising and falling edges are valid	1	0	Effective immediately	Set when running	PST
P02	13	DI3 terminal logic selection	Input polarity: 0-4 0-low level effective 1-high level effective 2-rising edge is valid 3-falling edge is valid 4- rising and falling edges are valid	1	0	Effective immediately	Set when running	PST
P02	21	DO1 terminal function selection	Output code：1～25 0：no definition 1～25：FunOUT.1～25 refer to DIDO function selection code definition	1	1	Effective immediately	Setting after motor stop	PST
P02	22	DO2 terminal function selection	The DO can only be set to FunOUT11 (Lock release signal output)	1	11	Effective immediately	Setting after motor stop	PST
P02	31	DO1 terminal logic level selection	Output polarity inversion setting: 0-1 0: conduct when active (normally open) 1: not conduct when active (normally closed)	1	0	Effective immediately	Setting after motor stop	PST
P02	32	DO2 terminal logic level selection	Output polarity inversion setting: 0-1 0: conduct when active (normally open) 1: not conduct when active (normally closed)	1	0	Effective immediately	Setting after motor stop	PST

P04 group: position control parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P04	00	Main position command A source	0-low speed pulse command 1- high speed pulse command 2-step amount given 4-multi-segment position command given 5-given through communication	1	0	Effective immediately	Setting after Motor stop	P
P04	02	Step amount	-9999～9999	1Unit	50	Effective immediately	Setting after Motor stop	P
P04	03	Position command smoothing filter	0 ～65535	0.1ms	0	Effective immediately	Setting after Motor stop	P
P04	04	Position command FIR filter	0～1280	0.1ms	0	Effective immediately	Setting after motor stop	P
P04	05	Number of instruction unit required for one round (32-bit) (Only PTP)	16～1073741824	1Unit/Turn	0	Re-power	Setting after motor stop	P
P04	07	Electronic gear 1 numerator (32 bit)	1～1073741824	1	Motor resolution	Effective immediately	Set when running	P
P04	09	Electronic gear 1 denominator (32 bit)	1～1073741824	1	10000	Effective immediately	Set when running	P
P04	11	Electronic gear 2 numerator (32 bit)	1～1073741824	1	Motor resolution	Effective immediately	Set when running	P
P04	13	Electronic gear 2 denominator (32 bit)	1～1073741824	1	10000	Effective immediately	Set when running	P
P04	21	Pulse shape	0-pulse + direction, positive logic.(Defaults) 1-direction + pulse, negative logic 2-A phase + B phase orthogonal pulse, positive logic 3-A phase + B phase orthogonal pulse, negative logic 4- CCW+CW, positive logic 5- CCW+CW, negative logic	1	0	Re-power	Setting after motor stop	P
P04	22	Position deviation clear function	0- Clear position deviation pulse when fault occurs or servo OFF 1- Clear the position deviation pulse only when a fault occurs 2- Cleared by DI input function (PERR-CLR)	1	0	Effective immediately	Setting after motor stop	P
P04	23	Positioning complete (COIN) output	0-Output when the absolute position deviation value is smaller than positioning completion range. 1- Output when the absolute position deviation value is smaller than positioning completion range and the filtered position command is 0. 2- Output when the absolute position deviation value is smaller than positioning completion range and position command is 0.	1	0	Effective immediately	Setting after motor stop	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P04	24	Positioning completion range	1～65535	1P	2500 line motor: 7P 17-bit motor: 100P 23-bit motor: 1000P	Effective immediately	Setting after motor stop	P
P04	25	Positioning close range	1～65535	1P	65535	Effective immediately	Setting after motor stop	P

P05 group: speed control parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P05	00	Main speed command A source	0-digit value(P0503) 1- reserved 2- reserved	1	0	Effective immediately	Setting after motor stop	S
P05	01	Auxiliary speed command B source	0-digit value(P0503) 1- reserved 2- reserved 3-multi-speed command	1	3	Effective immediately	Setting after motor stop	S
P05	02	Speed command selection	0-main speed command A source 2-auxiliary speed command B source 3-A/B switching	1	0	Effective immediately	Setting after motor stop	S
P05	03	Speed command keyboard setting	-9000～9000	1rpm	200	Effective immediately	Setting when running	S
P05	04	Jog speed setting	0～9000	1rpm	200	Effective immediately	Setting when running	S
P05	05	Speed command acceleration time	0～10000	1ms	50	Effective immediately	Reserved	S
P05	06	Speed command deceleration time	0～10000	1ms	50	Effective immediately	Reserved	S
P05	07	Speed command limit selection	0-forward P05.08, reverse P05.09, internal limit (default) 1- reserved 2- reserved	1	0	Effective immediately	Setting when running	S
P05	08	Forward speed limit	0～9000	1rpm	6000	Effective immediately	Setting when running	S
P05	09	Backward speed limit	0～9000	1rpm	6000	Effective immediately	Setting when running	S
P05	14	Speed direction selection	0-direction unchanged 1-direction reversal 2-direction is determined by DI function 25 3-direction is determined by DI function 40/41	1	2	Effective immediately	Setting after motor stop	S
P05	15	Zero fixed speed fixed value	0～6000	1rpm	10	Effective immediately	Setting when running	S
P05	16	Speed threshold of motor running signal output	0～1000	1rpm	20	Effective immediately	Setting when running	PS
P05	17	Speed uniform signal width	0～100	1rpm	10	Effective immediately	Setting when running	PS
P05	18	Speed reaches the specified value	0～6000	1rpm	1000	Effective immediately	Setting when running	PST
P05	20	Zero speed judgment threshold	0～6000	1rpm	10	N/A	Setting when running	PST

P06 group: torque control parameters

Function code	Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P06	00	Main torque command A source	0-Digit given (P06-05) 1-Reserved	1	0	Effective immediately	Setting after motor stop	T
P06	02	Torque command selection	0-Main torque command A source 1-Auxiliary speed command B source 2- Command A+B source 3-A/B switching	1	0	Effective immediately	Setting after motor stop	T
P06	04	Torque command filter time in torque mode	0～65535	0.01ms	0	Effective immediately	Set when running	T
P06	05	Torque command keyboard setting	-3000～3000(based on rated motor torque)	0.1%	0	Effective immediately	Set when running	T
P06	06	Torque limit source	0-Positive and negative internal torque limit(default) 1-Positive and reverse external torque limit (selected with P_CL, N_CL)	1	0	Effective immediately	Set when running	PST
P06	08	Forward internal torque limit	0~5000 (based on rated motor torque）	0.1%	3000	Effective immediately	Set when running	PST
P06	09	Backward internal torque limit	0~5000 (based on rated motor torque）	0.1%	3000	Effective immediately	Set when running	PST
P06	10	Forward side external torque limit	0~5000 (based on rated motor torque）	0.1%	3000	Effective immediately	Set when running	PST
P06	11	Backward side external torque limit	0~5000 (based on rated motor torque）	0.1%	3000	Effective immediately	Set when running	PST
P06	13	Speed limit source selection for torque control	0-Internal speed limit (P06.15, P06.16 setting) 1-Reserved	1	0	Effective immediately	Set when running	T
P06	15	Positive speed limit During torque control	0～9000	1	3000	Effective immediately	Set when running	T
P06	16	Negetive speed limit During torque control	0～9000	1	3000	Effective immediately	Set when running	T
P06	17	Torque arrival command reference value	0～5000 (1000 corresponds to the rated torque of the motor)	0.1%	0	Effective immediately	Set when running	PST
P06	18	Torque arrival effective offset threshold	0～5000 (1000 corresponds to the rated torque of the motor)	0.1%	200	Effective immediately	Set when running	PST
P06	19	Torque arrival ineffective offset threshold	0～5000 (1000 corresponds to the rated torque of the motor)	0.1%	100	Effective immediately	Set when running	PST
P06	20	Speed limit window in torque mode	1~900	1ms	50	Effective immediately	Set when running	PST
P06	21	Multi-segment torque command 1	-3000～3000 (based on motor rated torque）	0.1%	0	Effective immediately	Set when running	T
P06	22	Multi-segment torque command 2	-3000～3000(based on motor rated torque）	0.1%	0	Effective immediately	Set when running	T
P06	23	Multi-segment torque command 3	-3000～3000 (based on motor rated torque）	0.1%	0	Effective immediately	Set when running	T

P07 group: gain parameter

Function code	Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P07	00	Position loop gain 1	10～20000	0.1HZ	320	Effective immediately	Set when running	P
P07	01	Speed loop gain 1	10～20000	0.1HZ	180	Effective immediately	Set when running	PS
P07	02	Speed loop integral time 1	15～512	0.01ms	3100	Effective immediately	Set when running	PS
P07	03	Speed detection filter 1	0～200	0.01ms	20	Effective immediately	Set when running	PST
P07	04	Torque command filtering 1	0～10000	0.01ms	126	Effective immediately	Set when running	PST
P07	05	Position loop gain 2	10～20000	0.1HZ	380	Effective immediately	Set when running	P
P07	06	Speed loop gain 2	10～20000	0.1HZ	180	Effective immediately	Set when running	PS
P07	07	Speed loop integral time 2	15～51200	0.01ms	51200	Effective immediately	Set when running	PS
P07	08	Speed detection filter 2	0～200	0.01ms	0	Effective immediately	Set when running	PST
P07	09	Torque command filtering 2	0～10000	0.01ms	126	Effective immediately	Set when running	PST
P07	10	DI function GAINSWITCH switching action selection	0-Speed loop regulator P (1) / PI(0) switch, gain is fixed to first group. 1-First gain (0), second gain (1) switching.	1	0	Effective immediately	Set when running	PS
P07	11	Gain switching mode	0-First gain fixed 1-Second gain fixed 2-Using DI input (GAINSWITCH) 3-Torque command greater 4-Speed command change greater 5-Speed command greater 6-Position deviation greater (P) 7-Receive position command (P) 8-Positioning is not completed (P) 9-Actual speed is greater (P) 10-Receive position command and actual speed (P) 11-Speed loop controller with PDFF control (PS) 12-Reserved 13-Speed loop controller with improved PI control (PS)	1	0	Effective immediately	Set when running	PS
P07	12	Gain switching delay	0~10000	0.1ms	50	Effective immediately	Set when running	PS
P07	13	Gain switching level	0~20000（unit: according to the gain switching mode description）	1	50	Effective immediately	Set when running	PS
P07	14	Hysteresis time of gain switching	0~20000（unit: according to the gain switching mode description）	1	33	Effective immediately	Set when running	PS
P07	15	Position gain switching time	0~10000	0.1ms	33	Effective immediately	Set when running	PS
P07	16	Speed regulator PDFF coefficient	0~1000	0.1%	700	Effective immediately	Set when running	PS
P07	17	Improved speed PI control level	2~9	1	5	Effective immediately	Set when running	PS
P07	18	Anti-integration saturation coefficient	0~1000	0.001	820	Effective immediately	Set when running	PS
P07	19	Speed feedforward control selection	0-No speed feedforward 1-Internal speed feedforward	1	1	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P07	20	Speed feedforward gain	0～1000	0.1%	0	Effective immediately	Set when running	P
P07	21	Speed feedforward filter time parameter	0～6400	0.01ms	50	Effective immediately	Set when running	P
P07	22	Torque feedforward selection	0-No torque feedforward 1-Internal torque feedforward	1	1	Effective immediately	Set when running	PS
P07	23	Torque feedforward gain	0～1000	0.1%	0	Effective immediately	Set when running	PS
P07	24	Torque feedforward filter time parameter	0～6400	0.01ms	50	Effective immediately	Set when running	PS

P08 group: advanced adjustment parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P08	00	Adaptive filter mode	0~5	1	0	Effective immediately	Set when running	PST
P08	01	Resonance frequency	0~65535	1Hz	0	N/A	Display parameters	PST
P08	02	The 1st notch filter frequency (manual)	10~4000	1Hz	4000	Effective immediately	Set when running	PST
P08	03	The 1st notch filter width	0~8	1	8	Effective immediately	Set when running	PST
P08	04	The 1st notch filter depth	0~100	1	50	Effective immediately	Set when running	PST
P08	05	The 2nd notch filter frequency (manual)	10~4000	1Hz	4000	Effective immediately	Set when running	PST
P08	06	The 2nd notch filter width	0~8	1	8	Effective immediately	Set when running	PST
P08	07	The 2nd notch filter depth	0~100	1	50	Effective immediately	Set when running	PST
P08	15	Damping filter switch	0-OFF 1-ON	1	0	Effective immediately	Setting after motor stop	PS
P08	16	Damping filter frequency	10~2000	0.1Hz	2000	Effective immediately	Setting after motor stop	PS
P08	17	Damping filter selection	0- Damping filter A 1- Damping filter B	1	1	Effective immediately	Setting after motor stop	PS
P08	18	Filter A width	0~200	1	4	Effective immediately	Setting after motor stop	PS
P08	19	Filter B gain	0~100	1	100	Effective immediately	Setting after motor stop	PS
P08	20	Offline inertia Identification maximum speed	200 ~1000	1rpm	500	Effective immediately	Setting after motor stop	PST
P08	21	Offline inertia Identification	50~800	1ms	100	Effective immediately	Setting after	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
		acceleration and deceleration time					motor stop	
P08	22	Waiting time after a single offline inertia identification is completed	100~10000	1ms	800	Effective immediately	Setting after motor stop	PST
P08	23	Inertia identification mode selection	0-Offline inertia identification: speed command is positive and negative triangle wave 1-Offline inertia identification: Speed command is JOG mode	1	0	Effective immediately	Setting after motor stop	PST
P08	24	Motor rotation round for complete single offline inertia identificatio	0~65535	0.01R	83	N/A	Display parameter	PST
P08	26	High frequency vibration suppression control switch	0-OFF 1-ON	1	0	Effective immediately	Setting after motor stop	PS
P08	27	High frequency vibration suppression frequency	10~4000	1 Hz	4000	Effective immediately	Set when running	PS
P08	28	High frequency vibration suppression gain compensation	0~1000	0.01	100	Effective immediately	Set when running	PS
P08	29	High frequency vibration suppression damping coefficient 1	0~1000	0.01	180	Effective immediately	Set when running	PS
P08	30	High frequency vibration suppression frequency compensation 1	-1000~1000	1 Hz	0	Effective immediately	Set when running	PS
P08	31	High frequency vibration suppression frequency compensation 2	-1000~1000	1 Hz	0	Effective immediately	Set when running	PS
P08	32	High frequency vibration suppression damping coefficient 2	0~1000	0.01	0	Effective immediately	Set when running	PS
P08	33	Anti-disturbance compensation switch	0-OFF 1-ON	1	0	Effective immediately	Setting after motor stop	PS
P08	36	Disturbance compensation factor	0~10000	0.1%	0	Effective immediately	Set when running	PS
P08	39	Momentary speed compensation switch	0-OFF 1-ON	1	0	Effective immediately	Setting after motor stop	PS
P08	40	Instantaneous speed compensation gain	0~1000	1 Hz	0	Effective immediately	Set when running	PS
P08	41	Instantaneous speed compensation gain compensation	0 ~1000	0.01	600	Effective immediately	Set when running	PS

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P08	45	Model compensation switch	0 - Turn off model compensation 1 - Rigid model 2-Second order vector model	1	0	Effective immediately	Set when running	PS
P08	46	Model compensation gain	10~20000	0.1/s	300	Effective immediately	Setting after motor stop	PS
P08	48	Model Compensation Forward Compensation Coefficient	0~10000	0.1%	1000	Effective immediately	Setting after motor stop	PS
P08	49	Model Compensation Reverse Compensation Coefficient	0~10000	0.1%	1000	Effective immediately	Setting after motor stop	PS
P08	50	Model compensation vibration suppression frequency A	0~2500	0.1Hz	500	Effective immediately	Setting after motor stop	PS
P08	51	Model compensation vibration suppression frequency R	0~2500	0.1Hz	500	Effective immediately	Setting after motor stop	PS
P08	52	Model Compensation Speed Compensation Coefficient	0~10000	0.1%	1000	Effective immediately	Setting after motor stop	PS

P09 group: failure and protection

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P09	02	Undervoltage detection delay	100 ~20000	0.1ms	700	Effective immediately	Set when running	PST
P09	04	Motor out of control protection	0- Protection open 1- Protection off	1	0	Effective immediately	Set when running	PST
P09	05	Overload warning value	1～100	1%	90	Effective immediately	Set when running	PST
P09	06	Motor overload protection factor	10 ～300	1%	100	Effective immediately	Set when running	PST
P09	07	Undervoltage protection point	50 ～100（100 corresponds to the default undervoltage point）	1%	100	Effective immediately	Set when running	PST
P09	08	Overspeed fault point	50 ～ 120 （ 100 corresponds to maximum motor speed）	1%	120	Effective immediately	Set when running	PST
P09	09	Position deviation excessive threshold (32bit)	1~1073741824	1P	2500 line motor: 32767P 17-bit motor: 393216 23-bit motor: 25165824	Effective immediately	Set when running	PST
P09	12	Low-speed command pulse input pin filter time	0-500	8.33ns	45	Re-power	Setting after motor stop	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P09	16	Average load rate overload threshold	100~3000	0.1%	1100	Effective immediately	Setting after motor stop	PST
P09	17	Multi-segment position/speed DI input delay detection time	0~1000	1ms	3	Effective immediately	Set when running	PS
P09	23	Motor stall over temperature protection time	10~65535	1ms	200	Effective immediately	Setting after motor stop	PST
P09	24	Motor stall over temperature protection enable	0 - Shield locked rotor over temperature protection monitoring detection 1- Turn on the motor stall and over-temperature protection monitoring and detection	1	0	Effective immediately	Set when running	PST
P09	25	Motor overload protection enable	0- Open motor overload and average load rate overload detection 1- Open motor overload, shield the average load rate overload detection 2- Shield motor overload, open average load rate overload detection 3- Shield motor overload and average load rate overload detection	1	0	Effective immediately	Setting after motor stop	PST
P09	29	Average load rate protection time	10~65535	160ms	300	Effective immediately	Setting after motor stop	PST

P10 group: communication parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P10	00	Communication address	0~247，0 is broadcast address	1	1	Effective immediately	Set when running	PST
P10	02	Modbus baud rate setting	0-2400 1-4800 2-9600 3-19200 4-38400 5-57600 6-115200	1	6	Effective immediately	Set when running	PST
P10	03	Modbus data format	0 -No parity, 2 stop bits 1-Even parity, 1 stop bit 2-Odd parity, 1 stop bit 3-No parity, 1 stop bit	1	0	Effective immediately	Set when running	PST
P10	04	Update writed communication function code to EEPROM?	0-Do not update EEPROM 1-Update to EEPROM except P11 group and P18 group	1	0	Effective immediately	Set when running	PST
P10	05	RS232 communication baud rate setting	0-2400 1-4800 2-9600 3-19200 4-38400 5-57600 6-115200	1	6	Effective immediately	Set when running	PST
P10	06	RS485 communication address selection	0- Dip switch Settings 1- Host computer communication Settings	1	0	Re-power	Setting after motor stop	PST

P11 group: auxiliary function parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P11	01	Fault reset	0-No operation 1-Fault reset	1	0	Effective immediately	Setting after motor stop	PST
P11	02	Soft reset	0-No operation 1-Fault reset	1	0	Effective immediately	Setting after motor stop	PST
P11	03	Inertia recognition function	Effective upon enter	1	0	Effective immediately	Set when running	PST
P11	06	Absolute encoder reset	0-No operation. 1-Absolute encoder related warning and error clearing. 2- Absolute encoder multiturn data reset.	1	0	Re-power	Setting after motor stop	PST
P11	07	Absolute value system soft limit setting	0 - no operation 1-The current position is set to the negative limit 2 - The current position is set as the positive limit	1	0	Effective immediately	Set when running	PST
P11	09	System initialization function	0-No operation 1-Restore factory setting (except P1 and P17 parameters) 2- Clear the fault record	1	0	Effective immediately	Setting after motor stop	PST
P11	10	DIDO forced input and output enable	0- No operation 1- Forced DI enable 2- Forced DO enable 3- Forced DIDO enabled	1	0	Effective immediately	Set when running	PST
P11	11	Set DI forced input	0-0x01FF	1	511	Effective immediately	Set when running	PST
P11	12	Set DO forced output	0-0x001F	1	0	Effective immediately	Set when running	PST
P11	13	Emergency stop settings	0 – No operation 1- Emergency stop	1	0	Effective immediately	Set when running	PST

P12 group: keyboard display parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P12	00	LED warning display selection	0-LED immediately outputs warning message 1-LED does not output warning message	1	0	Effective immediately	Set when running	PST
P12	01	Default display settings	0~100	1	1	Effective immediately	Set when running	PST
P12	03	Speed display filter time	0~10000	0.1ms	50	Effective immediately	Set when running	PST
P12	11	Non-standard version number	VV.B.DD	1	0	N/A	Display parameter	
P12	12	Software version number	VV.B.DD	1	0	N/A	Display parameter	
P12	13	FPGA version number	VV.B.DD	1	0	N/A	Display parameter	
P12	14	Product series code	PP.XXX	1	3000	N/A	Display parameter	

P13 group: multi-segment position

Function code	Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P13	00	Multi-segment position operation mode	0-Stop after the end of single operation (P13.01 for segment number selection) 1-Cycle operation (P13.01 for segment number selection) 2-DI switching operation (selected by DI)	1	1	Effective immediately	Setting after motor stop	P
P13	01	Specify starting segment	1～16	1	1	Effective immediately	Setting after motor stop	P
P13	02	Specify end segment	1～16	1	16	Effective immediately	Setting after motor stop	P
P13	03	Interrupted position handle setting	Valid in the other three modes except DI mode 0: Continue to run the segment that has not finished 1: Restart from the first segment	1	0	Effective immediately	Setting after motor stop	P
P13	04	Waiting time unit	0-ms 1-s	1	0	Effective immediately	Setting after motor stop	P
P13	05	Position control method	0-Incremental position control 1-Absolute position control	1	0	Effective immediately	Setting after motor stop	P
P13	08	Segment 1: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	10	Segment 1: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	11	Segment 1: acceleration/ deceleration time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	12	Segment 1: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	13	Segment 2: movement displacement (32bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	15	Segment 2: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	16	Segment 2: acceleration/ deceleration time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	17	Segment 2: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	18	Segment 3: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	20	Segment 3: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	21	Segment 3: acceleration/ deceleration time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	22	Segment 3: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P13	23	Segment 4: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	25	Segment 4: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	26	Segment 4: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	27	Segment 4: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	28	Segment 5: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	30	Segment 5: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	31	Segment 5: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	32	Segment 5: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	33	Segment 6: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	35	Segment 6: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	36	Segment 6: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	37	Segment 6: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	38	Segment 7: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	40	Segment 7: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	41	Segment 7: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	42	Segment 7: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	43	Segment 8: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	45	Segment 8: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P13	46	Segment 8: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	47	Segment 8: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	48	Segment 9: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	50	Segment 9: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	51	Segment 9: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	52	Segment 9: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	53	Segment 10: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	55	Segment 10: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	56	Segment 10: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	57	Segment 10: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	58	Segment 11: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	60	Segment 11: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	61	Segment 11: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	62	Segment 11: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	63	Segment 12: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	65	Segment 12: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	66	Segment 12: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	67	Segment 12: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P13	68	Segment 13: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	70	Segment 13: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	71	Segment 13: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	72	Segment 13: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	73	Segment 14: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	75	Segment 14: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	76	Segment 14: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	77	Segment 14: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	78	movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	80	Segment 15: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	81	Segment 15: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	82	Segment 15: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P
P13	83	Segment 16: movement displacement (32 bits)	-1073741824～1073741824	1Unit	10000	Effective immediately	Set when running	P
P13	85	Segment 16: maximum running speed	0～9000	1rpm	200	Effective immediately	Set when running	P
P13	86	Segment 16: acceleration/ deceler ation time	0~65535	1ms	1	Effective immediately	Set when running	P
P13	87	Segment 16: waiting time after completion	0～10000	1ms(s)	1000	Effective immediately	Set when running	P

P14 group Multi-speed command

Function code	Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P14	00	Multi-speed command operation mode	0- Stop after the end of single operation (P1401 for segment number selection) 1-Cycle operation (P1401 for segment number selection) 2-DI switching operation	1	1	Effective immediately	Setting after motor stop	S
P14	01	Speed command end segment selection	1～16	1	16	Effective immediately	Setting after motor stop	S
P14	02	Running time unit selection	0-s 1-min	1	0	Effective immediately	Setting after motor stop	S
P14	03	Acceleration/dec ele ration time 1	0-10000	1ms	0	Effective immediately	Setting after motor stop	S
P14	04	Acceleration/dec ele ration time 2	0-10000	1ms	0	Effective immediately	Setting after motor stop	S
P14	05	Acceleration/dec ele ration time 3	0-10000	1ms	0	Effective immediately	Setting after motor stop	S
P14	06	Acceleration/dec ele ration time 4	0-10000	1ms	0	Effective immediately	Setting after motor stop	S
P14	07	Speed segment 1	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	08	Running time of speed segment 1	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	09	Speed up and down time of speed segment 1	0- zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	10	Speed segment 2	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	11	Running time of speed segment 2	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	12	Speed up and down time of speed segment 2	0- zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	13	Speed segment 3	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	14	Running time of speed segment 3	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	15	Speed up and down time of speed segment 3	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	16	Speed segment 4	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	17	Running time of speed segment 4	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	18	Speed up and down time of speed segment 4	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	19	Speed segment 5	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P14	20	Running time of speed segment 5	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	21	Speed up and down time of speed segment 5	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	22	Speed segment 6	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	23	Running time of speed segment 6	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	24	Speed up and down time of speed segment 6	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	25	Speed segment 7	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	26	Running time of speed segment 7	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	27	Speed up and down time of speed segment 7	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	28	Speed segment 8	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	29	Running time of speed segment 8	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	30	Running time of speed segment 8	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	31	Speed segment 9	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	32	Running time of speed segment 9	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	33	Speed up and down time of speed segment 9	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	34	Speed segment 10	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	35	Running time of speed segment 10	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	36	Speed up and down time of speed segment 10	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	37	Speed segment 11	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	38	Running time of speed segment 11	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P14	39	Speed up and down time of speed segment 11	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	40	Speed segment 12	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	41	Running time of speed segment 12	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	42	Speed up and down time of speed segment 12	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	43	Speed segment 13	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	44	Running time of speed segment 13	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	45	Speed up and down time of speed segment 13	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	46	Speed segment 14	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	47	Running time of speed segment 14	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	48	Speed up and down time of speed segment 14	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	49	Speed segment 15	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	50	Running time of speed segment 15	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	51	Speed up and down time of speed segment 15	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S
P14	52	Speed segment 16	-9000～+9000	1rpm	0	Effective immediately	Setting after motor stop	S
P14	53	Running time of speed segment 16	0～65535	0.1s(min)	50	Effective immediately	Setting after motor stop	S
P14	54	Speed up and down time of speed segment 16	0-zero acc./dec. time 1- acc./dec. time 1 2- acc./dec. time 2 3- acc./dec. time 3 4- acc./dec. time 4	1	0	Effective immediately	Setting after motor stop	S

P16 group: special function parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P16	00	Interrupt fixed length enable	0-Disable 1-Enable	1	0	Effective immediately	Setting after motor stop	P
P16	01	Interrupt fixed length 1 displacement	0~1073741824	1 Unit	10000	Effective immediately	Set when running	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P16	03	Interrupt fixed length 1 constant running speed	0~9000	1rpm	200	Effective immediately	Set when running	P
P16	04	Interrupt fixed length acceleration time	0~1000	1ms	200	Effective immediately	Set when running	P
P16	05	Interrupt fixed length deceleration time	0~1000	1ms	200	Effective immediately	Set when running	P
P16	06	Fixed length lock release signal enable	0~1	1	1	Effective immediately	Set when running	P
P16	08	Homing enable control	0 - Turn off the Homing function. 1- Enable the Homing function by inputting the HomingStart signal through DI. 2-Start return to Home immediately after power-on. 3-Start return to Home immediately. 4- Take the current position as the Home. 5- Set the Home through DI trigger. 6- host computer homing	1	0	Effective immediately	Setting after motor stop	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P16	09	Homing mode	0-Forward homing, deceleration and home points are home switch; 1- Backward homing, deceleration and home points are home switch; 2- Forward homing, deceleration and home points are motor Z signal; 3- Backward homing, deceleration and home points are motor Z signal; 4- Forward homing, deceleration point is Home switch, home is motor Z signal; 5- Backward homing, deceleration point is Home switch, home is motor Z signal; 6- Forward homing, deceleration and home points are forward overtravel switch; 7- Backward homing, deceleration and home points are backward overtravel switch; 8- Forward homing, deceleration point is forward overtravel switch, home is motor Z signal; 9- Backward homing, deceleration point is backward overtravel switch, home is motor Z signal; 10- Forward homing, no deceleration point, home is home switch; 11-Backward homing, no deceleration point, home is home switch; 12- Forward homing, no deceleration point, home is motor Z signal; 13- Backward homing, no deceleration point, home is motor Z signal; 14- Borward homing, no deceleration point, home is forward overtravel switch; 15- Backward homing, no deceleration point, home is backward overtravel switch; 16-Absolute value homing; 17- Indexing absolute value homing;	1	0	Effective immediately	Setting after motor stop	P
P16	10	High-speed for searching home switch signal	10-3000	1rpm	100	Effective immediately	Setting after motor stop	P
P16	11	Low-speed for searching home switch signal	10 -1000	1rpm	10	Effective immediately	Setting after motor stop	P
P16	12	Limit the accel. /decel. time for searching home	0-65535	1ms	1000	Effective immediately	Setting after motor stop	P
P16	13	Limit time for finding home	0-65535	1ms	10000	Effective immediately	Setting after motor stop	P
P16	14	Mechanical home offset (32 bits)	-1073741824 ~ 1073741824	1Unit	0	Effective immediately	Setting after motor stop	P

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P16	28	The origin corresponds to the absolute position of the encoder	0~4294967295	1inc	0	Effective immediately	Set when running	P
P16	30	The origin corresponds to the number of encoder turns	0~32767	1inc	0	Effective immediately	Set when running	P
P16	31	The return to zero process position reaches wait count	0~65535	1ms	500	Effective immediately	Set when running	P
P16	37	Interrupt fixed length 2 displacement	-1073741824~1073741824	1 Unit	0	Effective immediately	Set when running	P
P16	39	Interrupt fixed length 2 constant running speed	0~9000	1rpm	200	Effective immediately	Set when running	P

P18 group: display parameters

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P18	00	Servo status	Rdy 、Run 、Err.00~99（ error ） ,AL.00~10(warning）	1	-	N/A	Display parameter	PST
P18	01	Motor speed feedback (32 bit)	-9000~9000	1rpm	-	N/A	Display parameter	PST
P18	02	average load rate	0~3000	0.1%	-	N/A	Display parameter	PST
P18	03	Speed command	-9000~9000	1rpm	-	N/A	Display parameter	PST
P18	04	Internal torque command (relative to rated torque)	-5000~5000	0.1%	-	N/A	Display parameter	PST
P18	05	Phase current RMS	0~10000	0.01A	-	N/A	Display parameter	PST
P18	06	Bus voltage value	0~10000	0.1V	-	N/A	Display parameter	PST
P18	07	Absolute position counter (32 bit)	-1073741824 ~ 1073741824	1Unit	-	N/A	Display parameter	PST
P18	09	Electrical angle	0~3600	0.1°	-	N/A	Display parameter	PST
P18	10	Mechanical angle (relative to encoder zero)	0~3600	0.1°	-	N/A	Display parameter	PST
P18	11	Bus encoder communication check error counter	-	1	-	N/A	Display parameter	PST
P18	12	Corresponding speed information of input position command	-9000~9000	1rpm	-	N/A	Display parameter	PST
P18	13	Position deviation counter (32 bit)	-1073741824 ~ 1073741824	1P	-	N/A	Display parameter	PST
P18	15	Input command Pulse counter (32 bit)	-1073741824 ~ 1073741824	1Unit	-	N/A	Display parameter	PST
P18	17	Feedback pulse counter (32 bit)	-1073741824 ~ 1073741824	1P	-	N/A	Display parameter	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P18	19	Position deviation counter command unit (32 bit)	-1073741824 ~ 1073741824	1Unit	-	N/A	Display parameter	PST
P18	21	Digital input signal monitoring	-	-	-	N/A	Display parameter	PST
P18	22	Record the U-phase zero current value during shutdown	-32768~32767	0.01A	-	N/A	Display parameter	PST
P18	23	Digital output signal monitoring	-	-	-	N/A	Display parameter	PST
P18	24	Record the V-phase zero current value during shutdown	-32768~32767	0.01A	-	N/A	Display parameter	PST
P18	25	Total power-on time (32-bit)	0-4294967295	0.1s	-	N/A	Display parameter	PST
P18	27	U-Phase zero sampling value	0~65535	1	-	N/A	Display parameter	PST
P18	28	V-Phase zero sampling value	0~65535	1	-	N/A	Display parameter	PST
P18	29	U-phase current sampling value	0~65535	1	-	N/A	Display parameter	PST
P18	30	V-phase current sampling value	0~65535	1	-	N/A	Display parameter	PST
P18	31	Module temperature value	-	1°C	-	N/A	Display parameter	PST
P18	32	Absolute encoder single-round data	-	pulse	-	N/A	Display parameter	PST
P18	34	Absolute encoder multi-round data	-	turn	-	N/A	Display parameter	PST
P18	35	Maximum load rate	0~3000	0.1%	-	N/A	Display parameter	PST
P18	40	Display of fault record	0- Current fault 1- The first fault before current fault 2- The second fault before current fault ..... 9- The 9th fault before current fault	1	0	Effective immediately	Set when running	PST
P18	41	Error code	-	-	-	N/A	Display parameter	PST
P18	42	Selected fault timestamp (32 bit)	-	0.1s	-	N/A	Display parameter	PST
P18	44	In-time speed at selected fault	-	1rpm	-	N/A	Display parameter	PST
P18	45	In-time current U at Selected fault	-	0.01A	-	N/A	Display parameter	PST
P18	46	In-time current V at selected fault	-	0.01A		N/A	Display parameter	PST
P18	47	Bus voltage at selected fault	-	0.1V	-	N/A	Display parameter	PST
P18	48	Input terminal status at selected fault	-	-	-	N/A	Display parameter	PST

Function code		Name	Setting range	Min. unit	Default setting	Effective time	Type	Mode
P18	49	Output terminal status at selected fault	-	-	-	N/A	Display parameter	PST

Appendix A: Alarm light information description

Fault	Decription	Red light state			
		Flicker frequency	Flicker number	Flicker times	Cyclic flicker
Voltage fault	Overvoltage ,Undervoltage (The main circuit)	1HZ	1	3 seconds	YES
Current fault	Overcurrent (Overcurrent A, Overcurrent B)	1HZ	2	3 seconds	YES
Overload/ Motor stall	Drive overload, Motor overload, Motor stall	1HZ	3	3 seconds	YES
Parameter setting error	DI/DO terminal parameter setting fault，Soft limit setting erro	1HZ	4	3 seconds	YES
Encoder error	Encoder matching fault， Encoder disconnection，Encoder check exception	1HZ	5	3 seconds	YES
Motor verification erro	Moto data verification failure	1HZ	6	3 seconds	YES
Remaining faults	Faults other than the above faults	Stay on			
No faults	-	Stay off			

Appendix B: RS-485 Communication address calculation description

1.1 Communication signal port

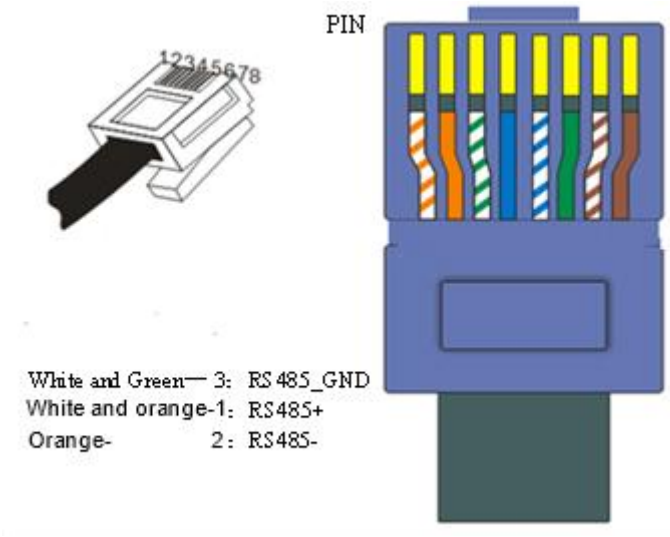
DS3 series servo crystal port Physical interface, support standard Modbus protocol, CN3 and CN4 all can communicate, crystal port pin description is as follows:

Pin	Definition	Description
3	RS485_GND	RS485 signal PE
1	RS485+	RS485 signalling+
2	RS485-	RS485 signalling -
Connector housing	PE	ground connection

1.2 Wiring instructions and EMC precautions

1.2.1 This section describes how to make RS485 network cables

If the network cable crystal head is made according to the 568A or 568B standard, only the blue and white and blue cables are drawn out at the other end, and it can be used as an RS485 communication cable (using twisted-pair network cables, the communication distance is up to 300m). The specific wiring diagram is as follows：



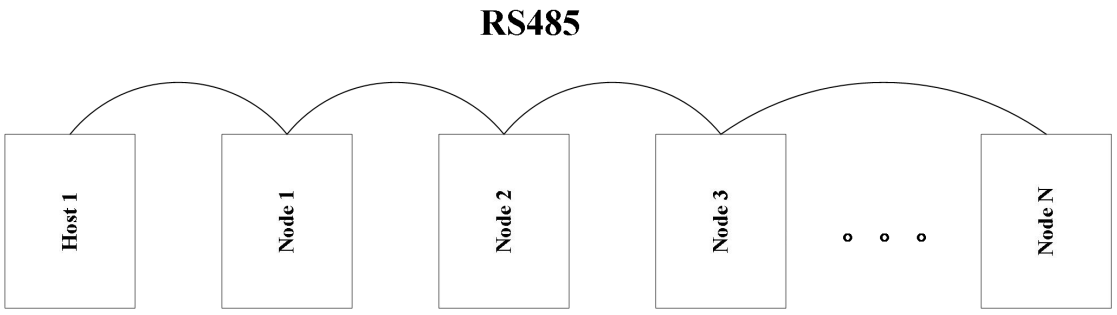
Network plug 568B serial connection method

1.2.2 Multi-computer Communication

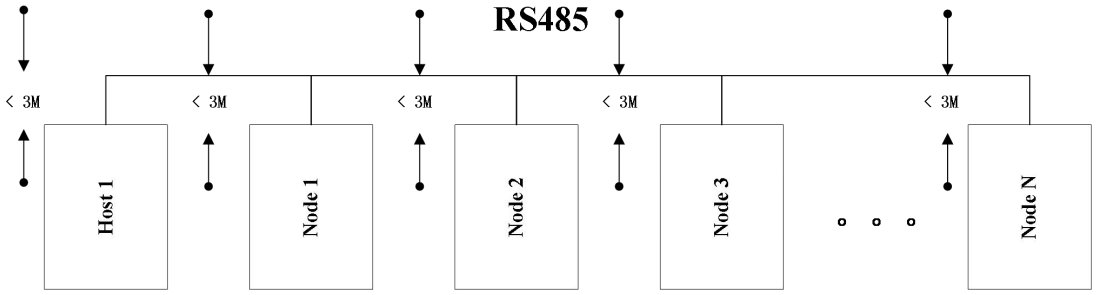
In the project engineering environment, when multiple servos need to communicate at the same time, 485 bus is recommended to use the hand-in-hand bus structure.

The shorter the branch length between the bus and the node, the shorter the better. It is recommended that the branch length not exceed 3m. The general bus structure is as follows:

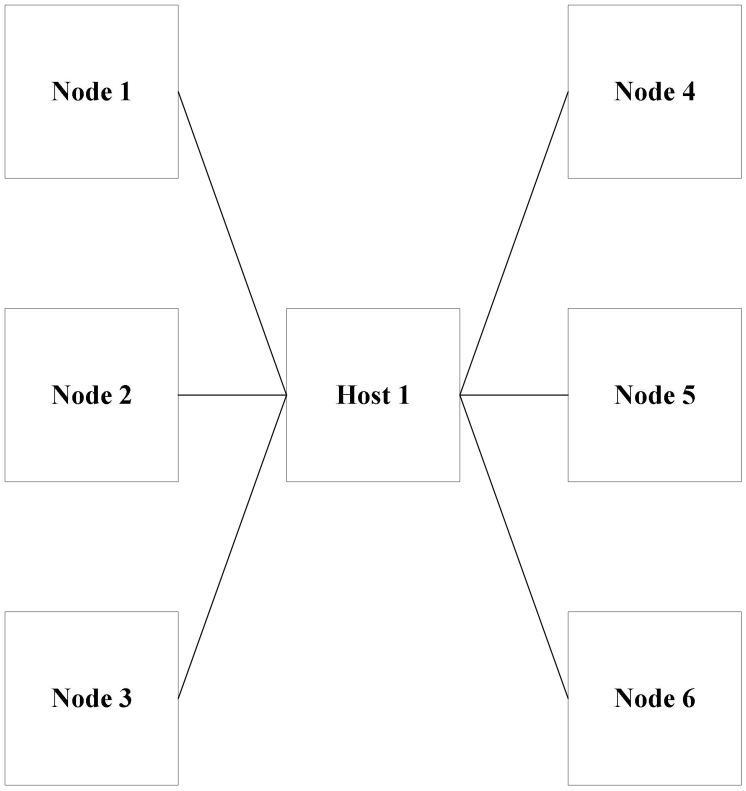
a) Recommended scheme: Join the structure hand in hand



b) General scheme: branch line connection structure

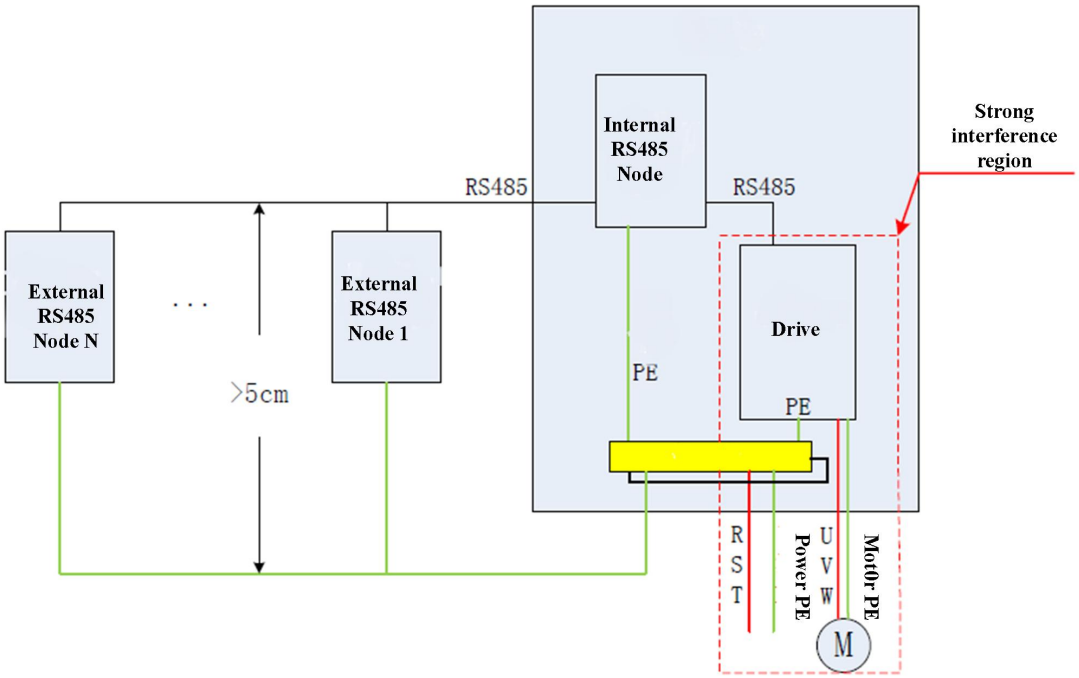


c) Error scheme: Star join structure



1.2.3 EMC layout requirements

1) Layout requirements: Interference sources are isolated from sensitive devices



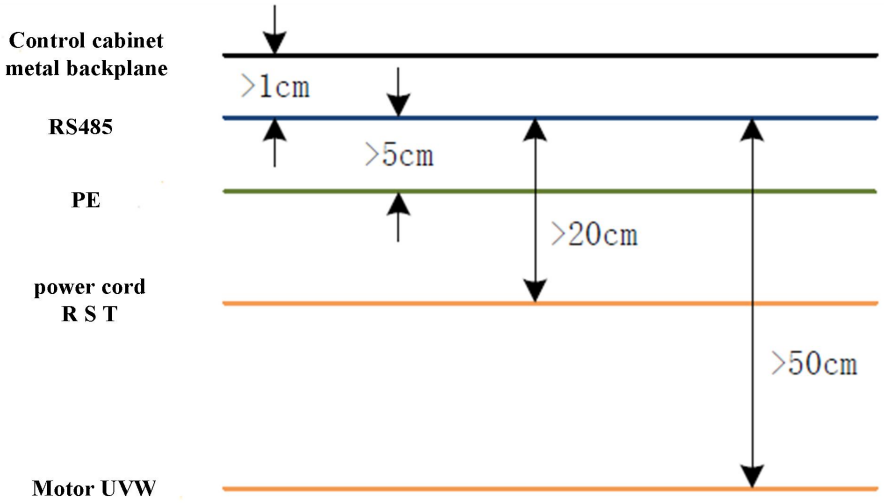
3) Ground PE connection requirements:

a. The ground cable must be thicker than the AWG12.

b. Ground cable The PE connects to the ground terminal of a node or the ground bar of the cabinet where the node is located

c. The distance between the PE and the bus is greater than 5cm

4) Cable layout requirements：



a. Keep the RS485 bus at least 1cm away from the metal backplane of the control cabinet.

b. Keep the RS485 bus at least 5cm away from the ground cable.

c. Keep a distance of at least 20cm between the RS485 bus and the field strong current cable.

d. The distance between the RS485 bus and the field motor power line should be at least 50cm.

1.3 Communication related parameter

Function code	Name	Description	Setting range	Default setting
P1000	Communication address	0 : broadcast address 1~247: Slave address	0～247	1
P1002	Modbus baud rate setting	0-2400 1-4800 2-9600 3-19200 4-38400 5-57600 6-115200	0～6	6

P1003	Modbus data format	0 -No parity, 2 stop bits 1-Even parity, 1 stop bit 2-Odd parity, 1 stop bit 3-No parity, 1 stop bit	0～3	0
P1004	Update writed communication function code to EEPROM?	0-Do not update EEPROM 1-Update to EEPROM except P11 group and P18 group	0～1	0

Explain:

1) In the case of long transmission cable and serious interference, the servo communication line should be shielded twisted pair cable, and the Modbus data format should be set to odd check or even check as far as possible.

2) The servo can only do the slave station, the 0 communication address is used for the master station broadcast, the servo communication address is prohibited to set to 0.

When multiple servos communicate on the network, each drive has a unique communication address, otherwise the communication will be abnormal. RS485 communication, can connect 32 servos at the same time, if you want to connect more servos, you must add amplifier, up to 247 servos.

5) The baud rate of servo communication must be consistent with that of host computer, otherwise it is easy to have abnormal communication.

6) Servo provides three types of data verification methods: odd check (P1003:2), even check (P1003:1), no check (P1003:0 \3)) :

A) Odd check or even check

The actual transmission of data per frame is 11 bits, 1 start bit, 8 data bit, 1 check bit, and 1 end bit.

10	9	8～1	0
End bit	Check bit	Data bit	Start bit

B）No check: The data frame has two formats

① The actual transmission of data per frame is 11 bits, 1 starting bit, 8 data bits, and 2 end bits.

10～9	8～1	0
End bit	Data bit	Start bit

② The actual transmission of data per frame is 11 bits, 1 starting bit, 8 data bits, and 1 end bits.

10	9	8～1	0
Invalid bit	End bit	Data bit	Start bit

The data frame format of the upper computer must conform to the above format, otherwise it cannot communicate with the driver.

5） The P1004 address parameter takes effect immediately, and its change value is saved to EEPROM. During the communication task, if no power failure is required to save parameters, set P1004 to 0. Otherwise, if the function code value is changed in large quantities for a long time and saved to the EEPROM, the EEPROM will be damaged and the server Er.04(parameter storage) will fail.

1.4 ModBus Protocol

DS3 series servo driver embedded standard Modbus RTU communication protocol, support Modbus master station read and write a single, multiple parameters operation; Among them, the three most commonly used commands can satisfy the controller's all-round control of the servo drive, and the specific function code is as follows

Function code（CMD)	Decription

16#03	Read single or multiple driver register parameters
16#06	Read single driver register parameters
16#10	Read multiple driver register parameters

The Modbus communication address corresponding to the internal parameter group of the servo meets the following relation:P_{MmNn} =Mm*256+Nn, This data is a decimal number, Mm indicates the parameter group and Nn indicates the corresponding server address under the parameter group. For example, the P1801 address in the servo corresponds to the Modbus communication address of 4609(18*256+01), which is converted to the hexadecimal address of 16#1201.

When PLC and touch screen MODBUS programming of some manufacturers, the register address is not equal to the actual register address, but is equal to the actual register address plus 1 ,Because the standard MODBUS instruction register starts at 1, the device register address starts at 0. Therefore, when this kind of PLC and touch screen are addressed by MODBUS communication with the servo, the programmer must clarify the addressing method to correctly read and write the servo address parameters, and when it is unclear, you can consult the product manufacturer.

1.4.1 Read parameter（16#03）

The following request frame represents reading 2-word data starting with the P1801 (motor speed feedback) parameter from a drive with address 01.

Request frame：

Format	Slave Adr	CMD	Start Adr H	Start Adr L	No. of Regs H	No. of Regs L	CRC	
	1Byte	1Byte	2Bytes		2Bytes		2Bytes	
Example	16#01	16#03	16#12	16#01	16#00	16#02	XXXX	XXXX

Slave Adr	Drive communication address, that is, the P1000 communication address set value, which needs to be converted to a hexadecimal number.
CMD	Function code, 16#03 is read function code operation.

Start Adr H/L	The start address must be converted to a hexadecimal number. In this example, P1801 corresponds to the decimal Modbus address number of 4609 and is converted to the hexadecimal number of 16#1201. Therefore, the high and low values are 16#12 and 16#01.
No.of Regs H/L	The number of read parameters is high or low, and the start address value must be converted to a hexadecimal number. In this example, there are two parameters (16#0002), so the high value is 16#00, and the low value is 16#02.
CRC	CRC check word.

If the read is successful, the reply frame:

Format	Slave Adr	CMD	Data Length	Data 0	Data1	...	Data n×2-2	Data n×2-1	CRC	
	1Byte	1Byte	1Byte	2Bytes		...	2Bytes		2Bytes	
Example	16#01	16#03	16#04	16#00	16#00	/	16#00	16#00	XXX	XXX

Slave Adr	Drive communication address, that is, the P1000 communication address set value, which needs to be converted to a hexadecimal number.
CMD	Function code, 16#03 is read function code operation.
Data Length	Data byte length, equal to the number of read parameters (No.of Regs)*2.
Data0/Data1	Read the start parameter value 8 bits higher/Start parameter value 8 bits lower.
…	
Data n*2-1	Read the last parameter value 8 bits lower.
CRC	CRC check word.

1.4.2 Write single parameter（16#06）

Case requirements under the frame, said to the correspondence address of 01 drive, a single

parameter P0503 (speed instruction set keyboard) writing new value in 500.

Request frame：

Format	Slave Adr	CMD	Reg Adr H	Reg Adr L	Preset Data H	Preset Data L	CRC	
	1Byte	1Byte	2Bytes		2Bytes		2Bytes	
Example	16#01	16#06	16#01	16#44	16#01	16#F4	XXXX	XXXX

Slave Adr	Drive communication address, that is, the P1000 communication address set value, which needs to be converted to a hexadecimal number.
CMD	Function code, 16#06 is to write a single number of operations.
Reg Adr H/L	The value of the start address must be converted to a hexadecimal number. In this example, P0503 corresponds to the decimal Modbus address number of 1283 and is converted to 16#0503 in hexadecimal. Therefore, the high value is 16#05 and the low value is 16#03.
Preset Data H/L	When the data is written with high or low bytes, the data must be converted into hexadecimal numbers. In this example, 500 is converted into 16#01F4 in hexadecimal, so the high byte is 16#01 and the low byte is 16#F4.
CRC	CRC check word.

If read successfully, reply frame:

Format	Slave Adr	CMD	Reg Adr H	Reg Adr L	Preset Data H	Preset Data L	CRC	
	1Byte	1Byte	2Bytes		2Bytes		2Bytes	
Example	16#01	16#06	16#01	16#44	16#01	16#F4	XXXX	XXXX

1.4.3 Write multiple arguments（16#10）

The following request frame indicates that new values 200 and 300 are continuously written to the two parameters P0505/P0506(speed instruction acceleration and deceleration time) to the communication address of drive 01.

Request frame：

Format	Slave Adr	CMD	Start Adr H	Start Adr L	No. of Regs H	No. of Regs L	Data Length	Data 0	Data 1	…	Data n×2-2	Data n×2-1	CRC	
	1Byte	1Byte	2Bytes		2Bytes		1Byte	2Bytes		…	2Bytes		2Bytes	
Example	16#01	16#10	16#05	16#05	16#00	16#02	16#04	16#00	16#C8	/	16#01	16#2C	XXXX	XXXX

Slave Adr	Drive communication address, that is, the P1000 communication address set value, which needs to be converted to a hexadecimal number.
CMD	Function code, 16#10 is to write multiple parameter operations.
Start Adr H/L	The value of the start address must be converted into a hexadecimal number. In this example, P0505 corresponds to the decimal Modbus address number of 1285, and is converted into a hexadecimal number of 16#0505, so the high value is 16#05 and the low value is 16#05.
No.of Regs H/L	The number of write parameters is high/low, and the start address value must be converted to a hexadecimal number. In this example, there are two parameters (16#0002), so the high value is 16#00, and the low value is 16#02.
Data Length	The length of data bytes, equal to the number of write parameters (No.of Regs)*2, must be converted to hexadecimal.
Data0/Data1	The start parameter value is 8 bits higher or the start parameter value is 8 bits lower. 200 corresponds to 16#C8.
…	
Datan*2-1	Read the last parameter value is 8 bits lower; 300 corresponds to 16#012C.
CRC	CRC check word.

If multiple writes are successful, the reply frame:

Format	Slave Adr	CMD	Start Adr H	Start Adr L	No.of Regs H	No.of Regs L	CRC	
	1Byte	1Byte	2Bytes		2Bytes		2Bytes	
Example	16#01	16#10	16#05	16#05	16#00	16#02	XXXX	XXXX

Slave Adr	Drive communication address, that is, the P1000 communication address set value, which needs to be converted to a hexadecimal number.
CMD	Function code, 16#10 is to write multiple parameter operations.
Start Adr H/L	The value of the start address must be converted into a hexadecimal number. In this example, P0505 corresponds to the decimal Modbus address number of 1285, and is converted into a hexadecimal number of 16#0505, so the high value is 16#05 and the low value is 16#05.
No.of Regs H/L	The number of write parameters is high or low. The start address must be converted to a hexadecimal number. In this example, there are two parameters (16#0002), so the high value is 16#00 and the low value is 16#02.
CRC	CRC check word.

1.4.4 Error code

The driver receives the MODBUS communication message from the master station without transmission error, but the slave station does not answer correctly. In this case, the slave station will reply with an abnormal response message.

Exception response frame format: In this example, indicates the exception response frame when writing a single drive register parameter.

Format	Slave Adr	CMD	Error H	Error L	CRC	
	1Byte	1Byte	2Bytes		2Bytes	
Example	16#01	16#86	16#00	16#02	XXXX	XXXX

Slave Adr	Drive communication address, the P1000 communication address set value, which needs to be converted to a hexadecimal number.
CMD	Function code: The highest bit (7th bit) of the function code of the master station is set to 1 in the abnormal response frame. In this example, the request frame of the master station is written into a single parameter bit 16#06, and the highest position of the function code of the master station is set to 1 in the abnormal response frame, which is 16#86.
Error H/L	The error code takes 2 bytes.
CRC	CRC check word.

Error codes are described in the following table:

Error code	Name	Decription
16#0001	Illegal command code	The function code command is incorrect.
16#0002	Illegal data address	The write parameter range is incorrect or contains reserved parameters, and the parameters are read-only.
16#0003	Illegal data	The parameter group address exceeds the range that can be set by the server.
16#0004	The slave station device is faulty	The communication interface of the slave server is abnormal and the CRC check error occurs.
16#0006	Busy, refused to execute.	The servo slave is performing EEPROM storage, or is on enable.

1.4 .5 Function code with signed number

When writing single or multiple parameters, the data may be signed. In this case, the written data needs to be converted to the corresponding complement.

1）16 bit function code

a) When the data is positive or 0, complement = source code.

b) For negative data bits, the complement =0xFFFF- the complement of the absolute value of the

data +0x0001.

For example, in the 16-digit 100 code bit 0x0064, the complement is: 0x0064；

16-digit -100 complement: 0xFFFF-0x0064+0x0001=FF9C

2）32-bit function code

a) When the data is positive or 0, complement = source code.

b) For negative data bits, the complement =0xFFFFFFFF- the complement of the absolute value of the data +0x00000001

For example, if the 32-bit 100 code bit is 0x00000064, the complement is: 0x00000064；

32 bits -100 complement: 0xFFFFFFFF-0x00000064+0x00000001=FFFFFF9C.

1.4.6 Addresses of parameters that can be read and written by MODBUS

The server address that can be read and written through the MODBUS communication task, see the parameter group address table;

Parameter group address table:

Parameter group number	Parameter group description
P00	Basic control parameters
P01	Servo motor parameters
P02	Digital input and output parameters
P04	Position control parameters
P05	Speed control parameters
P06	Torque control parameters
P07	Gain parameters
P08	Advanced adjustment parameters
P09	Fault and protection parameters
P10	Communication parameters
P11	Auxiliary function parameters
P12	Keyboard display parameters
P13	Multi-segment position function parameters
P14	Multi-speed function parameters
P16	Special function parameters
P17	Drive parameters
P18	Display parameters

If the communication task is to write multiple data in a row, pay attention to whether the parameter

addresses in the write task are consecutive addresses in the servo parameter group. If they are not consecutive, the communication task will be affected. In this case, you need to split the task. Whether the address is continuous can be seen in the detailed servo address parameter table or through the servo host computer.

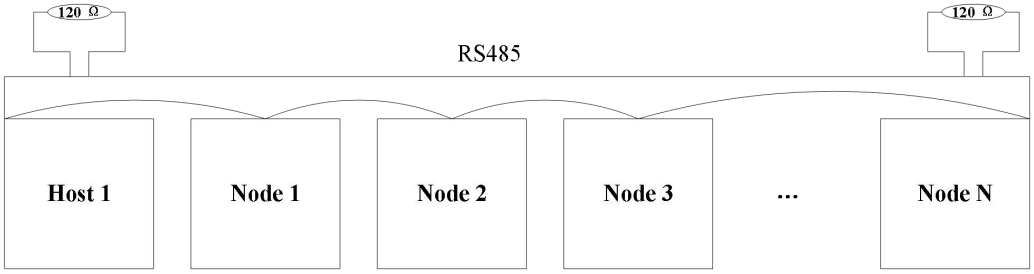
If you write data to P02 group parameters, write 12 numbers starting from P0201 to P0212, the write task fails. In the P02 servo parameter group, the addresses P0201 to P0212 are discontinuous, and the hidden address P0209 affects the communication task. Add a write task from P0201 to P0208 as a write task and P0210 to P0212 as another write task.

When using PLC to write the MODBUS communication task program, we should pay special attention to the above phenomenon, otherwise there will be a communication task failure.

1.5 485 Precautions for onsite communication

1.5.1 Terminal resistance treatment

When the 485RS communication is unstable, the access terminal resistance can be considered to eliminate the reflected signal in the communication cable.



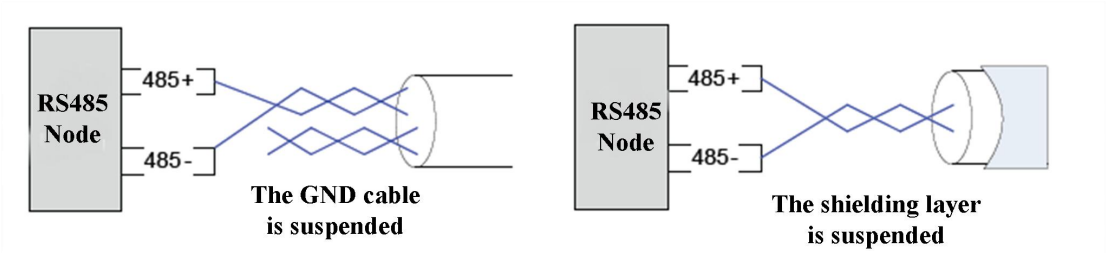
a. Only the resistors matched on both ends of the communication network can be connected.

b. It is recommended that the master station be placed at one end of the bus.

c. In the case of power failure, after connecting the terminal resistance, you can use a multimeter ohm to measure the resistance between 485 buses, and the normal measurement value is 60Ω. If the displayed value is less than 50 ohms, check whether other nodes have added matching resistors and disconnect them. If 0 Ohm is displayed, check whether the node is short-circuited or damaged.

1.5.2 GND wiring point processing

For nodes that do not have a GND connection point, refer to the following methods for handling communication cables:



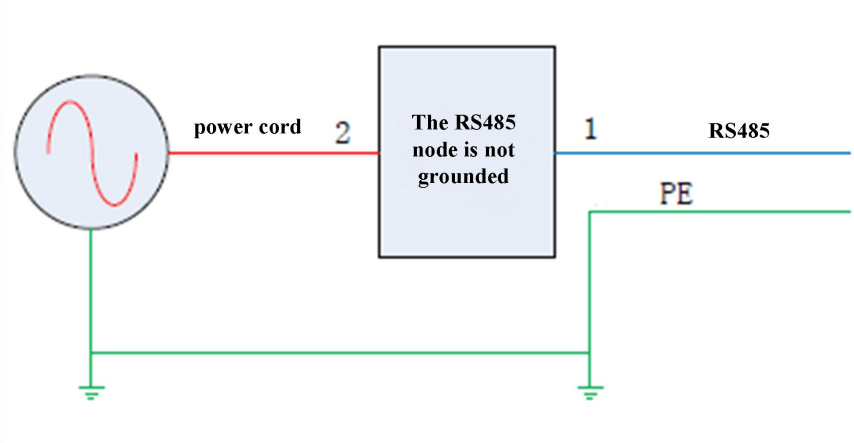
a. Find out whether there is a reference ground shared with the 485 circuit on other ports of this node. If there is, GND is connected to this reference ground. Note that the shield layer cannot be connected to the reference ground, otherwise the 485 port will be damaged.

b. Check whether there is a reference ground shared with the 485 circuit on this node board. If there is, GND is connected to this reference ground. Note that the shield layer cannot be connected to the reference ground, otherwise the 485 port will be damaged.

c. If the reference ground of the 485 circuit cannot be found, suspend the GND cable and ensure that the ground cable is reliably connected to the PE.

1.5.3 System external interference processing

To effectively suppress the impact of external interference on the communication, you can take the following measures.

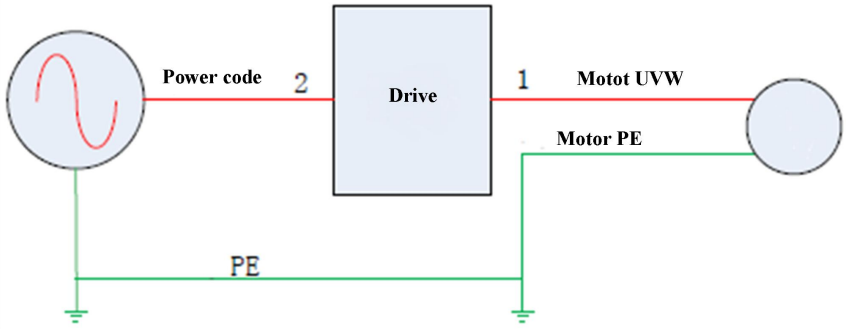


a. Wrap a magnetic loop at position 1 to suppress external interference from the system;

b. Wrap a magnetic loop at position 2 to suppress external interference from the system.

1.5.4 Driver interference processing

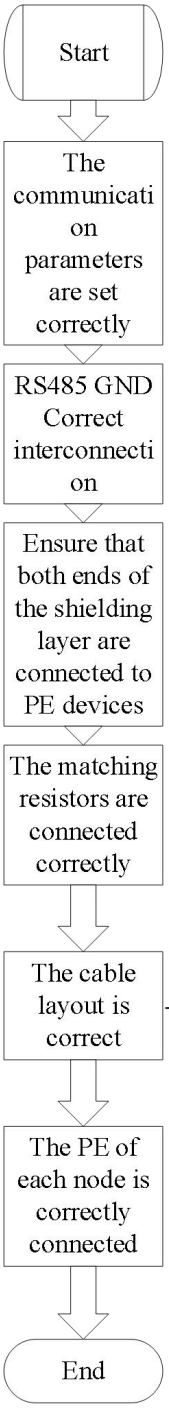
When the driver is running, it will also affect the communication. In order to effectively suppress the interference of the driver itself, you can refer to the following measures.



a. Add a filter magnetic ring at position 1. Three UVW wires (excluding ground wire PE) pass through the magnetic ring at the same time. Three turns are recommended.

b. Wrap a magnetic ring at position 2, add a filter magnetic ring at position 2, and the power cable (excluding ground PE) passes through the magnetic ring at the same time. It is recommended that you wind three turns.

Process for locating onsite problems:



- Communication address P1000: Unique for each node
- Communication baud rate Set P1002: Each node maintains one baud rate
- Communication data format P1003: One for each node

- The reliable connection of RS485 GND can effectively improve the EMC effect

- The shielding layer of the network cable can only be suspended or connected to PE at both ends. Do not connect to RS485GND, otherwise the port will be damaged.

- Matching resistors can only be connected at both ends

- The distance between the 485 bus and the backplane of the metal cabinet must be at least 1cm
- The distance between the 485 bus and the ground cable PE should be at least 5cm
- Keep a distance of at least 20cm between the 485 bus and the strong-current cable
- The distance between the 485 bus and the U, V and W cables of the motor should be at least 50cm

Appendix C: Brake resistor configuration

When the servo motor is in the state of deceleration braking, the motor will be in the state of generating electricity and feed the electric energy back to the servo driver bus circuit. When the driver bus voltage detection value is higher than the set threshold, the driver will report the overvoltage warning, and the energy generated by the motor needs to be consumed by an external brake resistor. Connect the brake resistance through RB+ and RB- at the power end, and correctly set the brake resistance value and brake resistance power. The resistance value of the optional brake resistance cannot be lower than the recommended resistance value. The recommended parameters of the brake resistance are shown in the following table:

Mode	Brake resistance value (Ω)	Brake resistance power（W）
DSY-200RS	10	100
DSY-400RS	10	100
DSY-750RS	5	100
DSY-1200RS	3.5	200

Appendix D: Version change log

Date	Modified version	Content of change
2024.12	V1.0	First release